A LOCAL APPROXIMATION CHARACTERIZATION OF OKA MANIFOLDS

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Abstract. We characterize Oka manifolds by the local approximation property (LAP), a planar approximation operation that preserves holomorphic dependence on prescribed parameter polydiscs. We prove that LAP yields convex approximation in every source dimension and propagates across sets of zero $(2n-1)$-dimensional Hausdorff measure in an n-dimensional target. Consequently, strong dominability characterizes Oka manifolds, convex approximation in the target dimension suffices, the Oka property is local in the Euclidean topology, and it descends under surjective holomorphic maps. Local families of entire curves give geometric applications: smooth projective rationally connected manifolds are Oka, and deleting an algebraic subset of codimension at least two from any such manifold preserves the Oka property. Structure theorems further show that compact Kähler manifolds with nonnegative holomorphic sectional curvature or nef tangent bundle are Oka. Finally, the complement of a smooth degree-d hypersurface in $\mathbb{P}^n$ is Oka exactly when $d \leq n+1$.

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1. Introduction

The Oka principle concerns the existence of holomorphic solutions to problems on Stein spaces when there is no topological obstruction, provided the target is flexible enough. It is not a single theorem but a program: one looks for geometric conditions on a complex manifold Y under which continuous maps from Stein spaces into Y can be deformed to holomorphic maps, with approximation, interpolation, and parametric control. For broad accounts, see the survey of Forstnerič and Lárusson [30] and Forstnerič's recent exposition [28]. The subject began with Oka's solution of the second Cousin problem [55] and Grauert's theorem for holomorphic fiber bundles with homogeneous fibers over Stein spaces [38]. Gromov placed it in the framework of the h-principle and introduced ellipticity through dominating holomorphic sprays [39], and Forstnerič introduced subellipticity and proved the Oka principle for subelliptic manifolds [18]. Forstnerič then established approximation, interpolation, and finite-order jet interpolation [20, Proposition 1.2 and Corollary 1.3], identified convex approximation with the Oka property with approximation [21, Theorem 0.1], and proved that convex approximation implies the full parametric Oka property [22, Theorem 1.1]. This is the notion of an Oka manifold used below [22, Definition 1.2]; see also Forstnerič's monograph [25].

Here we show that this global holomorphic flexibility is detected by a local approximation operation: planar approximation with holomorphic parameters characterizes Oka manifolds. The characterization also shows that the Oka property propagates across proper analytic subsets and descends under surjective holomorphic maps from Oka manifolds, leading to applications to projective manifolds, complements, and spaces of rational maps.

Definition 1.1 ([22, Definition 1.2])**.** A complex manifold Y is an *Oka manifold* if it has the *convex approximation property* CAP: for every $m \geq 1$, compact convex set $K \subset \mathbb{C}^m$, holomorphic map $f : U \to Y$ on an open neighborhood U of K, and $\epsilon > 0$, there is $F \in \mathscr{O}(\mathbb{C}^m, Y)$ such that

$$\sup_{z \in K} d_Y(F(z), f(z)) < \epsilon,$$

where d_Y is any distance inducing the topology of Y.

Unless explicitly stated otherwise, complex manifolds are connected. Forstnerič's work shows that convex approximation implies the approximation and interpolation forms of the Oka principle for maps from reduced Stein spaces [21, 25].

The *local approximation property* (LAP), denoted by $\mathcal{L}_n$ for an n-dimensional target, asks for approximation while enlarging a planar source domain and retaining holomorphic dependence on every smaller parameter polydisc prescribed in advance. The given maps take values in a

fixed neighborhood of a target point, and their approximants take values in the whole target. Definition 2.1 gives the precise formulation.

Theorem 1.2. *A complex manifold is Oka if and only if it has the local approximation property.*

The implication LAP $\Rightarrow$ CAP proceeds from planar enlargement to polydisc approximation and then to arbitrary compact convex sets by induction across real faces. At each stage, the retained parameter domain is prescribed in advance so that earlier source and spray variables can serve as passive parameters. This quantifier order distinguishes LAP from the local spray approximation property introduced by Alarcón and Forstnerič, for which the retained radius may depend on the spray [1, Definition 7.11 and Remark 7.12].

An n-dimensional manifold Y is *dominable at* y if there is a holomorphic map $H : \mathbb{C}^n \to Y$ with $H(0) = y$ and dH_0 invertible, and is *strongly dominable* if this holds at every point. Dominability at y gives LAP at y by a local inverse of H (Proposition 2.2), while every Oka manifold is strongly dominable by Forstnerič's result [25, Proposition 7.1.9(a)]. Hence Theorem 1.2 settles the strong-dominability question raised by Forstnerič and Lárusson [32, the paragraph following Theorem 1.11] and subsequently emphasized by Forstnerič [28, Section 3.3].

Theorem 1.3. *A complex manifold is Oka if and only if it is strongly dominable.*

The next three corollaries concern source dimension, stratifications, and open covers. For $d \geq 1$, write CAP_d for convex approximation on compact convex subsets of $\mathbb{C}^d$.

Corollary 1.4. *Let X be an n-dimensional complex manifold, where $n \geq 1$. Then X is Oka if and only if it satisfies CAP_n.*

Forstnerič and Lárusson introduced a *finite Oka stratification*: a finite decreasing filtration by closed complex subvarieties, ending in the empty set, whose successive differences are manifolds with Oka connected components [31, Definition 1]. They proved that such a stratification implies strong dominability [31, Theorem 2].

Corollary 1.5. *Every complex manifold with a finite Oka stratification is Oka.*

Corollary 1.6. *If every point of a complex manifold has an open Oka neighborhood, then the manifold is Oka.*

Corollary 1.4 resolves Forstnerič's stabilization question [21, Problem 6.2]; see also [25, Problem 7.1.2(a)]. Corollary 1.5 shows that a finite Oka stratification gives an Oka manifold, a question raised by Forstnerič and Lárusson [31, p. 3716]. Kusakabe proved localization for Zariski-open covers and asked whether Euclidean-open covers suffice [47, Theorem 1.4 and Problem 1.6]; Corollary 1.6 settles his question. The proofs are completed in Section 3.

Kusakabe's Ell_1 characterization requires every holomorphic map $f : S \to Y$ from a Stein manifold to admit a dominating spray $F : S \times \mathbb{C}^N \to Y$ for some N, with $F(\cdot, 0) = f$; sprays over maps near compact convex sets also suffice [47, Theorems 1.3 and 2.2]. Theorem 1.3 obtains the Oka property from the pointwise case, with S a singleton. Kusakabe also characterized Oka manifolds by dense entire curves in spaces of holomorphic maps and by universal maps [44].

Let E be a subset of an n-dimensional complex manifold Y, $n \geq 1$, with $\mathcal{H}^{2n-1}(E) = 0$. If Y has LAP at every point of $Y \setminus E$, then it has LAP everywhere (Lemma 4.4). Dominability outside such a set therefore implies the full Oka property (Corollary 4.5). This strengthens the Oka-1 conclusion of Alarcón and Forstnerič [1, Theorem 2.2 and Corollary 2.5], which concerns maps from open Riemann surfaces. Forstnerič and Lárusson developed further properties of Oka-1 manifolds and of LSAP [32]. For analytic exceptional sets we obtain:

Theorem 1.7. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If $X \setminus A$ is Oka, then X is Oka.*

Theorem 1.7 settles Forstnerič's union problem [25, Problem 7.6.4] and the first part of Question D of Forstnerič and Lárusson [30]: one nonempty Zariski-open Oka subset suffices. More generally, Corollary 4.7 assumes dominability at every point of $X \setminus A$ by entire maps into X.

The preceding criterion for exceptional sets of zero $(2n-1)$-dimensional Hausdorff measure also gives descent along arbitrary surjective holomorphic maps from Oka manifolds (Theorem 5.1). Together with Forstnerič's surjectivity theorem [26, Theorem 1.1], this characterizes positive-dimensional Oka manifolds by surjective entire maps and by the basic Oka property with surjectivity (Corollary 5.3), which settles Forstnerič's Problem 1.4(a) [26].

For the geometric applications we use the criterion of Corollary 4.9, formulated in terms of families of entire curves. Outside a proper closed analytic subset of Y, suppose every point has a neighborhood V with finitely many holomorphic maps

$$h_j : V \times \mathbb{C} \longrightarrow Y, \qquad h_j(y,0) = y,$$

whose vectors $\partial_t h_j(y,0)$ span T_yY. Then Y is Oka. The full product domain preserves holomorphic dependence on y, and the images of the families may meet the exceptional subset; Corollary 4.6 then extends LAP across that subset.

Our analytic constructions arise from recent work on Oka K3 surfaces. Xie and Zhao constructed the first examples of Oka K3 surfaces and developed a period-theoretic strategy for studying their Oka loci; in that work they conjectured that every complex K3 surface is Oka [61]. Wang, Xie, and Zhao treated genus-one K3 surfaces and generalized Kummer surfaces [59]. The conjecture was proved by OpenAI [56] following the Xie–Zhao strategy; that proof supplies the surface case generalized here. The polynomial source coordinates, polydisc extension, induction across real affine hyperplanes, and gluing across their interfaces are extended from surfaces to targets of arbitrary dimension, with n chart-spray variables retained as passive parameters. For surfaces, LAP is precisely property $\mathcal{L}$ of [56, Definition 2.1]. Lemma 3.1 handles the real-parameter complexification step by a different argument. For propagation across analytic subsets, we use n recentered scalar families and adapt the scalar-substitution method developed by An and Xie [3, Lemma 4.3 and Section 4.4]; polynomial approximation of the substitutions permits their use on expanding scalar discs.

Geometric applications. A smooth complex projective manifold X is *rationally connected* if a general pair of its points lies on the image of a map $\mathbb{P}^1 \to X$, and is *Fano* if K_X^{-1} is ample. Deformations of very free rational curves give:

Theorem 1.8. *Every smooth complex projective rationally connected manifold is Oka. In particular, every smooth complex Fano manifold is Oka.*

Theorem 1.8 settles the rationally connected case of Forstnerič's Problem 11.3 [27, Problem 11.3]. In particular, every smooth hypersurface of degree $d \le n$ in $\mathbb{P}^n$ is Oka, which gives the affirmative answer to the hypersurface part of Problem E of Forstnerič and Lárusson [30, pp. 33–34] throughout the Fano range.

Campana and Winkelmann constructed dense entire curves in rationally connected projective manifolds whose images contain prescribed countable subsets [5]. Alarcón and Forstnerič formulated the Oka-1 conjecture for these manifolds [1, Conjecture 9.1], and Du, Guo, and Xie proved it [14, Corollary 1.8]. Related analytic constructions of rational curves were developed by Du, Guo, and Xie [15]. Among earlier full Oka results are the smooth quartic hypersurfaces of dimension at least three [40] and all smooth Fano threefolds [3]. Section 6 proves the general rationally connected case directly by constructing families of entire curves with independent tangent directions.

Corollaries 6.2 and 6.3 show that compact Kähler manifolds with nonnegative holomorphic sectional curvature or nef tangent bundle are Oka. Forstnerič and Kusakabe proved that, for

$n \geq 2$ and $k \geq 1$, the unit disc bundle defined by a semipositive Hermitian metric on $\mathscr{O}_{\mathbb{P}^n}(k)$ is Oka [29, Theorem 1.1].

For complements, Kusakabe's theorem covers compact polynomially convex sets in $\mathbb{C}^n$, $n > 1$ [51, Theorem 1.2 and Corollary 1.3]; Forstnerič and Wold treated broader classes of unbounded convex sets [36], and further examples come from the periodic-transport constructions of Du, Guo, Wang, and Xie [13]. In the projective setting, Section 7 gives a criterion for $X \setminus D$ using a rational curve $f : \mathbb{P}^1 \to X$ with $f^{-1}(D) \subseteq \{\infty\}$ and ample $f^*T_X(-\log D)$, for a reduced simple normal crossings divisor D. Chen and Zhu's theory of $\mathbb{A}^1$-curves on log smooth varieties [8], combined with Corollary 7.4, gives applications to ample rationally connected divisors and complete intersections. The same deformation method also shows that deleting an algebraic subset of codimension at least two from a smooth projective rationally connected manifold preserves the Oka property (Corollary 7.3).

A *holomorphic spray* on Y is a holomorphic map $s : E \to Y$ from a holomorphic vector bundle $E \to Y$ such that $s(0_y) = y$. It is *dominating* if $d(s|_{E_y})_{0_y}$ is surjective for every y; a manifold admitting such a spray is *holomorphically elliptic*. Gromov's theorem implies that every holomorphically elliptic manifold is Oka [39]. For smooth hypersurface complements we obtain:

Theorem 1.9. *Let $D \subset \mathbb{P}^n$ be a smooth hypersurface of degree $d \geq 1$, where $n \geq 1$. The following conditions are equivalent:*

(i) $\mathbb{P}^n \setminus D$ *is Oka;*
(ii) $\mathbb{P}^n \setminus D$ *is holomorphically elliptic;*
(iii) $d \leq n + 1$.

For complements, Kusakabe treated the quadric case [47, Corollary 4.9(1)] and Xie the smooth cubic case [60]; the plane cubic case answered Problem B of Forstnerič and Lárusson [30, p. 33]. Theorem 1.9 resolves Forstnerič's Problem 5.3(a) for smooth hypersurfaces [27, Problem 5.3(a)] and settles the complement part of Problem E of Forstnerič and Lárusson [30, pp. 33–34] for every smooth hypersurface, with the exact threshold $d \leq n + 1$. At the critical degree $d = n + 1$, successive polar equations produce families of lines meeting D in at most two distinct points, whose complements are parametrized by entire curves. For $d \geq n + 2$, the rank obstruction of Carlson and Griffiths [7, Corollary (6.11)] excludes the Oka property.

Theorem 1.7 also applies to spaces of rational maps: the Euclidean algorithm exhibits an Oka open set in a normalized evaluation fiber. The evaluation fiber bundle then gives:

Proposition 1.10. *For every $d \geq 1$, the manifold $\mathrm{Hol}_d(\mathbb{P}^1, \mathbb{P}^1)$ is Oka and holomorphically elliptic.*

Hanysz asked whether these spaces are Oka in every degree, and proved that the cubic space is strongly dominable and strongly $\mathbb{C}$-connected [41]; Xie later showed that the cubic space is holomorphically elliptic [60, Theorem 1.3]. Proposition 1.10 settles Hanysz's question for every positive degree.

Organization. The analytic core has two stages. Planar enlargement (Appendix A) and polynomial source coordinates give polydisc approximation (Appendix B); induction across real affine hyperplanes then yields CAP in Section 3. Scalar substitution propagates LAP from the boundary of a coordinate disc to its center in Section 4, leading to the criterion for exceptional sets of zero $(2n-1)$-dimensional Hausdorff measure and to propagation across analytic subsets. Section 5 treats surjective holomorphic images, product complements, and open group orbits. The criterion using families of entire curves connects these analytic results to the geometric applications in Sections 6–9.

2. The local approximation property

2.1. **Definition and elementary consequences.** For a compact set K, the notation $\operatorname{Op} K$ denotes an unspecified open neighborhood of K, and $\mathscr{O}(\operatorname{Op} K, Y)$ denotes maps to Y holomorphic on such a neighborhood. Let d_Y be a complete distance inducing the topology of the target manifold Y; its existence follows from the complete Riemannian metric theorem of Nomizu–Ozeki [54, Theorem 1]. Completeness enters the exhaustion at the end of Appendix B.

For $b > 0$, we write

$$\Delta_b = \{\zeta \in \mathbb{C} : |\zeta| < b\}, \qquad \Delta = \Delta_1, \qquad \Delta_b^n = \{v \in \mathbb{C}^n : \|v\|_\infty < b\}.$$

Here $|\cdot|$ denotes the Euclidean norm and $\|\cdot\|_\infty$ the maximum norm. Throughout the analytic arguments, n denotes the target dimension, m the total complex source dimension, and q the number of passive variables when these are displayed separately. In CAP_n, by contrast, the subscript specifies the source dimension at which convex approximation is assumed. For a subset E of a complex Euclidean space, $\operatorname{int} E$ denotes its interior in that space; $E \Subset U$ means that $\overline{E}$ is compact and contained in U. The closed polydisc $\overline{\Delta}_b^m$ consists of the points $z \in \mathbb{C}^m$ with $|z_j| \le b$ for $1 \le j \le m$. In planar approximation, *passive parameters* are the variables other than the planar variable whose domain is being enlarged: holomorphic dependence on these variables is preserved on a prescribed smaller polydisc. The dimension of each such parameter polydisc is specified at its introduction.

Definition 2.1 (Local approximation property (LAP))**.** A *special planar disc pair* $D \subset D'$ consists of closed topological discs $D, D' \subset \mathbb{C}$ with piecewise smooth boundaries such that $D' = D \cup E$ for another closed disc E, where $D \cap E$ is a proper boundary arc of both D and E. An n-dimensional complex manifold Y has the *local approximation property* (LAP), denoted by $\mathcal{L}_n$, at y if there is an open neighborhood $V \ni y$ such that, for every special planar disc pair $D \subset D'$, every integer $q \ge 0$, every pair of closed polydiscs

$$P_0 \Subset \operatorname{int} P \subset \mathbb{C}^q$$

and $f \in \mathscr{O}(\operatorname{Op}(D \times P), V)$, the map f can be approximated uniformly on $D \times P_0$ by maps in $\mathscr{O}(\operatorname{Op}(D' \times P_0), Y)$. Only the given map is required to take values in V; the approximating maps take values in Y and may leave V. The set V is a *witness neighborhood* for LAP at y. A witness neighborhood contained in a target coordinate chart is called a *coordinate witness neighborhood*. The manifold Y has LAP if it has $\mathcal{L}_n$ at every point.

By shrinking a witness neighborhood about its distinguished point, we may always assume that it is a coordinate witness neighborhood.

The Euclidean model. The simplest instance of Definition 2.1 is $Y = \mathbb{C}^n$. Indeed, if

$$f \in \mathscr{O}(\operatorname{Op}(D \times P), \mathbb{C}^n),$$

then $D \times P$ is polynomially convex. The Cartan–Oka–Weil theorem, applied componentwise, therefore approximates f uniformly on $D \times P$ by entire maps $\mathbb{C}^{1+q} \to \mathbb{C}^n$ [25, Theorem 2.8.4]. These maps are, in particular, holomorphic near $D' \times P_0$. Thus $\mathbb{C}^n$ has LAP.

Proposition 2.2. *If Y is dominable at y, then Y has LAP at y.*

Proof. If $H : \mathbb{C}^n \to Y$ dominates at y, the inverse function theorem gives a biholomorphic restriction $H|_W : W \to V$. Lift an admissible map into W, apply the Euclidean model, and compose the approximants with the globally defined map H. □

LAP is a parameter-uniform strengthening of the local spray approximation property (LSAP) [1, Definition 7.11]. The version of LSAP in [1, Remark 7.12(B)] omits the domination condition. Its retained ball has a radius that may depend on the spray, whereas LAP retains

every prescribed $P_0 \Subset \operatorname{int} P$. With the planar pair and the initial parameter domain fixed, the quantifier order is schematically

$$\begin{aligned} \text{LAP}: &\quad \forall P_0 \Subset \operatorname{int} P\ \forall f\ \forall \epsilon > 0\ \exists g, \\ \text{LSAP}: &\quad \forall F\ \exists r \in (0,1)\ \forall \epsilon > 0\ \exists G. \end{aligned}$$

Here f and F range over the admissible maps and sprays, and approximation is required on $D \times P_0$ and $D \times r\overline{B^N}$, respectively, with B^N the unit ball in $\mathbb{C}^N$. In the iteration below, source and spray variables already in use become passive parameters for the next planar approximation step, so the smaller parameter polydisc P_0 must be prescribed before approximation.

The auxiliary parameters in LAP are complex variables on which the maps depend holomorphically. The usual parametric Oka principle, with topological parameter spaces, follows once CAP has been established [22, Theorem 1.1]. For $n = 2$, Definition 2.1 is precisely property $\mathcal{L}$ of [56, Definition 2.1]; the subscript records its direct extension to targets of arbitrary dimension.

Dominability also implies LSAP [1, paragraph following Remark 7.12]. For surfaces, Proposition 2.2 is a special case of [56, Proposition 2.2], obtained by taking both strip maps from one dominating entire map and setting the displacement to zero. The implication from LAP to CAP is the following.

Theorem 2.3. *Let Y be an n-dimensional complex manifold. If Y has LAP, then Y has* CAP.

The following subsections establish the polydisc approximation step; the passage to CAP is completed at the end of Section 3. For $n = 2$, the result is [56, Theorem 3.1]. The polynomial source coordinates and polydisc extension are taken from [56, Lemma 3.3 and Proposition 3.4], while the splitting across a real hyperplane is taken from [56, Lemmas 4.4–4.5]. For targets of dimension n, the same source-side constructions use chart sprays with n fiber variables.

2.2. **Chart sprays and planar splitting.** Our gluing construction needs holomorphic fiber coordinates along the graph of a map. They are supplied by the following standard local thickening lemma.

Lemma 2.4. *Let $f : S \to Y$ be holomorphic, where S is Stein, f^*TY is holomorphically trivial on S, and $K \Subset S$. There are an open neighborhood $U \subset S$ of K, a number $\beta > 0$, and a holomorphic map*

$$f^\sharp : U \times \Delta^n_\beta \longrightarrow Y, \qquad f^\sharp(z,0) = f(z),$$

such that

$$(z,v) \longmapsto (z, f^\sharp(z,v))$$

is biholomorphic onto its image. For every $K' \Subset U$ and $0 < \beta' < \beta$, there is an open neighborhood of the image of $K' \times \overline{\Delta}^n_{\beta'}$ on which the inverse is holomorphic.

A precursor of this local-spray construction appears in [21, Lemma 3.2]; the graph-chart form is [56, Lemma 2.6]. In the applications below, the source is either a contractible Stein domain or a product of a planar domain with a polydisc. A planar domain has the homotopy type of a one-dimensional CW complex, and taking its product with a polydisc does not change that homotopy type. Complex vector bundles on these sources are therefore topologically trivial. The Oka–Grauert principle gives the required holomorphic triviality, as in the proof of the cited lemma; the graph-chart construction then applies with n fiber variables.

In these fiber coordinates, the linear step in Cartan-type gluing is an additive Cousin splitting. A related bounded additive splitting in the Cartan-pair setting appears in [19, Lemma 4.6]; see also [25, Lemma 5.8.2]. The precise planar statement used here, including passive holomorphic parameters and an operator bound independent of their number and range, is [56, Lemma 2.4]. We record the quantitative consequence needed below.

For an open set $U \subset \mathbb{C}^m$, let

$$H^\infty(U, \mathbb{C}^N) = \{f \in \mathscr{O}(U, \mathbb{C}^N) : \sup_{z \in U} \|f(z)\|_\infty < \infty\}$$

be the Banach space of bounded holomorphic maps, equipped with the supremum norm $\|f\|_{\infty,U} = \sup_{z\in U} \|f(z)\|_\infty$; matrix norms are the induced operator norms. In Lemmas 2.5 and 2.6, the passive parameter polydisc P is open. The closed parameter polydiscs used in applications are compactly contained in such a P.

Lemma 2.5. *Suppose that U_A, U_B are bounded planar open sets and that the closed sets $\overline{U_A \setminus U_B}$ and $\overline{U_B \setminus U_A}$ are disjoint. For their overlap $C = U_A \cap U_B$, every open passive parameter polydisc P, and every $N \geq 1$, there are bounded linear operators*

$$R_A : H^\infty(C \times P, \mathbb{C}^N) \longrightarrow H^\infty(U_A \times P, \mathbb{C}^N), \qquad R_B : H^\infty(C \times P, \mathbb{C}^N) \longrightarrow H^\infty(U_B \times P, \mathbb{C}^N)$$

such that

$$R_A u - R_B u = u, \qquad \|R_A u\|_{\infty,U_A\times P} + \|R_B u\|_{\infty,U_B\times P} \leq C_0 \|u\|_{\infty,C\times P}. \tag{2.1}$$

They preserve holomorphic dependence on all passive parameters. The constant C_0 depends only on the planar pair and the chosen cutoff; in particular, it is independent of N and of the radii and dimension of the passive parameter polydisc P.

This is [56, Lemma 2.4]. Its Cauchy-integral construction acts componentwise. If χ is the separating cutoff and $d \geq \operatorname{diam}(U_A \cup U_B)$, it gives

$$C_0 = 2 + 4d \|\partial_{\bar{s}}\chi\|_\infty,$$

independently of the range dimension and of the radii and dimension of P.

The next lemma turns a small transition error into exact agreement of two sprays, using the quantitative contraction estimate of [56, Lemma 2.8].

Lemma 2.6. *Let U_A, U_B, C, R_A, R_B, and C_0 be as in Lemma 2.5, let P be an open polydisc, and fix $0 < \rho < \rho_+$. Let*

$$F_A \in \mathscr{O}(U_A \times P \times \Delta^n_{\rho_+}, Y), \qquad F_B \in \mathscr{O}(U_B \times P \times \Delta^n_\rho, Y).$$

Assume that F_A is fiberwise injective and that the transition

$$\gamma(s,t,v) = F_A(s,t,\cdot)^{-1}(F_B(s,t,v)) = v + E(s,t,v) \tag{2.2}$$

is defined and holomorphic on $C \times P \times \Delta^n_\rho$, where the inverse is taken on the larger fiber chart $\Delta^n_{\rho_+}$. Set

$$e = \|E\|_\infty, \qquad \kappa = \|D_v E\|_\infty,$$

where both norms are taken on this product. If

$$2C_0 e < \rho, \qquad C_0\kappa < \tfrac{1}{2} \tag{2.3}$$

then there are

$$a \in H^\infty(U_A \times P, \mathbb{C}^n), \qquad b \in H^\infty(U_B \times P, \mathbb{C}^n)$$

such that

$$F_A(s,t,a(s,t)) = F_B(s,t,b(s,t)), \qquad \|a\|_{\infty,U_A\times P} + \|b\|_{\infty,U_B\times P} \leq C_0 e < \rho/2. \tag{2.4}$$

Consequently

$$F(s,t) = \begin{cases} F_A(s,t,a(s,t)), & s \in U_A, \\ F_B(s,t,b(s,t)), & s \in U_B \end{cases}$$

is holomorphic on $(U_A \cup U_B) \times P$. If $L_A \subset U_A \times P$ is compact, then

$$\sup_{L_A} d_Y(F, F_A(\cdot, 0)) \leq \omega_{A,L_A}(C_0 e),$$

where ω_{A,L_A} is a continuity modulus of F_A in its fiber variable.

Proof. On $X = H^\infty(C \times P, \mathbb{C}^n)$, consider

$$\mathcal{T}(u)(s,t) = E(s,t,R_B u(s,t)). \tag{2.5}$$

On the ball $\|u\|_{\infty,C\times P} \le 2e$ in X, the first inequality in (2.3) ensures that $R_B u$ remains in Δ_ρ^n, while

$$\|\mathcal{T}(u)\|_{\infty,C\times P} \le e, \qquad \|\mathcal{T}(u) - \mathcal{T}(\widetilde{u})\|_{\infty,C\times P} \le C_0\kappa \, \|u - \widetilde{u}\|_{\infty,C\times P}\,. \tag{2.6}$$

Hence $\mathcal{T}$ has a fixed point u with $\|u\|_{\infty,C\times P} \le e$. Set $a = R_A u$ and $b = R_B u$. Then

$$a - b = u = E(s,t,b), \qquad a = \gamma(s,t,b),$$

and Lemma 2.5 gives

$$\|a\|_{\infty,U_A\times P} + \|b\|_{\infty,U_B\times P} \le C_0 e.$$

Thus $F_A(s,t,a) = F_B(s,t,b)$ on the overlap, proving all the assertions. □

2.3. **Planar enlargement.** The next lemma uses the radial-corridor and sector-filling construction of [56, Proposition 2.9], with the retained parameter polydisc prescribed in advance. It combines Mergelyan approximation with the spray framework of Gromov [39, Section 1.1.B] and the local-spray and splitting techniques used in [21, Lemma 2.1, Proposition 3.1, and Lemma 3.2]. Appendix A gives the construction with n spray variables.

The disc case will be used to construct expanding families in Section 4; the annular case will extend the maps expressed in the source coordinates of Subsection 2.4.

Lemma 2.7. *Let $r > 0$,*

$$K = \overline{\Delta}_r \quad \text{or} \quad K = \{s \in \mathbb{C} : r_- \le |s| \le r\}, \qquad 0 < r_- < r,$$

and suppose that f is holomorphic near $K \times P$ for a closed polydisc $P \subset \mathbb{C}^q$, where $q \ge 0$. Suppose in addition that for every $\zeta \in \partial\Delta_r$ there is a coordinate witness neighborhood V_ζ such that

$$f(\{\zeta\} \times P) \Subset V_\zeta.$$

For every $R > r$, every closed polydisc $P_0 \Subset \operatorname{int} P$, and $\epsilon > 0$, there is

$$F \in \mathscr{O}(\mathrm{Op}(K_R \times P_0), Y), \qquad \sup_{K\times P_0} d_Y(F,f) < \epsilon,$$

where $K_R = \overline{\Delta}_R$ in the disc case and $K_R = \{r_- \le |s| \le R\}$ in the annular case.

Appendix A gives the proof.

2.4. **Polynomial source coordinates and polydisc approximation.** The passage to polydisc approximation follows [56, Proposition 3.4]. The polynomial coordinates in Lemma 2.8 have transverse fibers of small diameter on the joining annulus and give a change of coordinates close to the identity near the initial polydisc. Planar enlargement, gluing, and exhaustion then yield Proposition 2.9.

Lemma 2.8. *Fix $m \ge 2$ and*

$$1 < r_m < \cdots < r_2 < r_1, \qquad 0 < a < 1, \qquad r_j < a r_{j-1} \quad (2 \le j \le m).$$

Set

$$P_j(w) = (w/r_j)^d, \qquad H_d(x) = (ax_m, P_2(x_2) - ax_1, \dots, P_m(x_m) - ax_{m-1}).$$

This is a polynomial automorphism. For $y = H_d(x)$, its inverse is given recursively by

$$x_m = y_1/a, \qquad x_{j-1} = (P_j(x_j) - y_j)/a \quad (j = m, m-1, \dots, 2).$$

There exist $C \ge 1$, $q \in (0,1)$, and an integer $d_0 \ge 2$, depending only on the fixed data, such that for every pair of integers $d \ge d_0$ and $J \ge 1$, the set $E_{d,J} = H_d^{-J}(\overline{\Delta}_1^m)$ satisfies

$$\max_{j\ge 2} |P_j(x_j)| \le C \max(1, |x_1|). \tag{2.7}$$

If also $|x_1| > r_1$*, then all* x_j *are nonzero and*

$$\left|\frac{H_d(x)_j}{ax_{j-1}}\right| \le q \qquad (2 \le j \le m). \tag{2.8}$$

By (2.8)*, the numbers*

$$\xi_j = \log(P_j(x_j)/(ax_{j-1})) \qquad (2 \le j \le m)$$

are defined using the same logarithm branch centered at 1*, and satisfy* $|\xi_j| \le B = -\log(1-q)$*. With*

$$D = d^{m-1}, \qquad \rho_1 = 1, \qquad \rho_j = r_j(a\rho_{j-1})^{1/d},$$

there is a single-valued function s*, holomorphic on an open neighborhood of* $E_{d,J} \cap \{|x_1| > r_1\}$*, such that*

$$x_j = \rho_j s^{d^{m-j}} \exp\left(\sum_{k=2}^{j} \frac{\xi_k}{d^{j-k+1}}\right), \qquad x_1 = s^D. \tag{2.9}$$

The proof is in [56, Lemma 3.3]; for later use we recall the inverse formula and the two estimates from it.

The single-valued coordinate in (2.9) is

$$s = \frac{x_m}{\rho_m} \exp\left(-\sum_{k=2}^{m} \frac{\xi_k}{d^{m-k+1}}\right).$$

The next estimates quantify the variation in the transverse parameters on the joining annulus. Write $\Phi_d(s,\xi)$ for the right side of (2.9), now allowing ξ to vary independently. Given $\ell > 0$, fix

$$r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell$$

and $\Xi_2 = \{\xi : |\xi_j| \le B + 3\}$. There are constants C_1, C_2 depending only on the fixed data r_1, a, B, c_3, and independent of d, J, s, ξ, such that, for all sufficiently large d, on the annulus $c_0 \le |s|^D \le c_3$ and for $\xi \in \Xi_2$,

$$|\Phi_{d,j}(s,\xi)| \le r_j e^{C_1/d} \quad (j \ge 2), \tag{2.10}$$

$$\sup_{\xi,\widetilde{\xi}\in\Xi_2} \left\|\Phi_d(s,\xi) - \Phi_d(s,\widetilde{\xi})\right\|_\infty \le C_2/d. \tag{2.11}$$

Equations (2.10) and (2.11) are the estimates in [56, equation (3.8)], with the transverse polydisc fixed at Ξ_2.

The change of source coordinates is also close to the identity near the original polydisc. With

$$L(x) = (ax_m, -ax_1, \ldots, -ax_{m-1}),$$

first fix J with $a^J R_+ < 1$ for a prescribed $R_+ > 1$, and set

$$h_d = H_d^{-J} L^J. \tag{2.12}$$

On a fixed neighborhood of the unit polydisc, $h_d \to \mathrm{id}$. More precisely, choose $1 < b_0 < b_1 < r_m$ and $\theta = b_1/r_m < 1$. The estimate proving [56, equation (3.7)] gives, for large d,

$$\|h_d - \mathrm{id}\|_{\overline{\Delta}_{b_0}^m} \le \theta^d \sum_{i=1}^{J} a^{-i} \longrightarrow 0. \tag{2.13}$$

The constant J is fixed before d tends to infinity.

Combining these coordinates with Lemma 2.7 gives the polydisc approximation needed to start the induction across real affine hyperplanes in Section 3.

Proposition 2.9. *If Y has LAP at every point, then for every $m \geq 1$, every closed polydisc $Q \subset \mathbb{C}^m$, and $f \in \mathscr{O}(\operatorname{Op} Q, Y)$, the map f can be approximated uniformly on Q by maps in $\mathscr{O}(\mathbb{C}^m, Y)$.*

Appendix B gives the proof, adapting [56, Proposition 3.4] with the n chart-spray variables retained as passive parameters. It combines the source coordinates of Lemma 2.8, planar enlargement (Lemma 2.7), the foliation-preserving compositional splitting theorem [19, Theorem 4.1], and an exhaustion in which the successive approximation errors have a summable bound.

3. From LAP to convex approximation

We follow the induction in [56, Section 4, proof of Theorem 3.1]: the half-space constraints of an enclosing polytope are removed one at a time inside a larger polydisc. Lemmas 3.2–3.4 use [56, Lemmas 4.2–4.5] with n spray variables. Lemma 3.1 corresponds to [56, Lemma 4.1].

3.1. **Inductive setup and approximation on real faces.** For each integer $r \geq 0$, let A_r be the following approximation property: for every compact convex set $K \subset \mathbb{C}^r$ and every closed polydisc Q, each map to Y holomorphic near $K \times Q$ can be approximated uniformly there by maps entire in all source variables. Proposition 2.9 gives A_0. Approximation on real faces and gluing across a single hyperplane will give $\mathsf{A}_{r-1} \Rightarrow \mathsf{A}_r$.

On a manifold target, finite derivative accuracy is measured in fixed target charts around the compact image of a reference map. Take finitely many compact source pieces K_ν and target charts ψ_ν whose domains contain $H(K_\nu)$ with a positive margin. For a map G with $\psi_\nu \circ G$ defined on K_ν, write

$$\|G - H\|_{C^j(K)} = \max_\nu \max_{|\mu| \leq j} \sup_{K_\nu} |D^\mu(\psi_\nu \circ G - \psi_\nu \circ H)| \,.$$

Derivatives are taken in the displayed real and complex source variables; on a real face they are tangential. Different finite families of charts give equivalent local notions of convergence. For maps jointly holomorphic on a fixed complex neighborhood, uniform approximation on a slightly larger compact neighborhood gives approximation in any fixed finite number of derivatives on the smaller set by Cauchy estimates.

Smooth paths will join the separate slice approximations in Lemma 3.3. To turn the real-parameter families into holomorphic ones, we use the following arbitrary-target analogue of [56, Lemma 4.1]. The proof proceeds through a Stein neighborhood of the parameterized graph and Banach-valued polynomial approximation, followed by holomorphic retraction.

Lemma 3.1. *Let $I \subset \mathbb{R}$ be a compact interval and $C \subset \mathbb{C}^k$ a compact convex set, with open neighborhoods $J \subset \mathbb{R}$ and $U \subset \mathbb{C}^k$, respectively. Suppose that*

$$H : J \times U \longrightarrow Y$$

is jointly smooth and that $H_s = H(s, \cdot)$ is holomorphic for every real $s \in J$. For every integer $j \geq 0$ and every $\epsilon > 0$, there exist complex open neighborhoods $V \supset I$, $U_0 \supset C$ and

$$F \in \mathscr{O}(V \times U_0, Y), \qquad \|F - H\|_{C^j(I \times C)} < \epsilon.$$

More generally, the approximation may be required on any prescribed compact product $I_1 \times C_1$ satisfying

$$I \Subset \operatorname{int} I_1 \Subset J, \qquad C \Subset \operatorname{int} C_1 \Subset U,$$

where I_1 is an interval and C_1 is convex. The thickness of the complex neighborhood may depend on F.

Proof. Choose a compact interval $I^+ = [a, b] \Subset J$, a compact convex set C^+, and a bounded convex open set X such that

$$I \Subset \operatorname{int} I^+, \qquad C \Subset \operatorname{int} C^+ \Subset X \Subset U. \tag{3.1}$$

Apply [35, Corollary 2.2] with parameter set $I^+ \subset \mathbb{R}$, Stein base X, submersion $X \times Y \to X$, the compact $\mathscr{O}(X)$-convex set C^+, and the holomorphic sections $q \mapsto (q, H_s(q))$. This gives a Stein neighborhood Θ of the graph

$$\Sigma = \{(s, q, H_s(q)) : s \in I^+,\ q \in C^+\} \subset \mathbb{C} \times X \times Y. \tag{3.2}$$

The Remmert–Bishop–Narasimhan embedding theorem and the Docquier–Grauert retraction theorem, in the form used in [17, proof of Lemma 3], give

$$e : \Theta \hookrightarrow \mathbb{C}^N, \qquad e(\Theta) \text{ closed}, \qquad \rho : \Omega \longrightarrow e(\Theta), \quad \rho|_{e(\Theta)} = \mathrm{id}, \tag{3.3}$$

with $\Omega \subset \mathbb{C}^N$ open and ρ holomorphic.

Choose an open set U_1 with $C \Subset U_1 \Subset \operatorname{int} C^+$. Compactness of (3.2) gives

$$g(s, q) = e(s, q, H_s(q)), \qquad g \in C^\infty(I^+, H^\infty(U_1, \mathbb{C}^N)). \tag{3.4}$$

Write $\mathcal{B} = H^\infty(U_1, \mathbb{C}^N)$ and $L = j + 1$, so that the derivatives $\partial_s^\ell g$ with $0 \le \ell < L$ are exactly the s-derivatives of g up to order j. For $x = (s - a)/(b - a)$, form the $\mathcal{B}$-valued polynomial

$$Q_\nu(s) = \sum_{k=0}^{\nu} \partial_s^L g\Big(a + \frac{k(b-a)}{\nu}\Big) \binom{\nu}{k} x^k (1-x)^{\nu-k}. \tag{3.5}$$

The binomial weights in (3.5) have mean x and variance $x(1 - x)/\nu$. Split the sum at $|a + k(b - a)/\nu - s| = (b - a)\nu^{-1/4}$. Uniform continuity of $\partial_s^L g$ controls the terms within this distance, while the variance estimate makes the remaining total weight tend to zero. Thus, in the Banach norm,

$$\varepsilon_\nu := \sup_{I^+} \big\| Q_\nu - \partial_s^L g \big\|_{\mathcal{B}} \longrightarrow 0. \tag{3.6}$$

Integrating (3.5) L times and choosing the integration constants to match the derivatives of g at a of orders $0, \dots, L - 1$, we obtain a polynomial in s that is holomorphic in q:

$$P_\nu(s) = \sum_{\ell=0}^{L-1} \frac{(s-a)^\ell}{\ell!} \partial_s^\ell g(a) + \int_a^s \frac{(s-t)^{L-1}}{(L-1)!} Q_\nu(t)\, dt. \tag{3.7}$$

By (3.7), for $0 \le \ell < L$ the difference $\partial_s^\ell (P_\nu - g)$ is the $(L - \ell)$-fold primitive of $Q_\nu - \partial_s^L g$ that vanishes to order $L - \ell$ at $s = a$; hence (3.6) implies

$$\sup_{I^+} \big\| \partial_s^\ell (P_\nu - g) \big\|_{\mathcal{B}} \le \frac{(b-a)^{L-\ell}}{(L-\ell)!} \varepsilon_\nu. \tag{3.8}$$

Equation (3.8) and Cauchy estimates across the fixed margin $C \Subset U_1$ yield

$$\max_{\ell + |\beta| \le j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0. \tag{3.9}$$

For large ν, the case $\ell = 0$ of (3.8) places $P_\nu(I \times \overline{U}_0)$ in Ω for some open U_0 with $C \Subset U_0 \Subset U_1$. By compactness of I, there is then a complex neighborhood $V_\nu \supset I$ with $P_\nu(V_\nu \times U_0) \subset \Omega$. Consequently

$$F_\nu = \pi_Y \circ e^{-1} \circ \rho \circ P_\nu \in \mathscr{O}(V_\nu \times U_0, Y).$$

On fixed charts around the compact image, (3.3), (3.4), (3.9), and the chain rule give

$$\|F_\nu - H\|_{C^j(I \times C)} \le C_j \max_{\ell + |\beta| \le j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0.$$

The margins in (3.1) can be taken around any prescribed compact product inside $J \times U$. □

After complexifying the real parameter, we use [56, Lemma 4.2] to make the approximation entire in that variable as well. A Riemann map places the parameter in a polydisc factor; it is approximated by a polynomial after the entire target-valued approximant has been fixed.

Lemma 3.2. *Suppose that Y satisfies A_r for an integer $r \geq 0$. Let $I \subset \mathbb{R}$ be a compact interval, let $C \subset \mathbb{C}^r$ be compact and convex, and let $P \subset \mathbb{C}^q$ be a closed polydisc, where $q \geq 0$. For every $j \geq 0$ and $\epsilon > 0$, each map H holomorphic near $I \times C \times P$ admits an entire map $\widetilde{H}$ in all displayed complex variables such that*

$$\left\| \widetilde{H} - H \right\|_{C^j(I \times C \times P)} < \epsilon.$$

Proof. Write the map as $H(s, y, u)$. Choose a bounded simply connected neighborhood V of I on which it is holomorphic, a Riemann map $\varphi : V \to \Delta$, and $0 < \rho < 1$ with $\varphi(I) \Subset \Delta_\rho$. Apply A_r to

$$\widehat{H}(\zeta, y, u) = H(\varphi^{-1}(\zeta), y, u),$$

treating $\overline{\Delta}_\rho \times P$ as a passive polydisc. After retaining slightly larger compact domains, Cauchy estimates give an entire map $E(\zeta, y, u)$ for which $E(\varphi(s), y, u)$ approximates H with the prescribed finite derivative accuracy on $I \times C \times P$. Fix E. By Runge's theorem [17, Theorem 2], choose a polynomial p approximating φ, together with the required derivatives, on I. Then $E(p(s), y, u)$ is entire in all variables, and the chain rule shows that it approximates H as closely as desired. □

The convex slices of a real face vary with its real coordinate. We use the construction of [56, Lemma 4.3], which joins their separate approximants into one family on a common domain. Preserving the endpoint maps exactly makes these local paths fit together.

Lemma 3.3. *Assume that A_{r-1} holds, where $r \geq 1$. Let $F \subset \mathbb{R} \times \mathbb{C}^{r-1}$ be compact and convex and P a closed polydisc. Every holomorphic map defined near $F \times P$ can be approximated on this set by entire maps on $\mathbb{C}^r \times \mathbb{C}^{\dim P}$, with control of tangential derivatives of any fixed finite order.*

Proof. Let f be the given map and write the variables as (t, y, u), with t real on the face. The proof has four steps: approximate finitely many slices, join adjacent approximants on a common domain, repair the endpoints exactly, and complexify the assembled smooth family.

Slice approximations. Thicken F and P slightly within the original domain of definition. Denote the projection of F onto the t-axis by I and the slices by F_t. For each $t_0 \in I$, choose a compact convex neighborhood C_{t_0} of the whole slice F_{t_0} and a complex disc D_{t_0} about t_0 such that the original map is defined near $D_{t_0} \times C_{t_0} \times P$. The inclusion $F_{t_0} \subset \operatorname{int} C_{t_0}$ persists under perturbation: if not, there would be

$$t_\nu \longrightarrow t_0, \qquad (t_\nu, y_\nu) \in F, \qquad y_\nu \notin \operatorname{int} C_{t_0}.$$

Compactness gives a subsequence with $(t_\nu, y_\nu) \to (t_0, y) \in F$, whereas the closedness of $\mathbb{C}^{r-1} \setminus \operatorname{int} C_{t_0}$ gives

$$y \in F_{t_0} \setminus \operatorname{int} C_{t_0} = \varnothing.$$

A finite subdivision of I therefore gives products $D_i \times C_i \times P$ with reserved margins, such that on each subinterval the whole slice F_t lies uniformly inside its assigned C_i. Near each subdivision point, choose

$$F_t \Subset \operatorname{int} C^-, \qquad C^- \Subset \operatorname{int} C, \qquad C \Subset \operatorname{int}(C_i \cap C_{i+1}),$$

where C^-, C are compact convex intermediate sets with nonempty interior. At a subdivision point, choose a small complex disc D_b about that point so that $\overline{D}_b \times C \times P$ lies, with positive margins, in both adjacent domains of holomorphy. Also choose a common large polydisc B containing all the C_i.

Apply A_{r-1}, treating the t-disc and the u-polydisc as passive parameters, on fixed compact complex products containing the assigned portion of $F\times P$ and each adjacent product $\overline{D}_b\times C\times P$ in their interiors. The reserved margins allow these compact products to be enlarged slightly within $\operatorname{dom} f$. We obtain entire maps g_i with arbitrarily small uniform error on the larger products and hence, by Cauchy estimates, arbitrarily small error in every prescribed finite number of derivatives on the retained products. On a neighborhood of $\overline{D}_b \times C \times P$, two adjacent maps g_i, g_{i+1} therefore lie in the fixed inverse fiber charts of a chart spray of f and can be written as

$$g_i = f^\sharp(\cdot, a_i), \qquad g_{i+1} = f^\sharp(\cdot, a_{i+1}).$$

Choose a fixed smooth cutoff χ on $[0,1]$ with $\chi = 0$ near 0 and $\chi = 1$ near 1, and set

$$f_\tau = f^\sharp(\cdot, (1-\chi(\tau))a_i + \chi(\tau)a_{i+1}), \quad 0 \le \tau \le 1. \tag{3.10}$$

The family in (3.10) is defined near C, and equals the entire map g_i or g_{i+1} on the corresponding endpoint subintervals. For $v_\tau = (1-\chi(\tau))a_i + \chi(\tau)a_{i+1}$,

$$\partial_\tau^\ell v_\tau = \chi^{(\ell)}(\tau)(a_{i+1} - a_i) \quad (\ell \ge 1), \qquad \|v_\tau\|_{C^j} \le C_{j,\chi}(\|a_i\|_{C^j} + \|a_{i+1}\|_{C^j}). \tag{3.11}$$

Since $f^\sharp(\cdot, 0) = f$, the chain rule, (3.11), and uniform bounds for the derivatives of the fixed spray on each retained compact piece K give, for sufficiently accurate choices of g_i and g_{i+1},

$$\|f_\tau - f\|_{C^j([0,1]\times K)} \le C_j(\|a_i\|_{C^j(K)} + \|a_{i+1}\|_{C^j(K)}) \longrightarrow 0. \tag{3.12}$$

Paths on a common domain. Fix a closed disc $\overline{D}_b' \Subset D_b$ about the subdivision point and write $K_- = \overline{D}_b' \times C^- \times P$. We claim that, for every fixed j, the path f_τ admits arbitrarily close C^j approximation on $[0,1]\times K_-$ by a smooth path $\widehat{A}_\tau$, holomorphic in (t,y,u) near $\overline{D}_b' \times B \times P$, which equals g_i near $\tau = 0$ and g_{i+1} near $\tau = 1$. Rescaling and entire approximation replace the local path by one available on the common larger polydisc B. Choose $d_* \in \operatorname{int} C$ and

$$r_C = \operatorname{dist}(d_*, \partial C) > 0, \qquad R_B = \max_{y\in B}|y - d_*|, \qquad L > \max(1, R_B/r_C). \tag{3.13}$$

Equation (3.13) gives

$$d_* + L^{-1}(B - d_*) \Subset \operatorname{int} C.$$

Choose smooth functions $\kappa_0, \kappa_1 : [0,1] \to [0,1]$ with disjoint supports in the intervals where $f_\tau = g_i$ and $f_\tau = g_{i+1}$, respectively, and with $\kappa_0 = 1$ near 0, $\kappa_1 = 1$ near 1. The scaling function is

$$\lambda(\tau) = 1 + (L-1)(\kappa_0(\tau) + \kappa_1(\tau)), \qquad 1 \le \lambda(\tau) \le L. \tag{3.14}$$

Set

$$\widetilde{f}_\tau(t,y,u) = f_\tau(t, d_* + \lambda(\tau)(y - d_*), u). \tag{3.15}$$

It is well defined near C: by (3.14), the condition $\lambda > 1$ forces f_τ to be one of the entire endpoint maps. Fix g_i, g_{i+1}, λ, B and chart sprays of both endpoint maps on a neighborhood of $D_b \times B \times P$ before choosing the next approximation.

Apply Lemma 3.1 to $\widetilde{f}$ in (3.15), followed by Lemma 3.2 with A_{r-1}, treating the t-disc and u variables as passive. This gives a jointly entire approximation $E(\tau,t,y,u)$. Set

$$A_\tau(t,y,u) = E(\tau, t, d_* + \lambda(\tau)^{-1}(y - d_*), u). \tag{3.16}$$

Since E is entire, (3.16) makes A_τ available on all of B. For $y \in C^-$, set $y_\tau = d_* + \lambda(\tau)^{-1}(y - d_*)$. Then

$$y_\tau \in \operatorname{conv}(C^- \cup \{d_*\}) \Subset \operatorname{int} C, \qquad \widetilde{f}_\tau(t, y_\tau, u) = f_\tau(t,y,u). \tag{3.17}$$

Given a derivative order j, choose a compact product K_C in the approximation domain containing all (τ, t, y_τ, u) with $(t,y,u) \in K_-$. By (3.16), (3.17), and the chain rule,

$$\|A_\tau - f_\tau\|_{C^j([0,1]\times K_-)} \le C_j \left\|E - \widetilde{f}\right\|_{C^j(K_C)}, \tag{3.18}$$

where C_j depends on the fixed scaling function and the retained products. The right side of (3.18) can be made arbitrarily small. Together with (3.12), this makes A_τ arbitrarily close to the original map f on the retained product. On smaller compact endpoint intervals E_0, E_1, the scaling satisfies

$$\lambda(\tau) = L, \qquad S_L(y) = d_* + L^{-1}(y - d_*), \qquad S_L(B) \Subset \operatorname{int} C. \tag{3.19}$$

Writing $g^{(0)} = g_i$ and $g^{(1)} = g_{i+1}$, equations (3.15) and (3.19) give

$$\widetilde{f}_\tau(t, S_L(y), u) = g^{(\ell)}(t, y, u), \qquad \tau \in E_\ell, \quad y \in B, \quad \ell = 0, 1. \tag{3.20}$$

Choose K_C also to contain the compact products obtained from the retained endpoint domains by $(\tau, t, y, u) \mapsto (\tau, t, S_L(y), u)$. By (3.16) and (3.20),

$$\left\| A - g^{(\ell)} \right\|_{C^j(E_\ell \times \overline{D}'_b \times B \times P)} \le C_{j,L} \left\| E - \widetilde{f} \right\|_{C^j(K_C)} \longrightarrow 0. \tag{3.21}$$

Thus (3.21) places A_τ in the previously fixed inverse fiber charts of the endpoint sprays over B.

Exact endpoint matching. To concatenate the approximating paths, their endpoint values must agree exactly with g_i and g_{i+1}. On E_ℓ, the inverse fiber coordinates of the fixed endpoint spray give

$$A_\tau = (g^{(\ell)})^\sharp(\cdot, \beta_\tau^{(\ell)}), \qquad \ell = 0, 1.$$

Choose smooth cutoffs $\vartheta_\ell : E_\ell \to [0, 1]$ equal to 0 near the path endpoint $\tau = \ell$ and to 1 near the other, interior endpoint of E_ℓ. The corrected path is

$$\widehat{A}_\tau = \begin{cases} (g^{(\ell)})^\sharp(\cdot, \vartheta_\ell(\tau)\beta_\tau^{(\ell)}), & \tau \in E_\ell, \quad \ell = 0, 1, \\ A_\tau, & \tau \notin E_0 \cup E_1. \end{cases} \tag{3.22}$$

At the interior endpoint of each E_ℓ, equation (3.22) agrees with A_τ on a neighborhood, so the corrected path is smooth. At the endpoints of the full path interval it satisfies

$$\widehat{A}_\tau = g_i \quad \text{for } \tau \text{ near } 0, \qquad \widehat{A}_\tau = g_{i+1} \quad \text{for } \tau \text{ near } 1.$$

The corrections are available over B by (3.21) and are supported where $\lambda = L$. The endpoint sprays, their inverse charts, and the cutoffs are fixed. The chain rule therefore gives, on each retained endpoint product,

$$\left\| \widehat{A} - g^{(\ell)} \right\|_{C^j} \le C_j \left\| A - g^{(\ell)} \right\|_{C^j} \longrightarrow 0, \qquad \ell = 0, 1,$$

where the norms include derivatives in τ. Since $f_\tau = g^{(\ell)}$ on these endpoint intervals, this proves the claim.

Assembly and complexification. Write the subdivision intervals as $I_i = [t_{i-1}, t_i]$. Around each inner division point t_i, choose disjoint joining intervals $J_i = (t_i - \delta_i, t_i + \delta_i) \Subset D'_b \cap \mathbb{R}$ and fixed smooth functions $\rho_i : J_i \to [0, 1]$ equal to 0 near the left endpoint and 1 near the right endpoint. Denote the path just constructed between g_i and g_{i+1} by $\widehat{A}^{(i)}$. With $I_i^0 = I_i \setminus \bigcup_\ell J_\ell$, the family on the common domain is

$$H(t, y, u) = \begin{cases} \widehat{A}^{(i)}_{\rho_i(t)}(t, y, u), & t \in J_i, \\ g_i(t, y, u), & t \in I_i^0. \end{cases} \tag{3.23}$$

Near the ends of J_i, (3.22) makes the first expression in (3.23) equal to g_i and g_{i+1}, respectively. Thus H is smooth in t and holomorphic in (y, u) near $I \times B \times P$. For the compact pieces

$$\mathcal{K}_i = (F \cap (I_i \times \mathbb{C}^{r-1})) \times P, \qquad \mathcal{J}_i = (F \cap (\overline{J}_i \times \mathbb{C}^{r-1})) \times P,$$

and any prescribed derivative order j, the chain rule and $\rho_i(J_i) \subset [0, 1]$ give

$$\begin{aligned} \|H - f\|_{C^j(F \times P)} \le{}& C_j \max_i \|g_i - f\|_{C^j(\mathcal{K}_i)} \\ &+ C_j \max_i \left\| \widehat{A}^{(i)} - f \right\|_{C^j([0,1] \times \mathcal{J}_i)}, \end{aligned} \tag{3.24}$$

where f is independent of the path parameter and C_j depends on the fixed ρ_i. The two errors in (3.24) can be made arbitrarily small, the first by the local entire approximation and the second by (3.12), (3.18), (3.21), and (3.22). Extend H slightly beyond the endpoints of I by its first and last entire maps. Apply Lemma 3.1 again to obtain H_1, jointly holomorphic near a complex neighborhood of $I \times B \times P$. Since B is a polydisc, Lemma 3.2 applied to H_1, with A_{r-1} and the u-variables as passive parameters, now gives an entire map in all variables. When $r = 1$, there is no y-variable and the rescaling step is omitted. □

3.2. **Gluing across a real hyperplane and convex approximation.** For $b > 0$, let

$$B_b = \{v \in \mathbb{C}^n : |v| < b\}$$

denote the Euclidean ball in the fiber variables.

Fix $0 < \alpha < 1$. For a subset M of a Euclidean space and a vector-valued function u, write

$$\|u\|_{C^\alpha(M)} = \sup_M |u| + [u]_{\alpha,M}, \qquad [u]_{\alpha,M} = \sup_{\substack{z,w\in M\\ z\neq w}} \frac{|u(z)-u(w)|}{|z-w|^\alpha}. \tag{3.25}$$

For $Q \subset \mathbb{C}^m$ compact and convex, write

$$Q_r = \{z \in \mathbb{C}^m : \operatorname{dist}(z, Q) < r\}, \qquad r > 0. \tag{3.26}$$

The interface and its two sides are

$$M_r = Q_r \cap \{\operatorname{Im} z_1 = 0\}, \qquad Q_r^\pm = Q_r \cap \{\pm \operatorname{Im} z_1 > 0\}. \tag{3.27}$$

The norms used for traces and for the two sides are

$$\begin{aligned} \|u\|_{r,b,\alpha} &= \sup_{v\in B_b} \|u(\cdot, v)\|_{C^\alpha(M_r)}, \\ \|a\|^\pm_{r,b,*} &= \sup_{Q_r^\pm \times B_b} |a| + \sup_{v\in B_b} \|a(\cdot, v)|_{M_r}\|_{C^\alpha(M_r)}. \end{aligned} \tag{3.28}$$

The difference quotient in (3.25) always uses all base variables. The fiber variable v is subject to a supremum and is treated by holomorphic estimates on smaller balls.

It remains to turn close traces on the cutting face into maps that agree exactly there. The additive splitting of [56, Lemma 4.4] plays the role of Lemma 2.5 in the C^α norms (3.28), which include all base variables. Its nonlinear correction gives the following gluing theorem across a real hyperplane [56, Lemma 4.5], the counterpart of [56, Lemma 3.2] for this geometry. The corrections also control the change on the F_- side, allowing successive cuts with a prescribed total error.

Lemma 3.4. *For a compact convex $Q \subset \mathbb{C}^m$ with the geometry (3.26)–(3.27), fix $r, b > 0$, an outer fiber margin $\mu > 0$, losses $0 < \sigma < r/2$, $0 < \tau < b$, and $\eta > 0$. There exists $\delta \in (0, \mu)$ such that the following holds. Suppose that F_-, F_+ are holomorphic on $Q_r^- \times B_{b+\mu}$, $Q_r^+ \times B_b$, continuous up to their interface, and their traces satisfy on $M_r \times B_b$*

$$F_-(z, \gamma(z, v)) = F_+(z, v), \qquad \gamma(z, v) = v + c(z, v), \qquad \|c\|_{r,b,\alpha} < \delta.$$

The transition is holomorphic in the complex tangential variables and in v. Under these hypotheses there are holomorphic fiber reparametrizations $\alpha_\pm = \mathrm{id} + u_\pm$ on $Q^\pm_{r-\sigma} \times B_{b-\tau}$, with C^α traces and values in B_b, such that

$$\begin{gathered} \|u_-\|^-_{r-\sigma,b-\tau,*} + \|u_+\|^+_{r-\sigma,b-\tau,*} < \eta, \\ F_-(z, \alpha_-(z, v)) = F_+(z, \alpha_+(z, v)) \quad (z \in M_{r-\sigma}). \end{gathered}$$

Setting $v = 0$ in the corrected families gives maps that join holomorphically across the interface. Denote their union by

$$F(z) = \begin{cases} F_-(z, \alpha_-(z, 0)), & z \in Q^-_{r-\sigma} \cup M_{r-\sigma}, \\ F_+(z, \alpha_+(z, 0)), & z \in Q^+_{r-\sigma} \cup M_{r-\sigma}. \end{cases}$$

If $K \Subset Q^-_{r-\sigma} \cup M_{r-\sigma}$, *then*

$$\sup_K d_Y(F, F_-(\cdot, 0)) \le \omega_{-,K}(\eta), \tag{3.29}$$

where $\omega_{-,K}$ *is a continuity modulus of the fixed spray* F_- *in its fiber variable.*

Proof. We recall the quantitative steps of [56, Lemma 4.5]. The proof there is written for two-dimensional fiber variables, but its argument is componentwise. The additive solution operators act on each component with the same scalar operator norm; the C^α product and composition estimates hold for maps between arbitrary fixed finite-dimensional normed spaces; and the Cauchy bounds below depend on the fiber dimension only through their constants. Thus the same iteration applies to the present fiber $\mathbb{C}^n$ after allowing C and p to depend on n. For a current transition error of trace norm e, the additive splitting of [56, Lemma 4.4], applied in all base variables with base loss h, gives side corrections whose supremum and trace norms have sum at most $Ch^{-p}e$. Here C and p depend on the fixed geometry, dimensions, and Hölder exponent. Cauchy estimates across a fiber loss h control the first two fiber derivatives, with further fixed powers of h^{-1}. The C^α multiplication and composition estimates then give, after increasing p, the quadratic bound

$$e_{\mathrm{new}} \le Ch^{-p}e^2$$

for the corrected transition on the smaller domains.

Choose $0 < h_0 < 1$ and $h_i = h_0 2^{-i}$ with $\sum_i h_i < \min(\sigma, \tau)$. For sufficiently small initial error, the iteration of [56, Lemma 3.2] makes these errors decrease doubly exponentially. The side corrections and their fiber derivatives are summable, including the C^α trace norms in all base variables. By reducing δ, the limiting reparametrizations have values in B_b and the sum of their displacement norms is less than η. Their corrected traces agree. In a common target chart, continuity and Morera's theorem across the real hyperplane show that the maps obtained by setting $v = 0$ join holomorphically. Finally, their displacement on the F_- side at $v = 0$ is less than η, so the uniform continuity of the fixed F_- on each compact product gives (3.29). □

Proof of Theorem 2.3. We prove A_r by induction on r, following [56, Section 4.4, proof of Theorem 3.1]. The base case A_0 follows from Proposition 2.9. Assume that A_{r-1} holds.

Extension across one real affine hyperplane. Write the cut as $\ell \le b$, where ℓ is a nonconstant real affine function, and set

$$\ell(z) = \operatorname{Re} \sum_{j=1}^{r} a_j z_j + c, \qquad w_1 = i\left(\sum_{j=1}^{r} a_j z_j + c - b\right), \qquad \operatorname{Im} w_1 = \ell(z) - b. \tag{3.30}$$

Complete the nonzero linear part of w_1 to a basis of complex linear functionals and retain the translation in (3.30). After renaming the resulting affine coordinates,

$$Q_- = Q \cap \{\operatorname{Im} z_1 \le 0\} \subset Q \subset \mathbb{C}^r,$$

where Q is compact and convex and f is holomorphic near $Q_- \times P$. When Q_- has nonempty interior and $Q_- \ne Q$, the cutting hyperplane meets $\operatorname{int} Q$; this is the case treated here. Take a slightly larger passive polydisc $P^+ \ni P$. For small $r_* > 0$,

$$(\overline{Q_{r_*}} \cap \{\operatorname{Im} z_1 \le 0\}) \times P^+ \subset \operatorname{dom} f. \tag{3.31}$$

Indeed, otherwise points outside $\operatorname{dom} f$ with $\operatorname{Im} z_1 \le 0$ and distance to $Q \times P^+$ tending to zero would have a subsequence converging to a point of $Q_- \times P^+$. This contradicts the compact containment $Q_- \times P^+ \Subset \operatorname{dom} f$.

Lemma 2.4, applied on the convex lower-side domain supplied by (3.31), gives a fixed chart spray $f^\sharp(z, u, v)$ with fiber domain Δ^n_β. Before approximating it, fix $b, \mu > 0$, $0 < \tau < b$, and

$\beta_0 < \beta$ such that

$$\overline{B}_{b+\mu} \Subset \Delta^n_{\beta_0} \Subset \Delta^n_{\beta}.$$

Apply Lemma 3.3 on a slightly larger compact real face, with the reserved u-polydisc and $\overline{\Delta}^n_{\beta_0}$ as passive factors. It gives an entire map $G(z,u,v)$ approximating the trace of $f^\sharp$ with tangential derivative control through order two. The order-two accuracy and the Cauchy estimates on the reserved parameter margins give uniform first-derivative control on the smaller products used below. For the gluing, the compact convex base and its complex dimension are

$$\mathcal{Q} = Q \times P, \qquad m = r + \dim P, \tag{3.32}$$

with coordinates (z,u); only $v \in \mathbb{C}^n$ remains a fiber variable. Small Euclidean thickenings of $\mathcal{Q}$ lie in the reserved product neighborhoods. On the interface, invert the fixed fiber charts of $f^\sharp$ to obtain $\gamma = \mathrm{id} + c$. The chain rule and Cauchy estimates in u across the reserved margin give the following bounds, where the norms use all base variables (z,u), uniformly for $v \in B_b$, and D_0 is the diameter of the retained convex face:

$$\|c\|_{C^1} \le C \left\| G - f^\sharp \right\|_{C^1} \longrightarrow 0, \tag{3.33}$$

$$\|c\|_{C^\alpha} \le \|c\|_{C^0} + D_0^{1-\alpha} \|D_{z,u} c\|_{C^0} \longrightarrow 0, \tag{3.34}$$

$$\sup_{v \in B_b} \|c(\cdot, v)\|_{C^\alpha(\mathcal{Q}_{r_*} \cap \{\operatorname{Im} z_1 = 0\})} \longrightarrow 0. \tag{3.35}$$

Thus Lemma 3.4 applies on $\mathcal{Q}_{r_*}$ with the base (3.32), the fixed fiber margins above, and any prescribed loss $0 < \sigma < r_*/2$, by (3.35). Since the old chart spray is fiberwise injective, equality of the corrected traces is equivalent to the transition identity. The resulting corrections satisfy

$$\gamma(z,u,\alpha_+(z,u,v)) = \alpha_-(z,u,v) \quad \text{on the interface,} \qquad \|\alpha_- - \mathrm{id}\|_* + \|\alpha_+ - \mathrm{id}\|_* < \eta. \tag{3.36}$$

The resulting map on $\mathcal{Q}_{r_*-\sigma}$ is

$$\widetilde{f}(z,u) = \begin{cases} f^\sharp(z,u,\alpha_-(z,u,0)), & \operatorname{Im} z_1 \le 0, \\ G(z,u,\alpha_+(z,u,0)), & \operatorname{Im} z_1 \ge 0. \end{cases} \tag{3.37}$$

The two traces in (3.37) agree by (3.36), so Lemma 3.4 makes $\widetilde{f}$ holomorphic near $Q \times P$. Equation (3.29) gives

$$\sup_{Q_- \times P} d_Y(\widetilde{f}, f) \le \omega_{f^\sharp, Q_- \times P}(\eta) \longrightarrow 0 \quad (\eta \longrightarrow 0). \tag{3.38}$$

Successive removal of the half-space constraints. To obtain approximation on an arbitrary compact convex set $K \subset \mathbb{C}^r$, take a map f holomorphic near $K \times P$. After thickening K and P inside the original domain, choose a full-dimensional real convex polytope K' with

$$K \Subset \operatorname{int} K', \qquad K' \times P \Subset \operatorname{dom} f.$$

Choose R with $K' \Subset \operatorname{int} \overline{\Delta}^r_R$ and real affine inequalities $\ell_j(z) \le b_j$ such that

$$K' = \bigcap_{j=1}^{M} \{\ell_j \le b_j\}, \qquad Q_k = \overline{\Delta}^r_R \cap \bigcap_{j=k+1}^{M} \{\ell_j \le b_j\} \quad (0 \le k \le M), \tag{3.39}$$

so that

$$Q_0 = K', \qquad Q_M = \overline{\Delta}^r_R, \qquad Q_{k-1} = Q_k \cap \{\ell_k \le b_k\}. \tag{3.40}$$

Every Q_k in (3.39) is compact and convex and contains the full-dimensional set K', and whenever a step enlarges the set, its cutting hyperplane meets $\operatorname{int} Q_k$. By (3.30), the cut in (3.40) reduces to the case treated above. Applying (3.38) successively, starting with $f_0 = f$, gives f_k holomorphic near $Q_k \times P$ with

$$\sup_{Q_{k-1} \times P} d_Y(f_k, f_{k-1}) < \epsilon_k, \qquad \sum_{k=1}^{M} \epsilon_k < \epsilon/2. \tag{3.41}$$

Using (3.41), Proposition 2.9 gives an entire F with

$$\sup_{\overline{\Delta}_R^r \times P} d_Y(F, f_M) < \epsilon/2, \qquad \sup_{K \times P} d_Y(F, f) < \epsilon.$$

This proves A_r. Taking the passive polydisc P to be a point gives CAP. □

Theorem 2.3 and Proposition 2.2 now give the two principal equivalences.

Proof of Theorems 1.2 and 1.3. The zero-dimensional case is immediate. By Proposition 2.2, Theorem 2.3, and Definition 1.1, we have

$$\text{strong dominability} \Longrightarrow \mathcal{L}_n \Longrightarrow \text{CAP} \Longleftrightarrow \text{Oka}.$$

Conversely, every Oka manifold is strongly dominable [25, Proposition 7.1.9(a)]. This completes both equivalences. □

Proof of Corollary 1.4. Only the converse requires proof. Fix $x \in X$ and a coordinate parametrization $\phi : B_{2r} \to X$ with $\phi(0) = x$. By CAP_n, an entire map $G : \mathbb{C}^n \to X$ can approximate ϕ arbitrarily closely on $\overline{B}_{3r/2}$. Choose G so close that $G(\overline{B}_{3r/2}) \subset \phi(B_{2r})$ and $g = \phi^{-1} \circ G$ satisfies

$$\sup_{\overline{B}_r} |g(z) - z| < \frac{r}{2}, \qquad \sup_{\overline{B}_r} \|dg_z - I\| < \frac{1}{2}.$$

The derivative estimate follows from the Cauchy estimates. Hence $T(z) = z - g(z)$ is a contraction of $\overline{B}_r$ into itself. Its fixed point a satisfies $g(a) = 0$, and dg_a is invertible. Thus $H(z) = G(z + a)$ is an entire map with $H(0) = x$ and dH_0 invertible. The manifold X is strongly dominable, so Theorem 1.3 applies. □

Corollary 1.5 follows from [31, Theorem 2], which makes a manifold with a finite Oka stratification strongly dominable, together with Theorem 1.3.

For every $x \in X$, a dominating map into an Oka neighborhood of x, composed with the inclusion into X, dominates X at x. Thus X is strongly dominable, and Theorem 1.3 applies.

4. Propagation of local approximation

We first propagate LAP from the boundary of a coordinate disc to its center. This yields propagation across sets of zero $(2n-1)$-dimensional Hausdorff measure and proper analytic subsets, and proves the union theorem. For surfaces, propagation across analytic subsets was proved in [56, Corollary 2.10]. We combine the constructions using coordinate discs and planar enlargement from that proof with the scalar-substitution method of [3, Lemma 4.3 and Section 4.4], using n recentered scalar families that equal the identity at $t = 0$. These families are defined on scalar discs of increasing radius and converge for t near zero.

4.1. Propagation from the boundary of a coordinate disc.

For a complex manifold Y of dimension $n \geq 1$, let

$$\mathcal{G} = \{x \in Y : Y \text{ has LAP at } x\}.$$

The LAP locus $\mathcal{G}$ is open: every point of a witness neighborhood has that same witness neighborhood. We will show that $\mathcal{G} = Y$ if $Y \setminus \mathcal{G}$ has zero $(2n-1)$-dimensional Hausdorff measure, or is locally contained in a proper analytic subset.

In this section the fiber domains are the maximum-norm polydiscs

$$\Delta_r^n = \{w \in \mathbb{C}^n : \|w\|_\infty < r\};$$

vector norms are the maximum norm and matrix norms its induced operator norm. Closed polydiscs retain the notation $\overline{\Delta}_r^n$. All maps near compact products are restricted, when needed, to a product of an open planar neighborhood and an open polydisc. These sources satisfy the

triviality hypothesis of Lemma 2.4, so its fiber-preserving coordinates and uniform inverse-chart margins are available throughout this section.

The next lemma adapts the scalar-substitution argument of An–Xie [3, Lemma 4.3] to approximating families defined on scalar discs of increasing radius and converging near zero. It shows that holomorphic scalar substitution preserves approximability on the original source set. We first fix a polynomial approximation to the scalar function, then choose the family index so that its scalar domain contains the polynomial's range on the compact set used by the second family.

Lemma 4.1. *Fix a special planar disc pair $D \subset D'$ and a closed polydisc $P_* \subset \mathbb{C}^q$, where $q \geq 0$. Write*

$$K = D \times P_*, \qquad L = D' \times P_*,$$

write $s = (z, p)$ for a source point, and let $\mathscr{A}$ consist of the maps in $\mathscr{O}(\operatorname{Op} K, Y)$ uniformly approximable on K by maps in $\mathscr{O}(\operatorname{Op} L, Y)$. Suppose $f_0 \in \mathscr{A}$, Ω is an open neighborhood of K, and

$$G : \Omega \times \Delta_r^n \to Y, \qquad G(s, 0) = f_0(s),$$

is holomorphic, with $(s, w) \mapsto (s, G(s, w))$ biholomorphic onto its image. Suppose

$$T_\nu : \Omega \times \Delta_r^n \times \Delta_{R_\nu} \to Y, \qquad T_\nu(s, w, 0) = G(s, w), \qquad R_\nu \to \infty,$$

are holomorphic and converge locally uniformly on $\Omega \times \Delta_r^n \times \Delta_\tau$, for some $\tau > 0$, to a holomorphic map T. For every $\varphi \in \mathscr{O}(\operatorname{Op} K, \mathbb{C})$ with $\|\varphi\|_K < \tau$, the map

$$s \longmapsto T(s, 0, \varphi(s))$$

belongs to $\mathscr{A}$.

Proof. Initial approximation and planar domains. Fix $\epsilon > 0$ and numbers $\|\varphi\|_K < t_0 < t_1 < \tau$. Since $f_0 \in \mathscr{A}$, there is $g \in \mathscr{O}(\operatorname{Op} L, Y)$ so close to f_0 on K that

$$\rho(s) = G(s, \cdot)^{-1}(g(s))$$

is holomorphic near K, $\|\rho\|_K < r/16$, and

$$\sup_K d_Y\big(T(s, \rho(s), \varphi(s)), T(s, 0, \varphi(s))\big) < \epsilon/4. \tag{4.1}$$

The last assertion follows from uniform continuity on a compact neighborhood of $K \times \{0\} \times \varphi(K)$. Lemma 2.4 supplies a family $F(s, v)$ centered at g near L, available for $v \in \Delta_{2b}^n$, with inverse fiber coordinates Ψ_s.

We specify the planar geometry used for the two families. Write $D' = D \cup E$, where E is the attached disc. In E, a crosscut close to $D \cap E$, but disjoint from D, separates a small collar of D from the rest of E. A narrow closed neighborhood of that crosscut and small thickenings of the two sides give compact Jordan regions A, B such that

$$D \subset \operatorname{int} B, \qquad D' \subset \operatorname{int}(A \cup B), \qquad C = A \cap B \text{ is a closed Jordan disc disjoint from } D,$$
$$\overline{A \setminus B} \cap \overline{B \setminus A} = \varnothing.$$

The collar can be as thin as needed, so $B \times P_*$ lies in the common domain of G, ρ, φ, and F is available near $A \times P_*$. There are disjoint closed Jordan neighborhoods D^+, C^+ of D, C, a compact neighborhood B^+ of B, and polydiscs

$$P_* \Subset \operatorname{int} P^+ \Subset \operatorname{int} P^{++}$$

such that

$$D \Subset \operatorname{int} D^+, \quad C \Subset \operatorname{int} C^+, \quad D^+ \cup C^+ \subset B^+, \quad B \Subset \operatorname{int} B^+,$$

and $B^+ \times P^{++}$ stays in those common domains, with $\|\rho\| < r/8$ there. The family F is available near both $A \times P^{++}$ and $C^+ \times P^{++}$. All these choices precede the index ν and all polynomial approximations below.

Choose fixed open planar neighborhoods U_A, U_B of A, B that cover a neighborhood of D' and have separated fringes. They can be taken with

$$U_A \cap U_B \Subset \operatorname{int} C^+, \qquad U_B \Subset \operatorname{int} B^+,$$

and within the relevant family domains. A fixed open polydisc U_P satisfies $P_* \Subset U_P \Subset \operatorname{int} P^+$. Lemma 2.5 applies with a fixed constant C_0.

The sets $C^+ \times P^{++}$ and $(D^+ \cup C^+) \times P^+$ are polynomially convex, since the planar factors have connected complements and polynomial hulls commute with products. This allows a matrix approximation on the overlap and a scalar approximation with different data near D and C.

Uniform estimates and choice of parameters. On a neighborhood of $(s, \rho(s), 0)$ over $C^+ \times P^{++}$, consider

$$\mathcal{H}_\nu(s, w, u) = \Psi_s(T_\nu(s, w, u)).$$

The exact normalization at $u = 0$ gives

$$\mathcal{H}_\nu(s, \rho(s), 0) = 0, \qquad \mathcal{E}(s) := \partial_w \mathcal{H}_\nu(s, \rho(s), 0) = \partial_w(\Psi_s \circ G)(s, \rho(s)) \in \mathrm{GL}_n(\mathbb{C}). \tag{4.2}$$

In particular, both the center and $\mathcal{E}$ are independent of ν. The inverse chart of F contains a fixed neighborhood of the compact graph of g on this product. Since $T(s, \rho(s), 0) = g(s)$, continuity first gives a common smaller neighborhood in (w, u) for the limit. Local uniform convergence then places all sufficiently large T_ν in the same inverse chart there. On still smaller compact products, Cauchy estimates give uniform bounds for the first and second derivatives of $\mathcal{H}_\nu$. Likewise, on a fixed compact neighborhood of

$$K \times \overline{\Delta^n_{r/2}} \times \overline{\Delta_{t_1}},$$

the first derivatives of T_ν in w, u are uniformly bounded in fixed finitely many target charts containing compact pieces of the limit image with positive margins. The source compact sets and target charts are independent of ν, and the same charts contain the corresponding images of all sufficiently large T_ν.

Write

$$N = 1 + \left\| \mathcal{E}^{-1} \right\|_{C^+ \times P^{++}}.$$

Fix a transition tolerance $\varepsilon_{\mathrm{tr}} > 0$ so small that a transition within $\varepsilon_{\mathrm{tr}}$ of the identity on Δ^n_2 satisfies (2.3) on Δ^n_1 and gives corrections of norm less than $1/4$. Here the derivative bound follows from Cauchy estimates, and the splitting constant C_0 is already fixed. The uniform Taylor bounds above give the estimate (4.7) below with a constant $c > 0$ depending only on the fixed products, $\mathcal{E}$, N, and those derivative bounds. In particular, c is independent of the polynomial matrix chosen below and of its size on $B^+ \times P^+$.

Choose $0 < \eta \le 1$ with $c\eta < \varepsilon_{\mathrm{tr}}/3$. Oka–Weil approximation [17, Theorem 18] and Taylor truncation give a polynomial matrix L_B such that

$$\left\| L_B - \mathcal{E}^{-1} \right\|_{C^+ \times P^{++}} < \eta, \qquad \|L_B\|_{C^+ \times P^{++}} \le N.$$

Fix this matrix and set

$$M = \max\{1, \|L_B\|_{B^+ \times P^+}\} < \infty.$$

The derivative bounds on the fixed target charts and uniform continuity of their inverse coordinate maps give moduli ω_w, ω_u, independent of all sufficiently large ν, such that

$$\begin{aligned} d_Y(T_\nu(s, w, u), T_\nu(s, w', u)) &\le \omega_w(\|w - w'\|_\infty), \\ d_Y(T_\nu(s, w, u), T_\nu(s, w, u')) &\le \omega_u(|u - u'|) \end{aligned} \tag{4.3}$$

on $K \times \overline{\Delta^n_{r/2}} \times \overline{\Delta}_{t_1}$, with $\omega_w(h), \omega_u(h) \to 0$ as $h \to 0$. Choose $\delta > 0$ sufficiently small that the uniform overlap Taylor estimates apply, $c\delta < \varepsilon_{\mathrm{tr}}/3$, and

$$\delta < b/2, \qquad \delta M < r/8, \qquad \omega_w(\delta M/4) < \epsilon/4. \tag{4.4}$$

Choose $e_0 > 0$ within the fixed scalar margin for the overlap Taylor estimates and so small that

$$ce_0/\delta < \varepsilon_{\mathrm{tr}}/3.$$

Choose $e_1 > 0$ such that

$$e_1 < t_0 - \|\varphi\|_K, \qquad \omega_u(e_1) < \epsilon/4. \tag{4.5}$$

With these errors fixed, the Oka–Weil theorem [17, Theorem 18] gives a scalar polynomial u_B satisfying

$$\|u_B - \varphi\|_{D^+\times P^+} < e_1, \qquad \|u_B\|_{C^+\times P^+} < e_0. \tag{4.6}$$

Indeed, the data φ and 0 are holomorphic on disjoint neighborhoods of the two polynomially convex pieces. Equations (4.6) and (4.5) give $|u_B| < t_0$ on K.

Transition estimate and the family index. On the two sides consider the families

$$F_A(s,\zeta) = F(s,\delta\zeta), \qquad F_{B,\nu}(s,\zeta) = T_\nu(s,\rho(s) + \delta L_B(s)\zeta, u_B(s)).$$

The first is available for $\zeta \in \Delta_4^n$. On the second side,

$$\|\rho(s) + \delta L_B(s)\zeta\| < r/8 + 2\delta M < r/2 \qquad (\zeta \in \Delta_2^n).$$

On the overlap its transition is

$$\gamma_\nu(s,\zeta) = \delta^{-1}\Psi_s(F_{B,\nu}(s,\zeta)).$$

Taylor expansion at $(w,u) = (\rho(s),0)$, using (4.2), gives

$$\|\gamma_\nu - \mathrm{id}\|_{(U_A\cap U_B)\times U_P\times\Delta_2^n} \le c(\eta + \delta + e_0/\delta), \tag{4.7}$$

uniformly for large ν. Indeed, (4.2) gives the exact expansion

$$\begin{aligned} \mathcal{H}_\nu(s,\rho(s) + \delta L_B(s)\zeta, u_B(s)) &= \delta\mathcal{E}(s)L_B(s)\zeta + \mathcal{R}_\nu(s,\zeta), \\ \|\mathcal{R}_\nu\|_\infty \le C(\delta^2N^2 + e_0), &\qquad \|\mathcal{E}L_B - I\|_\infty \le \|\mathcal{E}\|_\infty\,\eta. \end{aligned} \tag{4.8}$$

The quadratic term in (4.8) uses the uniform second derivative bound in w, and the term Ce_0 uses the first derivative bound in u and (4.6). Dividing (4.8) by δ and using $\|\zeta\|_\infty < 2$ gives (4.7), with c depending on the fixed products, N, and the uniform derivative bounds.

By the preceding choices, the right side of (4.7) is less than $\varepsilon_{\mathrm{tr}}$. Cauchy estimates from Δ_2^n therefore give (2.3) on Δ_1^n, with corrections of norm less than $1/4$.

The increasing scalar radii enter here: the polynomial u_B is already fixed and has a finite norm on $B^+ \times P^+$. Choose ν last, so large that

$$R_\nu > 1 + \|u_B\|_{B^+\times P^+}$$

and

$$\sup_K d_Y\left(T_\nu(s,\rho(s),\varphi(s)), T(s,\rho(s),\varphi(s))\right) < \epsilon/4. \tag{4.9}$$

Thus $u_B(B^+ \times P^+) \subset \Delta_{R_\nu}$, so $F_{B,\nu}$ is available on the full fixed B side. On the overlap, (4.6) keeps u_B in the fixed small scalar disc on which (4.8) holds.

Gluing and the approximation error. Apply Lemma 2.6 with working radius 1 and the inverse chart of F_A on Δ_4^n. It supplies a, b on the two sides with

$$F_A(s,a(s)) = F_{B,\nu}(s,b(s)), \qquad \|a\|, \|b\| < 1/4.$$

These expressions give a map $\widetilde{g}$ holomorphic near L. On $K \subset B \times P_*$ it is

$$\widetilde{g}(s) = T_\nu(s,\rho(s) + \delta L_B(s)b(s), u_B(s)).$$

For $s \in K$, the triangle inequality gives

$$\begin{aligned} & d_Y\big(\widetilde{g}(s), T(s,0,\varphi(s))\big) \\ & \le d_Y\big(\widetilde{g}(s), T_\nu(s,\rho(s),u_B(s))\big) \\ & \quad + d_Y\big(T_\nu(s,\rho(s),u_B(s)), T_\nu(s,\rho(s),\varphi(s))\big) \\ & \quad + d_Y\big(T_\nu(s,\rho(s),\varphi(s)), T(s,\rho(s),\varphi(s))\big) \\ & \quad + d_Y\big(T(s,\rho(s),\varphi(s)), T(s,0,\varphi(s))\big) \\ & \le \omega_w(\delta M/4) + \omega_u(e_1) + \epsilon/4 + \epsilon/4 < \epsilon. \end{aligned}$$

The four bounds follow from (4.3), (4.4), (4.5), (4.9), and (4.1). Since $\widetilde{g} \in \mathscr{O}(\mathrm{Op}(D' \times P_*), Y)$, the map $T(\cdot, 0, \varphi)$ belongs to $\mathscr{A}$. □

We iterate scalar substitution in independent tangent directions, using exact intermediate maps as in [3, Section 4.4]. Composing the limiting families gives local coordinates at x, so a map with image in a sufficiently small neighborhood of x can be obtained from the constant map by successive substitutions.

Proposition 4.2. *Let $x \in Y$, $\dim Y = n \ge 1$. Suppose on one neighborhood $V \ni x$ there are holomorphic maps*

$$h_{j,\nu} : V \times \Delta_{R_\nu} \to Y, \qquad h_{j,\nu}(y,0) = y, \qquad 1 \le j \le n, \qquad R_\nu \to \infty.$$

Suppose these maps converge locally uniformly on $V \times \Delta_\tau$, for some $\tau > 0$, to holomorphic maps h_j, and that $\partial_t h_j(x,0)$, $1 \le j \le n$, form a basis of $T_x Y$. Then $x \in \mathcal{G}$.

Proof. The limits satisfy $h_j(y,0) = y$. For $a = (a_1, \dots, a_n)$ near zero, consider the exact compositions

$$\theta_0(a) = x, \qquad \theta_j(a) = h_j(\theta_{j-1}(a), a_j), \qquad \Theta(a) = \theta_n(a).$$

On a sufficiently small polydisc $Q \ni 0$, every intermediate value belongs to a fixed relatively compact subset of V, and $|a_j| < \tau$. Since $d_y h_j(y,0) = \mathrm{id}$, the columns of $d\Theta_0$ are $\partial_t h_j(x,0)$. The inverse function theorem gives a fixed neighborhood $W \ni x$ on which Θ^{-1} is available with image relatively compact in Q.

For arbitrary data $f \in \mathscr{O}(\mathrm{Op}(D \times P), W)$ in the definition of LAP, choose

$$P_0 \Subset \operatorname{int} P_* \Subset \operatorname{int} P, \qquad (a_1, \dots, a_n) = \Theta^{-1} \circ f,$$

and consider near $K = D \times P_*$ the maps

$$f_0 = x, \qquad f_j = h_j(f_{j-1}, a_j), \qquad 1 \le j \le n.$$

They are exact holomorphic maps, $f_n = f$, and all their values lie in V; also $\|a_j\|_K < \tau$. The constant map f_0 is in the approximation class $\mathscr{A}$ of Lemma 4.1.

If $f_{j-1} \in \mathscr{A}$, Lemma 2.4 gives coordinates $G_j(s,w)$ centered at f_{j-1}, with their whole image in V after restriction of the fixed base neighborhood and the fiber radius. The families

$$T_{j,\nu}(s,w,t) = h_{j,\nu}(G_j(s,w), t), \qquad T_j(s,w,t) = h_j(G_j(s,w), t)$$

satisfy every hypothesis of Lemma 4.1; in particular their value at $t = 0$ is exactly $G_j(s,w)$. With $\varphi = a_j$, that lemma yields $f_j \in \mathscr{A}$. Finite induction gives $f \in \mathscr{A}$, and restriction to P_0 proves LAP at x.

□

We obtain the required families from a coordinate disc whose boundary lies in $\mathcal{G}$. Lemma 2.7 enlarges the scalar domain; inverse branches then recenter the approximants so that the map at $t = 0$ is exactly the identity on one fixed target neighborhood. This normalization permits repeated application of Lemma 4.1. For $n = 2$, the conclusion below is the first assertion of [56, Corollary 2.10]; Proposition 4.2 gives it in arbitrary dimension.

Proposition 4.3. *Let $\vartheta : U \to \Omega \subset \mathbb{C}^n$ be a coordinate chart, $x \in U$, and $v \neq 0$. Suppose for some $r > 0$ that*

$$\{\vartheta(x) + \zeta v : |\zeta| \leq r\} \subset \Omega, \qquad \vartheta^{-1}(\{\vartheta(x) + \zeta v : |\zeta| = r\}) \subset \mathcal{G}.$$

Then $x \in \mathcal{G}$.

Proof. Independent directions and transverse discs. The closed disc is compactly contained in the chart, and its boundary is a compact subset of the open set $\mathcal{G}$. Hence all sufficiently small changes of the direction v preserve both properties. Choose n such directions $v_1, \ldots, v_n$ which are linearly independent. Their choice and nonzero determinant are fixed before any approximation.

For each j, complete v_j to a basis $(v_j, e_{j,2}, \ldots, e_{j,n})$ of $\mathbb{C}^n$ and consider

$$\psi_j(z, p) = \vartheta^{-1}\left(\vartheta(x) + zv_j + \sum_{k=2}^{n} p_{k-1} e_{j,k}\right).$$

A sufficiently small closed polydisc $P_j \subset \mathbb{C}^{n-1}$ makes this a coordinate parametrization near $\overline{\Delta}_r^1 \times P_j$. Moreover, for each $\zeta \in \partial\Delta_r$, the whole fiber $\psi_j(\{\zeta\} \times P_j)$ lies in a coordinate witness neighborhood for LAP. To obtain one P_j for all ζ, first cover the boundary of the central disc by finitely many witness neighborhoods, use continuity to obtain corresponding source arcs, and then shrink the transverse polydisc so that the compact arc products stay in those neighborhoods.

For each j, choose $P_j' \Subset \operatorname{int} P_j$ centered at zero. Fix one sequence $S_\nu > r$ with $S_\nu \to \infty$, to be used for all j. Applying Lemma 2.7 for each j and ν, with errors tending to zero, gives

$$\sigma_{j,\nu} \in \mathscr{O}(\operatorname{Op}(\overline{\Delta}_{S_\nu}^1 \times P_j'), Y), \qquad \sigma_{j,\nu} \longrightarrow \psi_j \quad \text{uniformly on } \overline{\Delta}_r^1 \times P_j'.$$

Recentering the approximants. On a smaller fixed product around $(0, 0)$, the images lie in the target chart, and Cauchy estimates give convergence of the first derivatives. Choose $\varrho_j > 0$ with $\overline{\Delta}_{\varrho_j}^n \Subset \Delta_r \times \operatorname{int} P_j'$. In the coordinates of ψ_j, write

$$A_{j,\nu} = \psi_j^{-1} \circ \sigma_{j,\nu} = \mathrm{id} + e_{j,\nu}, \qquad \|e_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} < \varrho_j/4, \quad \|De_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} < 1/2 \tag{4.10}$$

for all sufficiently large ν. Since there are only finitely many j, we discard one finite initial segment so that these estimates hold for every j. For $b \in \Delta_{\varrho_j/2}^n$, the map $a \mapsto b - e_{j,\nu}(a)$ satisfies, by (4.10),

$$\|b - e_{j,\nu}(a)\|_\infty < 3\varrho_j/4, \qquad \|e_{j,\nu}(a) - e_{j,\nu}(a')\|_\infty \leq \tfrac{1}{2}\|a - a'\|_\infty .$$

It is therefore a contraction of $\overline{\Delta}_{\varrho_j}^n$ into its interior. Its unique fixed point $a_{j,\nu}(b)$ satisfies

$$A_{j,\nu}(a_{j,\nu}(b)) = b, \qquad \|a_{j,\nu}(b) - b\|_\infty \leq \|e_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} \longrightarrow 0.$$

The invertibility of $DA_{j,\nu}$ in (4.10) makes $a_{j,\nu}$ holomorphic. On the fixed neighborhood

$$V = \bigcap_{j=1}^{n} \psi_j(\Delta_{\varrho_j/2}^n)$$

of x, this gives inverse branches

$$\ell_{j,\nu} = a_{j,\nu} \circ \psi_j^{-1} = (s_{j,\nu}, p_{j,\nu}), \qquad \sigma_{j,\nu} \circ \ell_{j,\nu} = \mathrm{id}_V .$$

Limiting scalar families. After another fixed restriction of V, the images of all these inverse branches lie in compact subproducts of $\Delta_r \times \operatorname{int} P_j'$. They converge to ψ_j^{-1}. Since j ranges over a finite set, there is $C > 0$ such that $|s_{j,\nu}(y)| < C$ for every j, every $y \in V$, and every retained index. Let $R_\nu = S_\nu - C$ and discard finitely many initial terms so that $R_\nu > 0$. The maps

$$h_{j,\nu}(y, t) = \sigma_{j,\nu}(s_{j,\nu}(y) + t, p_{j,\nu}(y))$$

are consequently holomorphic on the common product $V \times \Delta_{R_\nu}$ and satisfy

$$h_{j,\nu}(y,0) = y$$

exactly. The radii R_ν tend to infinity.

There is one $\tau > 0$ such that, for $y \in V$ and $|t| < \tau$, all translated source points remain in a fixed compact subproduct of the original convergence product. Thus these maps converge locally uniformly on $V \times \Delta_\tau$ to

$$h_j(y,t) = \psi_j(\psi_j^{-1}(y) + (t,0)).$$

Their derivatives $\partial_t h_j(x,0)$ are $(d\vartheta_x)^{-1}v_j$ and form a basis. Proposition 4.2 gives $x \in \mathcal{G}$. □

4.2. **Exceptional sets of zero Hausdorff measure and analytic subsets.** Proposition 4.3 gives propagation across sets of zero $(2n-1)$-dimensional Hausdorff measure. We use Hausdorff outer measure with respect to a smooth Riemannian metric; its null sets are independent of the choice of such a metric.

Lemma 4.4. *Let Y be a complex manifold of dimension $n \geq 1$, and let $E \subset Y$ satisfy $\mathcal{H}^{2n-1}(E) = 0$. If Y has LAP at every point of $Y \setminus E$, then Y has LAP everywhere and is Oka.*

Proof. Throughout this proof, $|\cdot|$ is the Euclidean norm on $\mathbb{C}^n$. Let $\mathcal{G}$ be the open LAP locus and fix $x \in Y$. Choose a chart $\vartheta : U \to \Omega \subset \mathbb{C}^n$ with $\vartheta(x) = 0$, a ball $\overline{B}(0,R) \Subset \Omega$, and numbers $0 < a < b < R$. Smooth coordinate changes are locally bi-Lipschitz, so

$$A = \vartheta(E \cap U) \cap \{z : a \leq |z| \leq b\} \quad \text{satisfies} \quad \mathcal{H}^{2n-1}(A) = 0.$$

Represent $\mathbb{P}^{n-1}$ by the rank-one Hermitian projections $P_z = zz^*/|z|^2$, where z^* denotes the conjugate transpose. On the closed annulus $a \leq |z| \leq b$, the map

$$q(z) = (P_z, |z|) \in \mathbb{P}^{n-1} \times [a,b]$$

is Lipschitz: the normalization map satisfies $\left|\frac{z}{|z|} - \frac{w}{|w|}\right| \leq 2|z-w|/a$, and hence $\|P_z - P_w\| \leq 4|z-w|/a$ in the Frobenius norm. Consequently $\mathcal{H}^{2n-1}(q(A)) = 0$. The manifold $\mathbb{P}^{n-1} \times (a,b)$ has real dimension $2n-1$ and positive $(2n-1)$-dimensional Hausdorff measure. Choose $(P_v, r) \notin q(A)$ with $|v| = 1$ and $a < r < b$. The fibers are the circles in complex lines

$$q^{-1}(P_v, r) = \{\zeta v : |\zeta| = r\},$$

parameterized by $\mathbb{P}^{n-1} \times (a,b)$. This circle avoids $\vartheta(E \cap U)$, whereas its filled disc lies in Ω. Its inverse image therefore has boundary in $\mathcal{G}$. Proposition 4.3 gives $x \in \mathcal{G}$. This works also for $n = 1$, when $\mathbb{P}^0$ is a point. Since x was arbitrary, Theorem 2.3 proves that Y is Oka. □

Corollary 4.5. *Let Y be an n-dimensional complex manifold, $n \geq 1$, and let $E \subset Y$ satisfy $\mathcal{H}^{2n-1}(E) = 0$. If Y is dominable at every point of $Y \setminus E$, then Y is Oka.*

This follows from Proposition 2.2 and Lemma 4.4.

A suitable coordinate line through a point of a proper analytic subset avoids the subset on a small punctured disc, providing the boundary condition in Proposition 4.3. The following is the arbitrary-dimensional counterpart of the second assertion of [56, Corollary 2.10].

Corollary 4.6. *Suppose every point of Y has a connected neighborhood on which $Y \setminus \mathcal{G}$ is contained in a proper complex analytic subset. Then $\mathcal{G} = Y$.*

Proof. Consider a point $x \notin \mathcal{G}$. In a connected coordinate neighborhood centered at x, let B be a proper analytic subset containing the local complement of $\mathcal{G}$. Its ideal has a nonzero holomorphic germ a at x. In centered coordinates, write its first nonzero homogeneous term as a_m. Choose $v \in \mathbb{C}^n$ with $a_m(v) \neq 0$. Then

$$a(\zeta v) = \zeta^m a_m(v) + O(\zeta^{m+1})$$

is not identically zero. On a sufficiently small punctured disc it has no zeros. The corresponding closed coordinate-line disc lies in the chart and its boundary avoids B, hence lies in $\mathcal{G}$. Proposition 4.3 gives $x \in \mathcal{G}$, a contradiction. □

The following consequence uses entire maps into the ambient manifold.

Corollary 4.7. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If X is dominable at every point of $X \setminus A$, then X is Oka.*

Proof. The case $\dim X = 0$ is immediate. For $n = \dim X \geq 1$, Proposition 2.2 gives

$$X \setminus A \subset \mathcal{G}, \qquad X \setminus \mathcal{G} \subset A.$$

Since X is connected and A is proper, the germ of A is proper at every point. Corollary 4.6 yields $\mathcal{G} = X$, and Theorem 2.3 gives CAP. Thus X is Oka by Definition 1.1. □

Proof of Theorem 1.7. By Theorem 1.3, the Oka manifold $X \setminus A$ is strongly dominable. Composing its dominating entire maps with the inclusion into X shows that X is dominable at every point of $X \setminus A$. Corollary 4.7 applies. □

4.3. **A criterion using families of entire curves.** The geometric applications use the following special case of Proposition 4.2.

Definition 4.8. An n-dimensional complex manifold Y admits *local families of entire curves spanning the tangent spaces* at $x \in Y$ if there are an open neighborhood $V \ni x$, an integer $N \geq 1$, and holomorphic maps

$$h_j : V \times \mathbb{C} \longrightarrow Y, \qquad h_j(y, 0) = y, \qquad 1 \leq j \leq N,$$

such that $\partial_t h_1(y, 0), \ldots, \partial_t h_N(y, 0)$ span $T_y Y$ for every $y \in V$. Each family is defined on the full product $V \times \mathbb{C}$; its image need not remain in V.

Corollary 4.9. *A complex manifold Y satisfying the condition in Definition 4.8 at every point has* CAP *and is Oka. More generally, the same conclusion holds if this condition, with target Y, holds at every point of $Y \setminus A$, where $A \subset Y$ is a proper closed complex analytic subset. The images of the families may meet A. For a disconnected manifold the assertion holds componentwise, provided A is proper in each connected component.*

Proof. The zero-dimensional case is immediate. At a point x with the stated families, choose $n = \dim Y$ of their tangent vectors that form a basis of $T_x Y$, and relabel the corresponding families $h_1, \ldots, h_n$. Their determinant in local coordinates is nonzero at x, so these vectors remain independent throughout a smaller neighborhood V. The restrictions $h_{j,\nu} = h_j|_{V \times \Delta_\nu}$ have radii tending to infinity, satisfy $h_{j,\nu}(y, 0) = y$, and agree with h_j on $V \times \Delta_1$. Proposition 4.2 therefore gives LAP at x. If the condition holds outside A, Corollary 4.6 gives LAP everywhere, applied on each connected component when necessary. Theorem 2.3 yields CAP. □

5. Surjective holomorphic images, complements, and group orbits

We apply Lemma 4.4 and the strong dominability criterion to holomorphic images, complements, and group actions.

5.1. **Surjective holomorphic images.**

Theorem 5.1. *Let $p : X \to Y$ be a surjective holomorphic map of complex manifolds. If X is Oka, then Y is Oka.*

Proof. Let $n = \dim Y$. The case $n = 0$ is immediate. For $n \geq 1$, let

$$C_p = \{x \in X : \operatorname{rank} dp_x < n\}, \qquad E = p(C_p).$$

Choose a locally finite stratification of C_p by complex submanifolds on each of which the restriction of p has constant rank [57, Chapter 2, Proposition 3]. Second countability makes this family countable. The ranks are at most $n-1$, and each stratum has a countable constant-rank coordinate cover. The holomorphic constant-rank theorem therefore covers E by countably many subsets of complex submanifolds of Y of dimension at most $n-1$. Thus the critical-value set $E = p(C_p)$ satisfies $\mathcal{H}^{2n-1}(E) = 0$ in the outer-measure sense.

For $y \in Y \setminus E$, choose $x \in p^{-1}(y)$. Then dp_x is surjective. Since X is Oka, Theorem 1.3 gives an entire map $F : \mathbb{C}^m \to X$, where $m = \dim X$, with $F(0) = x$ and dF_0 invertible. Choose a linear map $\iota : \mathbb{C}^n \to \mathbb{C}^m$ identifying $\mathbb{C}^n$ with a complement of $\ker d(p \circ F)_0$. The map $p \circ F \circ \iota$ dominates Y at y, so Corollary 4.5 applies. □

Theorem 5.1 answers the blow-down question in [31, Section 6, p. 3725].

Corollary 5.2. *Let $\pi : \widetilde{X} \to X$ be a proper bimeromorphic holomorphic map of complex manifolds. If $\widetilde{X}$ is Oka, then X is Oka. In particular, if $\operatorname{Bl}_Z X$ is Oka for a smooth closed complex submanifold $Z \subsetneq X$, then X is Oka.*

Indeed, π is surjective, so Theorem 5.1 applies.

Following [26, Definition 1.2], a complex manifold Y has the *basic Oka property with surjectivity* (BOPS) if every continuous map $S \to Y$ from a Stein manifold S with $\dim S \geq \dim Y$ is homotopic to a surjective holomorphic map.

Corollary 5.3. *For a complex manifold Y of dimension $n \geq 1$, the following are equivalent:*

(i) *Y is Oka.*
(ii) *There is a surjective holomorphic map $\mathbb{C}^n \to Y$.*
(iii) *There is a surjective holomorphic map $\mathbb{C}^N \to Y$ for some finite integer $N \geq 1$.*
(iv) *Y has BOPS.*

Proof. By [26, Theorem 1.1], (i) implies both (ii) and (iv). The implication (ii)⇒(iii) is immediate, and Theorem 5.1 applied to $\mathbb{C}^N$ gives (iii)⇒(i). Finally, applying BOPS to a constant map $\mathbb{C}^n \to Y$ gives (iv)⇒(ii). □

In the algebraic category, Kusakabe proved that every smooth algebraically subelliptic variety Y admits a surjective regular morphism $\mathbb{A}^{\dim Y+1} \to Y$ [50, Theorem 1.2].

5.2. **Product complements.**

Theorem 5.4. *Let X, Y be positive-dimensional Oka manifolds, and let $A \subsetneq X$ and $B \subsetneq Y$ be closed complex analytic subsets. If $Z \subset X \times Y$ is a closed complex analytic subset contained in $A \times B$, then $(X \times Y) \setminus Z$ is Oka.*

Proof. First take $Z = A \times B$. Fix (p, q) in its complement and, after interchanging the factors, assume $p \notin A$. Choose entire maps $f : \mathbb{C}^n \to X$ and $g : \mathbb{C}^m \to Y$, where $n = \dim X$ and $m = \dim Y$, with $f(0) = p$, $g(0) = q$, and both differentials invertible at zero. Since the image of g contains a neighborhood of q, there is $c \in \mathbb{C}^m$ with $g(c) \notin B$.

The closed analytic set $f^{-1}(A)$ avoids zero. Cartan's theorem A applied to its coherent ideal sheaf gives $h \in \mathscr{O}(\mathbb{C}^n)$ with $h|_{f^{-1}(A)} = 0$ and $h(0) = 1$; see [25, Section 2.6]. Set

$$\Phi(u, v) = \big(f(u), g(c + h(u)(v - c))\big).$$

If $f(u) \in A$, the second component is $g(c) \notin B$. Thus Φ takes values in the complement and $\Phi(0, 0) = (p, q)$. Its differential is block triangular at zero, with invertible diagonal blocks df_0 and dg_0. Theorem 1.3 applies.

For a general Z, write $W = (X \times Y) \setminus Z$ and $U = (X \times Y) \setminus (A \times B)$. The first part makes U Oka. It is connected, being the complement of a proper closed analytic subset in the connected complex manifold $X \times Y$, and dense in W, so W is connected. The subset $(A \times B) \cap W$ is proper and closed analytic in W, with complement U. Theorem 1.7 therefore makes W Oka. □

Corollary 5.5. *Let X, Y be positive-dimensional Oka manifolds, and let $A \subsetneq X$ and $B \subsetneq Y$ be closed complex analytic subsets.*

(i) *If $Z \subset X \times Y$ is a smooth closed complex submanifold, possibly disconnected, contained in $A \times B$, then $\mathrm{Bl}_Z(X \times Y)$ is Oka. This includes $Z = A \times B$ when A and B are smooth.*

(ii) *For every finite set $D \subset X \times Y$, both $(X \times Y) \setminus D$ and the blowup of $X \times Y$ at the distinct points of D are Oka.*

Proof. For a nonempty smooth center Z, the blowup is smooth and its exceptional-divisor complement is $(X \times Y) \setminus Z$. Apply Theorems 5.4 and 1.7; an empty center gives the Oka product.

For finite D, its projections $A_D = \mathrm{pr}_X(D)$ and $B_D = \mathrm{pr}_Y(D)$ are proper closed analytic subsets, and $D \subset A_D \times B_D$. Apply Theorem 5.4 with $A = A_D$, $B = B_D$, and $Z = D$ to obtain that the punctured product is Oka. The first part of this corollary, with the same choices, shows that the blowup along D is Oka. The case $D = \varnothing$ is immediate. □

For an algebraically Oka ambient manifold, Kusakabe proved that blowing up any smooth closed algebraic center gives an Oka manifold [45]. For algebraic X, Y, A, B, Z, the first assertion of Corollary 5.5 gives a special case of the blow-up problem [27, Problem 6.23].

Earlier finite-point results of Kusakabe include Oka complements in manifolds of dimension at least two that are Zariski locally biholomorphic to abelian complex Lie groups [48], and both Oka complements and Oka blowups of Hopf manifolds [46].

5.3. **Further complements and open group orbits.** The strong dominability criterion also recovers Kusakabe's theorem that $Y \setminus K$ is Oka when Y is a Stein manifold with the density property and $K \subset Y$ is compact and $\mathscr{O}(Y)$-convex [51]. Indeed, the Fatou–Bieberbach domains constructed through each point of $Y \setminus K$ by Forstnerič–Wold [37, Lemma 3.3] give strong dominability, so Theorem 1.3 applies. For compact polynomially convex $K \subset \mathbb{C}^n$, $n > 1$, the requisite pointwise construction is due to Forstnerič–Ritter [34, Proposition 4].

Du [12, Theorem 1.1] applies the LAP characterization and the propagation result for coordinate discs to compact complements with a Levi-index condition. Let M be a connected Stein manifold of dimension $n \geq 3$ with the density property, and let $K \Subset M$ be the closure of its interior with C^∞ boundary. If, at every boundary point, the tangential Levi form of a local defining function oriented by $K = \{\rho \leq 0\}$ has at least two nonnegative eigenvalues, then $M \setminus K$ is connected and Oka.

Corollary 5.6. *Let X be a connected complex manifold and $Z \subsetneq X$ a closed complex analytic subset such that $X \setminus Z$ is Oka. If $K \subset Z$ is closed in X, then $X \setminus K$ is Oka.*

Proof. The open manifold $Y = X \setminus K$ contains the connected dense set $X \setminus Z$, so it is connected. The subset $Z \cap Y$ is proper and closed analytic in Y, with Oka complement $X \setminus Z$. Apply Theorem 1.7. □

In particular, $\mathbb{C}^n \setminus K$ is Oka for every closed subset K of an affine complex hyperplane $H \subset \mathbb{C}^n$, $n \geq 1$, since $\mathbb{C}^n \setminus H \cong \mathbb{C}^{n-1} \times \mathbb{C}^*$ is a complex Lie group. This recovers the Oka property of Kusakabe's example $\mathbb{C}^n \setminus (T^2 \times \{0\}^{n-2})$, where $T = \{0\} \cup \{1/j : j \geq 1\}$ and $n \geq 3$; Kusakabe also proved that this manifold is not weakly subelliptic [46].

Corollary 5.7. *Let a complex Lie group G act holomorphically on a connected complex manifold X. If one G-orbit is open, then X is Oka.*

Proof. The zero-dimensional case is immediate. For $n = \dim X > 0$, choose a Lie-algebra basis $\xi_1, \dots, \xi_N$ and let $V_1, \dots, V_N$ be the associated holomorphic vector fields. They span the tangent spaces on the open orbit, so the set

$$A = \{x \in X : \operatorname{rank}(V_1(x), \dots, V_N(x)) < n\}$$

is proper and closed analytic, being the common zero set of their n-fold exterior products. The holomorphic maps $h_j : X \times \mathbb{C} \to X$, $h_j(x,t) = \exp(t\xi_j) \cdot x$, fix x at $t = 0$ and have derivatives $V_j(x)$ spanning T_xX for $x \notin A$. Corollary 4.9 applies. □

6. Rationally connected and Fano manifolds

Deformations of very free rational curves give the families required by Corollary 4.9 at every point of a rationally connected projective manifold, and these families are holomorphic on the whole product $V \times \mathbb{P}^1$.

6.1. **Rationally connected manifolds.** A nonconstant holomorphic map $u : \mathbb{P}^1 \to X$ is *very free* if u^*TX is ample; see [9, Definition 2.25]. By the Birkhoff–Grothendieck splitting theorem this means that

$$u^*TX \simeq \bigoplus_{\nu=1}^{\dim X} \mathscr{O}_{\mathbb{P}^1}(a_\nu), \qquad a_\nu \geq 1;$$

see [42, Definition 3.1].

Proof of Theorem 1.8. Let X be a smooth complex projective rationally connected manifold. The zero-dimensional case is immediate. We construct the families in Definition 4.8 as restrictions of holomorphic maps $h_j : V \times \mathbb{P}^1 \to X$. Let $n = \dim X \geq 1$ and $x \in X$, and let $e_1, \dots, e_n$ be a basis of T_xX, which specifies the tangent directions to be realized. Write t for the standard affine coordinate on $\mathbb{P}^1 \setminus \{\infty\} = \mathbb{C}$. The theorem on very free rational curves with prescribed tangent directions [10, Theorem 2.2(ii)] applies to X. Its single-point case, applied to the lines $\mathbb{C}e_j$ separately, gives holomorphic maps

$$u_j : \mathbb{P}^1 \longrightarrow X, \qquad u_j(0) = x, \qquad (du_j)_0 \left(\frac{\partial}{\partial t}\bigg|_0 \right) \in \mathbb{C}^* e_j, \qquad u_j^*TX \text{ ample.}$$

Here a source reparametrization sends the prescribed point to 0. Write $u_j^*TX \simeq \bigoplus_{\nu=1}^n \mathscr{O}_{\mathbb{P}^1}(a_{j\nu})$ with $a_{j\nu} \geq 1$. Thus

$$H^1(\mathbb{P}^1, u_j^*TX) = 0, \qquad H^1\Big(\mathbb{P}^1, u_j^*TX \otimes \mathscr{O}_{\mathbb{P}^1}(-[0])\Big) = 0,$$

since the corresponding line-bundle summands have degrees $a_{j\nu}$ and $a_{j\nu} - 1$, respectively. The notation $[0]$ denotes the divisor consisting of the point $0 \in \mathbb{P}^1$.

We use the local deformation theory of maps with the source $\mathbb{P}^1$ fixed. Applied in local analytic form, the deformation results of [9, §1.4, Proposition 1.6 and Theorem 1.9] and the vanishing $H^1(\mathbb{P}^1, u_j^*TX) = 0$ give, for each j, a complex manifold S_j, a point $p_j \in S_j$, and a holomorphic family

$$F_j : S_j \times \mathbb{P}^1 \longrightarrow X, \qquad F_j(p_j, t) = u_j(t).$$

The parameter derivative identifies $T_{p_j}S_j$ with $H^0(\mathbb{P}^1, u_j^*TX)$. Under this identification, the differential of evaluation $\mathrm{ev}_{0,j} : S_j \to X$, $s \mapsto F_j(s,0)$, is $\xi \mapsto \xi(0)$. The evaluation exact sequence and the vanishing $H^1(\mathbb{P}^1, u_j^*TX \otimes \mathscr{O}_{\mathbb{P}^1}(-[0])) = 0$ make this differential surjective. Hence $\mathrm{ev}_{0,j}$ is a holomorphic submersion near p_j and has a local holomorphic section through p_j. Taking a common neighborhood V of x for the finitely many sections gives

$$\sigma_j : V \longrightarrow S_j, \qquad \mathrm{ev}_{0,j} \circ \sigma_j = \mathrm{id}_V, \qquad \sigma_j(x) = p_j.$$

We therefore obtain holomorphic maps

$$h_j : V \times \mathbb{P}^1 \longrightarrow X, \qquad h_j(y,t) = F_j(\sigma_j(y), t).$$

They satisfy $h_j(y,0) = y$ and $h_j(x,t) = u_j(t)$. Their derivatives at $(x,0)$ lie in the lines $\mathbb{C}e_j$ and are nonzero. By continuity, after shrinking V we have

$$\frac{\partial h_1}{\partial t}(y,0) \wedge \cdots \wedge \frac{\partial h_n}{\partial t}(y,0) \neq 0 \qquad (y \in V).$$

Consequently these vectors span T_yX for every $y \in V$. Restricting the second variable to $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$ gives the maps required in Definition 4.8. Corollary 4.9 therefore shows that X is Oka. Every smooth complex Fano manifold is rationally connected by Kollár–Miyaoka–Mori [43, Theorem 0.1], which gives the Fano case. □

By birational invariance of rational connectedness [9, Remark 2.21], every smooth projective birational model of a rationally connected projective manifold is Oka.

Corollary 6.1. *Every smooth complex projective rationally connected manifold, and hence every smooth complex Fano manifold, is holomorphically elliptic.*

This follows from Theorem 1.8 and Forstnerič–Lárusson's theorem [33, Theorem 1.1].

Kusakabe and Matsumura [52, Theorem 1.1] extend this implication to weakly pseudoconvex Oka manifolds admitting a positive line bundle.

6.2. **Compact Kähler consequences.** Theorem 1.8 and structure theorems for compact Kähler manifolds give the following two consequences. Corollary 6.2 answers the compact case of [27, Problem 11.6(a)], and Corollary 6.3 answers [27, Problem 11.6(b)].

Corollary 6.2. *Every compact Kähler manifold admitting a Kähler metric with nonnegative holomorphic sectional curvature is Oka.*

Proof. Let X be such a manifold. By Matsumura's structure theorem [53, Theorem 1.1(b) and Remark 1.2], its universal cover is biholomorphic to $\mathbb{C}^q \times F$, where F is a smooth projective rationally connected manifold, possibly a point. Theorem 1.8 makes F Oka. Hence $\mathbb{C}^q \times F$ is Oka, and the Oka property descends along the unramified covering to X by [27, Theorem 3.15]. □

For a holomorphic vector bundle E, we use the projectivization $\mathbb{P}(E)$ of one-dimensional quotients. The bundle E is *nef* if $\mathscr{O}_{\mathbb{P}(E)}(1)$ is nef in the analytic sense of Demailly–Peternell–Schneider [11, §1.A and Definition 1.9].

Corollary 6.3. *Every compact Kähler manifold with nef tangent bundle is Oka.*

Proof. Let X be such a manifold. By the structure theorem of Demailly–Peternell–Schneider [11, Theorem 3.14], there is a connected finite étale cover $p : X' \to X$, chosen to maximize the irregularity among finite étale covers, such that the Albanese map

$$a : X' \longrightarrow \mathrm{Alb}(X')$$

is a surjective holomorphic submersion with connected Fano fibers. If the fibers are points, then a is an isomorphism and X' is a complex torus, hence Oka. Otherwise, $TX' = p^*TX$ is nef, so $-K_{X'} = \det TX'$ is nef, and the general fiber of a is weak Fano. Theorem 1.3 of Cao–Höring [6] therefore makes a locally trivial in the analytic topology. Its fibers are Oka by Theorem 1.8, and its base is a complex torus. The Oka fiber bundle theorem [27, Theorem 3.15] shows that X' is Oka. Applied to the finite étale covering $p : X' \to X$, the same theorem shows that X is Oka. □

7. Logarithmic rational curves and divisor complements

Very free logarithmic rational curves supply the entire-curve families needed for divisor complements.

7.1. **A criterion using logarithmic rational curves.** Let X be a connected complex projective manifold of dimension n and let $D = \sum_i D_i$ be a reduced divisor with irreducible components D_i. The divisor has *simple normal crossings* if each D_i is smooth and D is a union of coordinate hyperplanes in suitable local holomorphic coordinates. For a simple normal crossings divisor D, the holomorphic sections of the logarithmic tangent bundle $T_X(-\log D)$ are the holomorphic vector fields tangent to every component D_i. In coordinates $x_1, \dots, x_n$ with $D = \{x_1 \cdots x_k = 0\}$ this bundle has the frame

$$x_1 \frac{\partial}{\partial x_1}, \dots, x_k \frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_{k+1}}, \dots, \frac{\partial}{\partial x_n}.$$

Theorem 7.1. *Let X be a connected complex projective manifold, and let $D \subset X$ be a reduced simple normal crossings divisor. Suppose that a nonconstant holomorphic map $f : \mathbb{P}^1 \to X$ satisfies*

$$f^{-1}(D) \subseteq \{\infty\}, \qquad f^*T_X(-\log D) \text{ is ample.}$$

Then $X \setminus D$ is Oka.

We prove the theorem by deforming rational curves while keeping the pullback divisors fixed, so that their restrictions to $\mathbb{C}$ avoid the boundary. The logarithmic tangent bundle controls the values and first derivatives of the deformed curves.

Throughout this section, $D = \sum_{i=1}^r D_i$ has simple normal crossings, $U = X \setminus D$, and $n = \dim X$. We allow $r = 0$, in which case $D = 0$. Fix distinct points $0, \infty \in \mathbb{P}^1$ and an affine coordinate t on $\mathbb{P}^1 \setminus \{\infty\}$ with $t(0) = 0$. We write $g'(0) = dg_0(\partial/\partial t|_0) \in T_{g(0)}X$, and $[0]$ and $[\infty]$ denote the corresponding point divisors.

7.2. **Prescribed pullback divisors and first jets.** Fix f as in Theorem 7.1 and write $f^*D_i = m_i[\infty]$ with $m_i \geq 0$. We work with parametrized maps, keeping the source $\mathbb{P}^1$, the points $0, \infty$, and the coordinate t fixed. Fix a projective embedding of X and the degree $\deg f^*\mathscr{O}_X(1)$; the corresponding scheme of morphisms $\mathbb{P}^1 \to X$ is of finite type [9, §1.3]. Restrict it to the open-and-closed locus on which $\deg g^*\mathscr{O}_X(D_i) = m_i$ for all i, and let H be the open subset of maps whose image is contained in no D_i. Pulling back the divisors gives an algebraic morphism

$$c : H \longrightarrow \prod_{i=1}^r \mathrm{Sym}^{m_i}(\mathbb{P}^1), \qquad c(g) = (g^*D_i)_i.$$

Here $\mathrm{Sym}^{m_i}(\mathbb{P}^1)$ parametrizes effective divisors of degree m_i, equivalently unordered m_i-tuples of points counted with multiplicity, so c records the intersection points with each D_i and their orders of contact. By the choice of H, the pulled-back defining section of D_i is not identically zero on any $\mathbb{P}^1$-fiber. Its zero scheme is locally principal, and its restriction to each fiber is an effective Cartier divisor. Since $H \times \mathbb{P}^1 \to H$ is flat and of finite presentation, the fiberwise criterion [58, Tag 062Y, part (3)] shows that this zero scheme is a relative effective Cartier divisor, flat over H. It is thus a family of effective divisors of degree m_i and determines a morphism to $\mathrm{Sym}^{m_i}(\mathbb{P}^1)$. We take $\mathrm{Sym}^0(\mathbb{P}^1)$ and the empty product to be a point. Let

$$M = c^{-1}((m_i[\infty])_i)$$

be the scheme-theoretic fiber. M therefore parametrizes maps g with $g^*D_i = m_i[\infty]$ for every i. When all $m_i = 0$, we have $M = H$ and $g^*T_X(-\log D) \simeq g^*T_X$ for every $g \in M$; in particular,

$D = 0$ gives the usual fixed-source deformation space. Every $g \in M$ satisfies $g(\mathbb{A}^1) \subset U$, so the universal evaluation map restricts to a morphism

$$\mathcal{F} : M \times \mathbb{A}^1 \longrightarrow U. \tag{7.1}$$

Lemma 7.2. *Let $g \in M$ be a map for which $E_g = g^*T_X(-\log D)$ is ample. Then M is smooth at g, with tangent space $T_gM \simeq H^0(\mathbb{P}^1, E_g)$. The first-jet map*

$$J : M \longrightarrow TU, \qquad g \longmapsto (g(0), g'(0))$$

is smooth at g, so evaluation at 0 is smooth at g. On the fiber of evaluation at 0 over $g(0)$, derivative evaluation $\widetilde{g} \mapsto \widetilde{g}'(0) \in T_{g(0)}U$ has surjective differential at g.

Proof. The fixed-contact deformation space. The inclusion of the logarithmic vector fields in all vector fields gives an exact sequence

$$0 \longrightarrow E_g \longrightarrow g^*T_X \longrightarrow Q_g \longrightarrow 0, \qquad Q_g = \bigoplus_{i:m_i>0} g^*\mathscr{O}_X(D_i)|_{m_i[\infty]}. \tag{7.2}$$

To check exactness, choose simple normal crossings coordinates near $g(\infty)$, in which the components met by g are given by $x_i = 0$. In a source coordinate z centered at infinity,

$$x_i \circ g = z^{m_i} u_i(z), \qquad u_i(0) \neq 0.$$

The corresponding logarithmic tangent vector maps to $z^{m_i}u_i(z)\,\partial/\partial x_i$. Thus the map is injective and its cokernel is the stated torsion sheaf. Away from infinity it is an isomorphism.

Ampleness gives $E_g \cong \bigoplus_{\nu=1}^{n} \mathscr{O}_{\mathbb{P}^1}(a_\nu)$ with $a_\nu \geq 1$, and therefore

$$H^1(\mathbb{P}^1, E_g) = H^1(\mathbb{P}^1, E_g(-2[0])) = 0. \tag{7.3}$$

Restriction to the divisor $2[0]$ means taking sections modulo t^2 in a local trivialization; it records their value and first derivative at 0. Since Q_g is a torsion sheaf, (7.2) also gives $H^1(\mathbb{P}^1, g^*T_X) = 0$. Deformation theory for maps with source $\mathbb{P}^1$ fixed therefore shows that H is smooth at g, with $T_gH = H^0(\mathbb{P}^1, g^*T_X)$; see [9, Proposition 1.6 and Theorem 1.9].

For $Z_i = m_i[\infty]$, the tangent space to the symmetric power has the intrinsic description

$$T_{Z_i} \operatorname{Sym}^{m_i}(\mathbb{P}^1) \simeq H^0(Z_i, \mathscr{O}_{\mathbb{P}^1}(Z_i)|_{Z_i}) \simeq H^0(\mathbb{P}^1, g^*\mathscr{O}_X(D_i)|_{Z_i}),$$

where the second identification uses $g^*D_i = Z_i$; both sides are zero when $m_i = 0$. Under these identifications the differential of c is the quotient map

$$H^0(\mathbb{P}^1, g^*T_X) \longrightarrow H^0(\mathbb{P}^1, Q_g).$$

Locally it sends an infinitesimal variation of $x_i \circ g$ to its class modulo z^{m_i}, which is the infinitesimal displacement of the zero divisor of $x_i \circ g$. The vanishing $H^1(\mathbb{P}^1, E_g) = 0$ makes this differential surjective. Since the product of symmetric powers is smooth, c is smooth at g and its fiber M is smooth at g with the asserted tangent space.

First-jet evaluation. The chosen coordinate t identifies the space of first jets at 0 of germs of maps to X with TX. At the jet $J(g)$, the tangent space $T_{J(g)}TU$ is naturally identified with $H^0(\mathbb{P}^1, g^*T_X|_{2[0]})$. Since the divisor conditions are supported at ∞, we have $E_g|_{2[0]} \simeq g^*T_X|_{2[0]}$. Under this identification and $T_gM \simeq H^0(\mathbb{P}^1, E_g)$, the differential dJ_g is the restriction map on global sections in the exact sequence

$$0 \longrightarrow E_g(-2[0]) \longrightarrow E_g \longrightarrow E_g|_{2[0]} \longrightarrow 0.$$

It is surjective by (7.3), so $J : M \to TU$ is smooth at g. Composing with the bundle projection $TU \to U$ gives the assertion about evaluation at 0. Restricting J to the fiber of evaluation at 0 over $g(0)$ gives the assertion about derivative evaluation. □

Proof of Theorem 7.1. We construct the families in Definition 4.8, with target U, at every point of a nonempty Zariski open subset $U^+ \subset U$. Let $M^+ \subset M$ consist of the maps g for which E_g is ample; it contains f. On $\mathbb{P}^1$, ampleness is equivalent to $H^1(\mathbb{P}^1, E_g(-2[0])) = 0$. Let $e : M \times \mathbb{P}^1 \to X$ be the universal evaluation. The bundle

$$e^*T_X(-\log D) \otimes \mathrm{pr}_{\mathbb{P}^1}^* \mathscr{O}_{\mathbb{P}^1}(-2[0])$$

is locally free on $M \times \mathbb{P}^1$ and flat over M. Since the projection $M \times \mathbb{P}^1 \to M$ is proper, semicontinuity of the fiberwise first cohomology shows that M^+ is Zariski open. By Lemma 7.2 the morphism

$$\mathrm{ev}_0 : M^+ \longrightarrow U$$

is smooth, so its image U^+ is a nonempty Zariski open subset of U.

By Lemma 7.2, M^+ is smooth. We now regard M^+ and U as complex manifolds, so the smooth evaluation morphisms become holomorphic submersions. The evaluation family (7.1) restricts to a holomorphic map

$$\mathcal{F} : M^+ \times \mathbb{C} \longrightarrow U.$$

Fix $x \in U^+$ and $g \in M^+$ with $g(0) = x$. Derivative evaluation on $\mathrm{ev}_0^{-1}(x)$ is a submersion near g, so its image contains a nonempty Euclidean open subset of T_xU, and such a subset contains n linearly independent vectors. We may therefore choose maps $g_1, \dots, g_n \in M^+$ with $g_j(0) = x$ whose derivatives at 0 form a basis of T_xU.

For each j the holomorphic submersion ev_0 has a local holomorphic section $\sigma_j : V_j \to M^+$ with $\sigma_j(x) = g_j$. On the common neighborhood $V = \bigcap_j V_j$ of x set

$$h_j(y,t) = \mathcal{F}(\sigma_j(y), t), \qquad (y,t) \in V \times \mathbb{C}.$$

By (7.1) these maps are holomorphic on $V \times \mathbb{C}$ with values in U, and $h_j(y,0) = y$. Their derivatives $\partial_t h_j(y,0)$ form a basis at x, hence also on a smaller common neighborhood of x, so the maps satisfy Definition 4.8.

Since X is smooth and connected, it is irreducible; its nonempty open subset $U = X \setminus D$ is therefore connected. The complement $U \setminus U^+$ is a proper closed algebraic, hence complex analytic, subset of U. Corollary 4.9 therefore implies that U is Oka. □

The hypotheses on f in Theorem 7.1 allow $f(\mathbb{P}^1) \cap D = \varnothing$. Writing

$$f^*T_X(-\log D) \simeq \bigoplus_{\nu=1}^{n} \mathscr{O}_{\mathbb{P}^1}(a_\nu), \qquad a_\nu \geq 1,$$

the determinant identity $\det T_X(-\log D) = \mathscr{O}_X(-(K_X + D))$ gives the necessary numerical condition

$$-(K_X + D) \cdot f_*[\mathbb{P}^1] = \deg f^*T_X(-\log D) \geq n.$$

Here $f_*[\mathbb{P}^1] = (\deg(f : \mathbb{P}^1 \to f(\mathbb{P}^1)))[f(\mathbb{P}^1)]$ is the pushforward one-cycle.

7.3. **Complements of codimension at least two.** Kusakabe proved that a closed complex subvariety of codimension at least two in a locally flexible algebraic manifold has Oka complement if a holomorphic automorphism carries the subvariety into a proper closed algebraic subset [49, Corollary 1.7]. Deformations of very free rational curves avoiding a closed algebraic subset of codimension at least two answer the deletion question in [24, Section 2.3.3, p. 761] for smooth projective rationally connected manifolds. In dimension at least two, taking A to be a point also answers the second part of [30, Question D] for this class of manifolds.

Corollary 7.3. *Let X be a smooth connected complex projective rationally connected manifold, and let $A \subset X$ be a closed algebraic subset whose irreducible components have complex codimension at least two. Then $X \setminus A$ is Oka.*

Proof. If $A = \varnothing$, the assertion is Theorem 1.8. Otherwise, choose a very free map $f : \mathbb{P}^1 \to X$. In the finite-type scheme of morphisms of its fixed projective degree, let M be a nonempty smooth irreducible open subset consisting of very free maps. Such an M exists by the deformation theory used in the proof of Theorem 1.8 and the openness of ampleness. For every $g \in M$ and $t \in \mathbb{P}^1$, the bundle g^*TX is globally generated, so the universal evaluation

$$e : M \times \mathbb{P}^1 \longrightarrow X, \qquad e(g,t) = g(t),$$

is smooth. Consequently the incidence subset $I = e^{-1}(A)$ has dimension at most $\dim M - 1$. The projection $M \times \mathbb{P}^1 \to M$ is proper, so the image of I is a proper closed subset of M. Thus

$$M_A = \{g \in M : g(\mathbb{P}^1) \cap A = \varnothing\}$$

is a nonempty Zariski open subset of M. Write $U = X \setminus A$ and $n = \dim X$. For every $g \in M_A$, very freeness gives

$$H^1(\mathbb{P}^1, g^*TX(-2[0])) = 0.$$

Lemma 7.2 with $D = 0$, restricted to the open subset M_A, shows that evaluation $\mathrm{ev}_0 : M_A \to U$ is smooth and that derivative evaluation on each of its fibers is a holomorphic submersion. Hence $U_A = \mathrm{ev}_0(M_A)$ is a nonempty Zariski open subset of U.

Fix $x \in U_A$. The open image of derivative evaluation on $\mathrm{ev}_0^{-1}(x)$ contains n linearly independent vectors. Choose corresponding maps $g_1, \dots, g_n$ and local holomorphic sections $\sigma_j : V \to M_A$ of ev_0 with $\sigma_j(x) = g_j$, on a common neighborhood $V \subset U_A$ of x. The maps

$$h_j(y,t) = \sigma_j(y)(t), \qquad (y,t) \in V \times \mathbb{C},$$

are jointly holomorphic by universal evaluation, take values in U because every member of M_A avoids A on all of $\mathbb{P}^1$, and satisfy $h_j(y,0) = y$. After shrinking V, their initial derivatives form a basis of T_yU for every $y \in V$. Since U is connected and $U \setminus U_A$ is a proper closed algebraic subset, Corollary 4.9 shows that U is Oka. □

7.4. **Geometric applications.**

$\mathbb{A}^1$-connected complements. Following Chen–Zhu [8, Definition 1.2], a complex algebraic variety U is $\mathbb{A}^1$-connected if there are an algebraic variety T and an algebraic map $a : T \times \mathbb{A}^1 \to U$ such that every map $a_s : \mathbb{A}^1 \to U$ is nonconstant and proper and the map

$$T \times \mathbb{A}^1 \times \mathbb{A}^1 \longrightarrow U \times U, \qquad (s, t_1, t_2) \longmapsto (a(s,t_1), a(s,t_2))$$

has Zariski dense image.

Corollary 7.4. *Let X be a connected complex projective manifold, and let $D \subset X$ be a reduced simple normal crossings divisor. If $X \setminus D$ is $\mathbb{A}^1$-connected in the sense of Chen–Zhu, then $X \setminus D$ is Oka.*

Proof. Over $\mathbb{C}$, dominance of the two-point evaluation map implies separability. Thus $X \setminus D$ is separably $\mathbb{A}^1$-connected. Chen–Zhu [8, Proposition 1.13(2)] give a very free $\mathbb{A}^1$-curve in the log smooth pair (X, D). For the divisorial log structure associated with D, the log tangent bundle is $T_X(-\log D)$. By their Definitions 1.11–1.12, the underlying nonconstant map $f : \mathbb{P}^1 \to X$ satisfies $f^{-1}(D) = \{\infty\}$ and $f^*T_X(-\log D)$ is ample. Theorem 7.1 therefore shows that $X \setminus D$ is Oka. □

Ampleness of the boundary will also give holomorphic ellipticity in the applications below. If D is a nonzero ample effective Cartier divisor and $X \setminus D$ is Oka, choose $k > 0$ such that $\mathscr{O}_X(kD)$ is very ample. The linear system $|kD|$ embeds X in projective space, and the section with zero divisor kD cuts out a hyperplane. Then $X \setminus D$ is a closed subvariety of the affine complement of that hyperplane, so it is affine and therefore Stein. Every Stein Oka manifold is holomorphically elliptic [39, 3.2.A, p. 879]; see also [33, Introduction].

Theorem 7.5. *Let X be a connected complex projective manifold of dimension at least two, and let $D \subset X$ be a smooth irreducible ample rationally connected divisor. Then $X \setminus D$ is Oka and holomorphically elliptic. The conclusion holds in particular for any smooth irreducible ample divisor $D \subset X$ such that $-(K_X + D)$ is ample.*

Proof. The smooth rationally connected divisor D carries a very free rational curve $a : \mathbb{P}^1 \to D$. When $\dim D = 1$ we have $D \simeq \mathbb{P}^1$, and the identity map under this identification suffices. Since D is ample,

$$\deg a^* \mathscr{O}_D(D) > 0.$$

For the single boundary component D, the intersection matrix associated with a is the one-by-one matrix $\big(\deg a^* \mathscr{O}_D(D)\big)$. Its positive entry gives rank one, so D satisfies Chen–Zhu's fully free condition [8, Remark 1.4]. Chen–Zhu [8, Theorem 1.3] show that $X \setminus D$ is $\mathbb{A}^1$-connected. Corollary 7.4 therefore shows that $X \setminus D$ is Oka. Since D is a nonzero ample divisor, the affine-complement argument preceding the theorem also gives holomorphic ellipticity.

If $-(K_X + D)$ is ample, adjunction gives

$$-K_D = -(K_X + D)|_D.$$

Thus $-K_D$ is ample, so D is Fano and hence rationally connected [4, 43]. The first assertion applies. □

Corollary 7.6. *Let $X \subset \mathbb{P}^N$ be a smooth complete intersection of multidegree $(d_1, \dots, d_c)$ and dimension at least two. If D is a smooth hypersurface section of degree $e \geq 1$ and*

$$d_1 + \cdots + d_c + e \leq N,$$

then $X \setminus D$ is Oka and holomorphically elliptic.

Proof. Write H for the hyperplane class of the ambient projective space. Then

$$-(K_X + D) = (N + 1 - d_1 - \cdots - d_c - e)H|_X.$$

The stated inequality makes this class ample. Since $\dim X \geq 2$, ampleness of D, Serre duality, and Kodaira vanishing give

$$H^1(X, \mathscr{O}_X(-D)) \simeq H^{\dim X - 1}(X, \mathscr{O}_X(K_X + D))^* = 0.$$

The sequence $0 \to \mathscr{O}_X(-D) \to \mathscr{O}_X \to \mathscr{O}_D \to 0$ then gives $H^0(D, \mathscr{O}_D) = \mathbb{C}$, so D is connected. Since D is smooth, it is irreducible, and Theorem 7.5 applies. □

For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree $n + 1$, the determinant $\det T_{\mathbb{P}^n}(-\log D)$ is trivial. Its pullback to any rational curve therefore has degree zero, so the pulled-back logarithmic tangent bundle cannot be ample. Section 8 treats this degree using successive polar equations.

8. Smooth hypersurface complements

For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree d, adjunction gives $K_D = \mathscr{O}_D(d - n - 1)$. Thus Theorem 7.5 gives the positive conclusion of Theorem 1.9 for $d \leq n$ when $n \geq 2$. For every smooth hypersurface of degree $3 \leq d \leq n + 1$, the construction below uses a general projection center and lines whose intersection multiplicity with D at a point of D is at least $d - 1$. Degrees one and two are treated directly. The obstruction in high degree is due to Carlson–Griffiths [7, Corollary (6.11)].

Lemma 8.1. *Let F be a homogeneous polynomial of degree d on $\mathbb{C}^N$, where $N \geq d \geq 3$, and suppose that F defines a smooth projective hypersurface $X \subset \mathbb{P}^{N-1}$. There is a nonempty Zariski open subset $\Omega \subset \mathbb{P}^{N-1} \setminus X$ with the following properties.*

(i) *For every* $[a] \in \Omega$ *and any nonzero representative* $a = (a_1, \dots, a_N)$*, the common zero set*

$$Z_a = \{F = D_aF = \cdots = D_a^{d-2}F = 0\} \subset \mathbb{P}^{N-1}, \qquad D_a = \sum_{i=1}^{N} a_i \frac{\partial}{\partial v_i}$$

is a nonempty complex submanifold of dimension $N - d$*. The displayed equations have linearly independent differentials along* Z_a *in local projective coordinates, and cut out a projective complete intersection of multidegree* $(d, d-1, \dots, 2)$ *and degree* $d!$*.*

(ii) *For every* $[a] \in \Omega$*, no hyperplane in* $\mathbb{P}^{N-1}$ *contains* Z_a*.*

(iii) *The incidence set*

$$\mathcal{I} = \{([a], [v]) \in \Omega \times X : D_aF(v) = \cdots = D_a^{d-2}F(v) = 0\}$$

is a complex submanifold of $\Omega \times X$*, and the projection* $\pi : \mathcal{I} \to \Omega$ *is a surjective holomorphic submersion whose fibers have dimension* $N - d$*.*

Proof. We first obtain transverse corank-one projection strata and identify their local equations with the successive polars. We then establish the complete-intersection, linear nondegeneracy, and incidence statements.

Generic projection. Fix a hyperplane $L \simeq \mathbb{P}^{N-2}$. Projection $p_a : X \to L$ from a point $[a] \notin X \cup L$ is finite: a positive-dimensional fiber would contain the entire line through $[a]$, contradicting $[a] \notin X$. The kernel of its differential has dimension at most one, since the ambient projection has one-dimensional fibers. Mather's generic-projection theorem, in the formulation of Alzati–Ballico–Ottaviani [2, Theorem 1.1 and Section 2, step (2)], applies with ambient projective dimension $N - 1$, codimension one, and center dimension zero. For $1 \le k \le d - 2 \le N - 2$, let $I_k = (1, \dots, 1)$ have k entries, and let $J^k(X, L)$ be the bundle of k-jets of holomorphic maps from X to L. The theorem gives a nonempty Zariski open set of centers for which the k-jet map $j^k p_a : X \to J^k(X, L)$ is transverse to the Thom–Boardman stratum Σ^{I_k}, of codimension k in this equidimensional corank-one case. The repeated symbol records a one-dimensional kernel at each successive critical locus. For the fixed hypersurface X, the space of centers $\mathbb{P}^{N-1} \setminus (X \cup L)$ is irreducible, so intersecting these finitely many nonempty open sets gives a nonempty Zariski open set $\Omega \subset \mathbb{P}^{N-1} \setminus (X \cup L)$ on which all these transversality conditions hold.

Identification with the polar equations. We identify the local defining ideals of the successive projection loci with the ideals generated by the polar equations. Near a common zero $[v]$, choose a linear form ℓ with $\ell(a) = 0$ and $\ell(v) \neq 0$. On the affine chart $\ell = 1$, take coordinates (z, u, t), with $z \in \mathbb{C}^{N-3}$, so that translation by a is translation in t and the projection forgets t. The equation of X is $P(z, u, t) = 0$, where $\partial_t^j P = (D_a^j F)|_{\ell=1}$. Since $\partial_t P = 0$ at the point and X is smooth, we can choose u so that $\partial_u P \neq 0$ there and write $u = g(z, t)$. After shrinking the coordinate neighborhood,

$$A(z, t) = \partial_u P(z, g(z, t), t)$$

is nowhere zero. Set $q_j = (\partial_t^j P)(z, g(z, t), t)$ and $h_j = \partial_t^j g$. The derivatives of P are taken with u fixed before restriction to the graph. Repeated differentiation of $P(z, g(z, t), t) = 0$ gives

$$q_j + Ah_j \in (h_1, \dots, h_{j-1}), \qquad (q_1, \dots, q_j) = (h_1, \dots, h_j).$$

These relations hold in the local holomorphic ring at the point for $1 \le j \le d-2$. At a common zero, the differentials of the q_j and h_j are related by an invertible triangular matrix. Write $S_k = \{h_1 = \cdots = h_k = 0\}$ in the graph coordinates, where the projection has the adapted form $p_a(z, t) = (z, g(z, t))$. In the corresponding adapted jet coordinates, the local equations for Σ^{I_k} are the vanishing of the first k successive derivatives in the kernel direction. We identify their pullbacks inductively. For $k = 1$, the critical locus is $S_1 = \{h_1 = 0\}$; jet transversality gives $dh_1 \neq 0$ there, so S_1 is smooth and the kernel of dp_a is $\mathbb{C}\,\partial/\partial t$ on S_1.

Suppose that $1 \le k \le d-3$, that S_k is the pullback of Σ^{I_k}, and that $d(h_1, \dots, h_k)$ has rank k on S_k. For $p \in S_k$, the identities $\partial_t h_i = h_{i+1}$ give

$$\frac{\partial}{\partial t} \in T_p S_k \quad \Longleftrightarrow \quad h_{k+1}(p) = 0.$$

Since

$$\ker d(p_a|_{S_k})_p = \ker(dp_a)_p \cap T_p S_k$$

and $\ker(dp_a)_p$ has dimension at most one, the next corank-one locus is S_{k+1}. Its local defining functions in the adapted jet coordinates pull back to $h_1, \dots, h_{k+1}$, so transversality to $\Sigma^{I_{k+1}}$ gives $\operatorname{rank} d(h_1, \dots, h_{k+1}) = k+1$ on S_{k+1}. This proves by induction that $\operatorname{rank} d(h_1, \dots, h_k) = k$ on S_k for every $1 \le k \le d-2$, including $d = 3$, when only the initial step is needed. The triangular change of generators gives the same rank for $(q_1, \dots, q_k)$. Hence the polar equations have linearly independent differentials along their common zero set in X. By the holomorphic implicit function theorem, they cut out a complex submanifold of codimension $d-2$ in X.

Complete intersection and incidence. Let J be the ideal generated by $F, D_a F, \dots, D_a^{d-2} F$ in $S = \mathbb{C}[v_1, \dots, v_N]$. Its projective zero set is nonempty: otherwise $\sqrt{J}$ would be the irrelevant ideal of height N, contrary to the height theorem for an ideal generated by $d-1 < N$ elements. The dimension calculation above shows that $\operatorname{ht} J = d-1$. Thus the generators form a regular sequence, and S/J is Cohen–Macaulay of dimension $N - d + 1 \ge 1$.

Let $\mathfrak{m} = (v_1, \dots, v_N)$. The saturation J^{sat} consists of the polynomials $b \in S$ such that $\mathfrak{m}^r b \subset J$ for some r. A general linear form ρ is a nonzerodivisor on S/J. If $b \in J^{\mathrm{sat}}$, then $\rho^r b \in J$ for some r, whence $b \in J$; therefore $J = J^{\mathrm{sat}}$.

The projective scheme cut out by these equations is smooth, hence reduced. Its saturated homogeneous ideal is therefore the ideal of homogeneous polynomials vanishing on its points. Saturation rules out an additional component supported only at the cone vertex. In particular, a linear form vanishing at all its points belongs to J. But J has no element of degree one, because all its generators have degree at least two. Thus no hyperplane contains its points. Its degree as a complete intersection is $d(d-1)\cdots 2 = d!$.

Each equation $D_a^j F(v) = 0$ is bihomogeneous of bidegree $(j, d-j)$ in (a, v), so the incidence set is independent of the chosen projective representatives. For fixed $[a]$, the differentials of $D_a^j F$, $1 \le j \le d-2$, restricted to $T_{[v]} X$ are linearly independent at every $[v] \in Z_a$. The holomorphic implicit function theorem therefore makes $\mathcal{I}$ a complex submanifold and $\pi : \mathcal{I} \to \Omega$ a holomorphic submersion whose fibers have dimension $N - d$. The nonemptiness of every fiber makes π surjective. □

Proof of Theorem 1.9. Write $D = \{F = 0\}$ for a homogeneous polynomial F of degree d on $\mathbb{C}^{n+1}$, and write $Y = \mathbb{P}^n \setminus D$. The manifold Y is connected: the line joining two distinct points of Y is not contained in D, hence meets D in a finite set, whose complement in the line is connected. For $d = 1$, the complement is $\mathbb{C}^n$.

The function

$$E(\lambda, t) = \begin{cases} \dfrac{e^{\lambda t} - 1}{\lambda}, & \lambda \neq 0, \\ t, & \lambda = 0, \end{cases} \qquad E(\lambda, t) = \sum_{k=0}^{\infty} \frac{\lambda^k t^{k+1}}{(k+1)!},$$

is entire on $\mathbb{C}^2$, since the series converges locally uniformly, and satisfies $1 + \lambda E(\lambda, t) = e^{\lambda t}$. Its images are

$$E(0, \mathbb{C}) = \mathbb{C}, \qquad E(\lambda, \mathbb{C}) = \mathbb{C} \setminus \{-1/\lambda\} \quad (\lambda \neq 0).$$

Both remaining positive cases use the following entire-curve construction. Suppose that $V \subset Y$ is open and that holomorphic maps $a, v_1, \dots, v_n : V \to \mathbb{C}^{n+1} \setminus \{0\}$ satisfy $[a(y)] = y$, the classes of $v_1(y), \dots, v_n(y)$ form a basis modulo $\mathbb{C}a(y)$, and

$$F(a(y) + s v_j(y)) = F(a(y)) + s\, dF_{a(y)}(v_j(y)) \qquad (y \in V,\ s \in \mathbb{C}).$$

Write $c(y) = F(a(y))$ and $\lambda_j(y) = dF_{a(y)}(v_j(y))$. The maps

$$h_j(y,t) = \big[a(y) + c(y)E(\lambda_j(y),t)v_j(y)\big], \qquad (y,t) \in V \times \mathbb{C}, \tag{8.1}$$

satisfy

$$F\big(a(y) + c(y)E(\lambda_j(y),t)v_j(y)\big) = c(y)e^{\lambda_j(y)t} \neq 0.$$

Thus $h_j : V \times \mathbb{C} \to Y$ is holomorphic and $h_j(y,0) = y$. For projectivization $q : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}^n$, we have

$$\partial_t h_j(y,0) = dq_{a(y)}(c(y)v_j(y)),$$

since $\partial_t E(\lambda,0) = 1$. The kernel of $dq_{a(y)}$ is $\mathbb{C}a(y)$, so these velocities form a basis of T_yY. Hence the maps in (8.1) satisfy Definition 4.8 on V.

Degree two. For $d = 2$, the smooth quadric D spans $\mathbb{P}^n$; when $n = 1$, it consists of two distinct points, which span $\mathbb{P}^1$. At any $y_0 \in Y$, choose a local holomorphic lift $a(y)$ of y and nonzero constant vectors $v_1, \dots, v_n$ with $F(v_j) = 0$ whose classes form a basis of $\mathbb{C}^{n+1}/\mathbb{C}a(y_0)$. After shrinking a neighborhood V of y_0, their classes remain a basis modulo $\mathbb{C}a(y)$. The quadratic identity $F(a + sv_j) = F(a) + s\,dF_a(v_j)$ supplies the hypothesis of the construction above. Formula (8.1) therefore gives the required families at every point of Y, and Corollary 4.9 makes Y Oka. For $n = 1$, this is $\mathbb{P}^1$ minus two points, which is $\mathbb{C}^*$.

The range $3 \le d \le n+1$. Suppose now that $3 \le d \le n+1$. Apply Lemma 8.1 with $N = n+1$ and $X = D$. At the critical degree $d = n+1$, each Z_a consists of $d!$ distinct points and spans $\mathbb{P}^n$; their local holomorphic variation follows from the incidence submersion. Write $y_0 = [a_0] \in \Omega$ with $a_0 \neq 0$. Since Z_{a_0} spans $\mathbb{P}^n$, we can choose points $[v_1^0], \dots, [v_n^0] \in Z_{a_0}$ whose representatives have classes forming a basis of $\mathbb{C}^{n+1}/\mathbb{C}a_0$. The holomorphic submersion π has local holomorphic sections through the points $([a_0], [v_j^0]) \in \mathcal{I}$. Taking nonzero holomorphic lifts of their second components, together with a local holomorphic lift of the inclusion into $\mathbb{P}^n$, gives maps

$$a, v_1, \dots, v_n : V \longrightarrow \mathbb{C}^{n+1} \setminus \{0\}, \qquad [a(y)] = y, \quad [v_j(y)] \in Z_{a(y)},$$

on a common coordinate neighborhood $V \subset \Omega$ of y_0. After shrinking V, the classes of $v_1(y), \dots, v_n(y)$ modulo $\mathbb{C}a(y)$ form a basis for every $y \in V$.

For each fixed y, write $a = a(y)$ and $v = v_j(y)$. For $s \neq 0$, homogeneity and Taylor expansion give

$$F(a+sv) = s^d F(v + a/s) = \sum_{\ell=0}^{d} \frac{s^{d-\ell}}{\ell!} D_a^\ell F(v).$$

The resulting polynomial identity also holds at $s = 0$. In particular,

$$[s^k]F(a+sv) = \frac{D_a^{d-k}F(v)}{(d-k)!} = 0 \qquad (2 \le k \le d),$$

where $[s^k]$ denotes the coefficient of s^k. Consequently,

$$F(a+sv) = F(a) + s\,dF_a(v), \qquad F(ra+sv) = r^{d-1}\big(rF(a) + s\,dF_a(v)\big).$$

Let $L_{a,v} = \{[ra+sv] : [r:s] \in \mathbb{P}^1\}$ and let $\lambda = dF_a(v)$. Since $F(a) \neq 0$, this line is not contained in D, and the displayed factorization gives its intersection divisor:

$$D|_{L_{a,v}} = \begin{cases} (d-1)[v] + [\lambda a - F(a)v], & \lambda \neq 0, \\ d[v], & \lambda = 0. \end{cases}$$

The line therefore meets D in at most two distinct points.

When $\lambda = 0$, the line complement is $\mathbb{C}$, parametrized by $s = F(a)t$; when $\lambda \neq 0$, it is $\mathbb{C}^*$, and $s = F(a)E(\lambda,t)$ omits exactly the finite value $-F(a)/\lambda$. These formulas give entire maps onto the line complements and remain jointly holomorphic in λ and t at $\lambda = 0$. The linear identity for $F(a+sv)$ and the basis property of the $v_j(y)$ therefore give, by (8.1), the families of Definition 4.8 near every point of Ω.

Let $A = Y \setminus \Omega$. Then A is a proper closed algebraic, hence complex analytic, subset of the connected manifold Y. The general case of Corollary 4.9 shows that Y is Oka.

The range $d \geq n+2$. If $d \geq n+2$, then $\mathscr{O}_{\mathbb{P}^n}(K_{\mathbb{P}^n} + D) \cong \mathscr{O}_{\mathbb{P}^n}(d-n-1)$ is ample. The smooth divisor $D \in |\mathscr{O}_{\mathbb{P}^n}(d)|$ therefore satisfies the hypotheses of Carlson–Griffiths [7, Section 6(a), Corollary (6.11)], with $V = \mathbb{P}^n$ and $L = \mathscr{O}_{\mathbb{P}^n}(d)$. Their corollary implies that every entire map $\mathbb{C}^n \to \mathbb{P}^n \setminus D$ has rank less than n everywhere.

An Oka n-fold admits an entire map from $\mathbb{C}^n$ with differential of rank n at some point [20, Proposition 1.2 and Corollary 1.3]. This contradicts the rank bound, so $\mathbb{P}^n \setminus D$ is not Oka in this degree range.

Finally, since D is an ample effective divisor, the affine-complement argument in Section 7 shows that $\mathbb{P}^n \setminus D$ is holomorphically elliptic whenever it is Oka. Together with the implication from holomorphic ellipticity to the Oka property, this proves that the three stated conditions are equivalent. □

Singular hypersurfaces can have Oka complements in arbitrarily high degree. For $m \geq 3$, the reduced divisor

$$D_m = \{Z(YZ^{m-1} - X^m) = 0\} \subset \mathbb{P}^2$$

has degree $m+1 > 3$ and is singular at $[0:1:0]$. In the chart $Z = 1$, the polynomial shear $(x, y) \mapsto (x, y - x^m)$ identifies $\mathbb{P}^2 \setminus D_m$ with $\mathbb{C} \times \mathbb{C}^*$, which is Oka.

9. Rational maps from the projective line to itself

For $d \geq 1$, write $R_d = \mathrm{Hol}_d(\mathbb{P}^1, \mathbb{P}^1)$. Pairs of degree-$d$ homogeneous polynomials in two variables, modulo a common nonzero scalar, identify R_d with the open subset of $\mathbb{P}^{2d+1}$ on which the resultant is nonzero. The resultant is homogeneous of total degree $2d > 0$, so R_d is a smooth affine variety of dimension $2d+1$, and hence Stein. The space R_2 is homogeneous and hence Oka [41, Theorem 1.1 and Corollary 1.2]. Hanysz proved that R_3 is strongly dominable and strongly $\mathbb{C}$-connected [41, Theorems 1.7 and 1.8]; Xie subsequently proved that it is holomorphically elliptic [60, Theorem 1.3].

For arbitrary d, the Euclidean algorithm gives a dense Zariski-open Oka subset in the normalized evaluation fiber. Theorem 1.7 makes the fiber Oka, and the evaluation bundle then gives the result.

The fiber of evaluation at infinity over 0 is identified by $(q, p) \mapsto p/q$ with

$$\mathcal{R}_d = \{(q, p) : q \text{ monic of degree } d,\ \deg p < d,\ \mathrm{Res}(q, p) \neq 0\} \subset \mathbb{C}^{2d}.$$

Indeed, every degree-d rational map taking ∞ to 0 has a unique such coprime representation. Thus $\mathcal{R}_d$ is a smooth principal open subset.

Proof of Proposition 1.10. We use Cazanave's decomposition in the case of linear partial quotients, expressed using monic remainders [16, Lemma A.1]; the pointed function A/B there is our reciprocal q/p.

Coordinates from the Euclidean algorithm. For $b_1, \ldots, b_d \in \mathbb{C}$ and $a, c_1, \ldots, c_{d-1} \in \mathbb{C}^*$, set

$$r_d = 1, \qquad r_{d-1} = z - b_d, \qquad r_{i-1} = (z - b_i) r_i + c_i r_{i+1} \quad (i = d-1, \ldots, 1). \tag{9.1}$$

Backward induction shows that r_i is monic of degree $d - i$ and that adjacent remainders are coprime. Hence $(q, p) = (r_0, a r_1) \in \mathcal{R}_d$.

Let $U_d \subset \mathcal{R}_d$ consist of pairs with $\deg p = d - 1$ and, when $d \geq 2$, successive nonzero monic remainders of degrees $d-2, \ldots, 0$. Monic division recovers the parameters uniquely: set $a = \mathrm{lc}(p)$, $r_0 = q$, and $r_1 = p/a$, and for $i = 1, \ldots, d-1$ write

$$r_{i-1} = (z - b_i) r_i + \widetilde{r}_{i+1}, \qquad c_i = \mathrm{lc}(\widetilde{r}_{i+1}), \qquad r_{i+1} = \widetilde{r}_{i+1}/c_i.$$

Here lc denotes the leading coefficient, and $r_{d-1} = z - b_d$ recovers the remaining parameter. At each step the coefficients are regular on the preceding open set, and the required remainder degree is the nonvanishing of its coefficient of z^{d-i-1}. Thus U_d is Zariski open, and the recursion and division give algebraic inverses

$$U_d \cong \mathbb{C}^d \times (\mathbb{C}^*)^d. \tag{9.2}$$

For $d = 1$, the recursion is empty: $r_1 = 1$, $r_0 = z - b_1$, and $p = a \neq 0$, giving $U_1 = \mathcal{R}_1 \cong \mathbb{C} \times \mathbb{C}^*$. The open set U_d is nonempty and Oka, since complex Lie groups are Oka [23, Proposition 5.5.1]. As $\mathcal{R}_d$ is irreducible and connected, (9.2) exhibits U_d as a dense Zariski-open Oka subset; Theorem 1.7 therefore makes $\mathcal{R}_d$ Oka.

The evaluation bundle. Evaluation at infinity gives a holomorphic fiber bundle

$$\mathrm{ev}_\infty : R_d \longrightarrow \mathbb{P}^1 \tag{9.3}$$

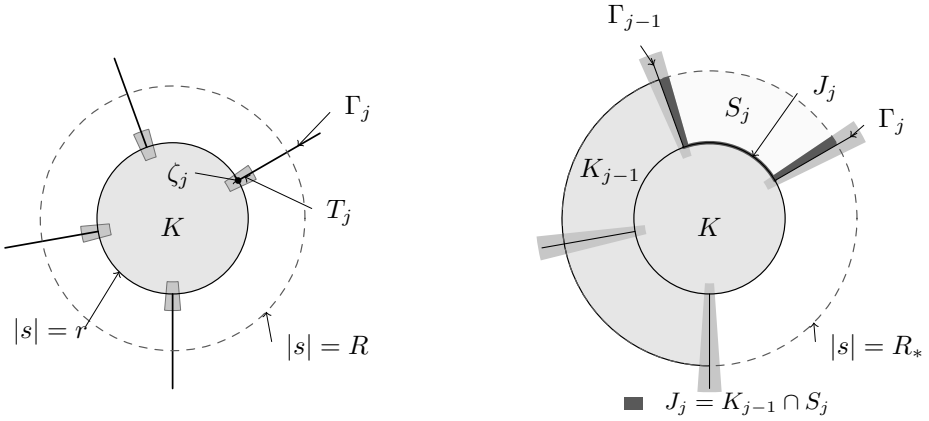
with fiber $\mathcal{R}_d$. Locally choose Möbius transformations g_b, holomorphic in b, with $g_b(0) = b$. The mutually inverse trivializations are

$$f \longmapsto (f(\infty), g_{f(\infty)}^{-1} \circ f), \qquad (b, h) \longmapsto g_b \circ h.$$

Both maps are holomorphic, and postcomposition preserves the degree. Since $\mathbb{P}^1$ is homogeneous and hence Oka [23, Proposition 5.5.1], the Oka fiber bundle theorem [22, Corollary 1.3] proves that R_d is Oka. Its Stein property then gives holomorphic ellipticity [39, 3.2.A, p. 879]. □

Appendix A. The planar enlargement construction

We follow the two geometric stages of [56, Proposition 2.9]: extending along finitely many radial corridors and then filling the intervening sectors (Figure 1).



(a) Extension along radial segments (b) Adding the sector S_j

Figure 1. Planar enlargement in the disc case, with $K = \overline{\Delta}_r$ and $r < R < R_*$. (a) The map is first extended along neighborhoods of the radial segments Γ_j; the polar rectangles T_j contain ζ_j and cross ∂K. (b) A sector S_j is added to the already constructed set K_{j-1}. The dark set $J_j = K_{j-1} \cap S_j$ comprises the base arc and the portions of both adjacent radial neighborhoods inside S_j. The radial segments extend beyond R_*. Widths are schematic.

Proof of Lemma 2.7. The image of the outer circle has a finite cover by the witness neighborhoods: choose $P_0 \Subset \operatorname{int} \widehat{P} \Subset \operatorname{int} P$ and a finite cyclic cover of the outer circle by slightly enlarged arcs $I_1, \dots, I_\ell$ such that there are coordinate witness neighborhoods $V_1, \dots, V_\ell$ with

$$f(I_j \times \widehat{P}) \Subset V_j.$$

Extension along radial corridors. By a radial corridor we mean a thin neighborhood of one of the radial segments Γ_j, with the rectangle T_j forming its inner part. We first extend the map along finitely many pairwise disjoint such corridors. The proof glues in $N = \ell + 1$ operations: one radial gluing followed by one operation for each sector. Choose a chain of parameter domains and positive errors with

$$P_0 \Subset \operatorname{int} P^{(N)} \Subset \cdots \Subset \operatorname{int} P^{(0)} \Subset \operatorname{int} \widehat{P}, \qquad \epsilon_k > 0, \qquad \sum_{k=1}^{N} \epsilon_k < \epsilon. \tag{A.1}$$

The intermediate polydiscs $\widehat{P}$ and $P^{(k)}$ are chosen concentric with P; we translate the parameter coordinates so that their common center is 0. Choose division points $\zeta_j \in I_j \cap I_{j+1}$, with cyclic indices, so that the arc from ζ_{j-1} to ζ_j lies in I_j and has angular width less than π. Then

$$f(\{\zeta_j\} \times \widehat{P}) \Subset V_j \cap V_{j+1}.$$

By Lemma 2.4, choose a chart spray $f^\sharp$ near $K \times \widehat{P}$; the planar source times a polydisc satisfies its triviality hypothesis. In applying LAP to this spray, the original parameters $t \in \mathbb{C}^q$ and the fiber variables $v \in \mathbb{C}^n$ together form the parameter tuple $(t, v) \in \mathbb{C}^{q+n}$. Only the planar domain is enlarged. After reducing its fiber radius, choose mutually disjoint thin filled polar rectangles T_j crossing the outer circle at ζ_j so that

$$f^\sharp(T_j \times \widehat{P} \times \overline{\Delta}^n_\beta) \Subset V_j \cap V_{j+1}.$$

Attach to the outer edge of T_j a radial segment L_j ending beyond radius R, and denote its attachment point by s_j^*. Set

$$E_j = T_j \cup L_j, \qquad w = (t, v).$$

Let Γ_j be the radial segment from ζ_j to the outer endpoint of L_j. Choose fixed radii $r < a < b$ in the initial portions of these segments such that

$$\Gamma_j \cap \{a \le |s| \le b\} \Subset \operatorname{int} T_j.$$

Fix a radial cutoff on $\{a < |s| < b\}$ and a disc containing all the planar domains used below. Lemma 2.5 then gives a common bound C_* depending only on this fixed data, independently of the subsequent polynomial approximations and narrowings. Choose a target coordinate chart ψ_j whose domain contains V_j, and extend the spray coordinates by

$$u_j(s, w) = \begin{cases} \psi_j(f^\sharp(s, w)), & s \in T_j, \\ \psi_j(f^\sharp(s_j^*, w)), & s \in L_j. \end{cases}$$

The set E_j has connected complement. Let $W = P^{(0)} \times \overline{\Delta}^n_{\beta/2}$, and fix radii $0 < \rho_0 < \widetilde{\rho}_0 < \beta/2$. Since the overlap portions lie in the fixed rectangles T_j and the splitting bound is C_*, choose positive numbers η_j^{sp} sufficiently small for the transition estimates, the error bound ϵ_1 on $K \times P^{(1)}$, and the finitely many target-chart margins used below. Uniform continuity of ψ_j^{-1} on a fixed compact coordinate neighborhood then gives $\delta_j > 0$ such that coordinate approximation within δ_j implies target approximation within η_j^{sp} and preserves half of the coordinate-image margin.

Truncate the Taylor expansion in w uniformly on $E_j \times W$ with coordinate error less than $\delta_j/2$. Each of the resulting coefficient functions is continuous on E_j and holomorphic on its interior. Mergelyan's theorem [17, Theorem 3], applied to these finitely many coefficients, therefore gives a polynomial map

$$p_j(s, w) = \sum_{|\nu| \le M} p_{j,\nu}(s) w^\nu, \qquad \|u_j - p_j\|_{E_j \times W} < \delta_j. \tag{A.2}$$

The compact coordinate image has a positive margin inside $\psi_j(V_j \cap V_{j+1})$. Thus, by (A.2), if $\|u_j - p_j\|_{E_j \times W}$ is less than half that margin, then p_j maps $E_j \times W$ into $\psi_j(V_j \cap V_{j+1})$. The same holds on $U_j \times W$ for an open neighborhood $U_j \supset E_j$ thin enough. The map $G_j = \psi_j^{-1} \circ p_j$

is then holomorphic on a neighborhood of $U_j \times W$, maps this product into $V_j \cap V_{j+1}$, and satisfies

$$\sup_{T_j \times W} d_Y(G_j, f^\sharp) < \eta_j^{\mathrm{sp}}. \tag{A.3}$$

Take a bounded open neighborhood O of $K \cup \bigcup_j \Gamma_j$ such that

$$\begin{aligned} \overline{O \cap \{a \le |s| \le b\}} &\subset \bigcup_j \operatorname{int} T_j, \\ O \cap \{|s| > a\} &\subset \bigcup_j U_j. \end{aligned} \tag{A.4}$$

Shrink O so that $f^\sharp$ is holomorphic on a neighborhood of $(O \cap \{|s| < b\}) \times W$; the neighborhoods U_j are pairwise disjoint. By (A.4), set

$$A = O \cap \{|s| < b\}, \quad B = O \cap \{|s| > a\}, \qquad F_A = f^\sharp, \quad F_B|_{B \cap U_j} = G_j.$$

Consequently

$$A \cup B = O, \qquad A \cap B = O \cap \{a < |s| < b\}, \qquad K \subset A,$$
$$\overline{A \setminus B} \cap \overline{B \setminus A} = \varnothing.$$

Choose an open polydisc $\widetilde{P}^{(1)}$ such that

$$P^{(1)} \Subset \widetilde{P}^{(1)} \Subset \operatorname{int} P^{(0)}.$$

After shrinking this parameter polydisc if necessary,

$$A \times \widetilde{P}^{(1)} \times \Delta^n_{\widetilde{\rho}_0} \subset \operatorname{dom} F_A,$$
$$(B \cap U_j) \times \widetilde{P}^{(1)} \times \Delta^n_{\widetilde{\rho}_0} \subset \operatorname{dom} F_B \qquad (1 \le j \le \ell).$$

Apply Lemma 2.5 to (A, B) with passive parameter domain $\widetilde{P}^{(1)}$, using the fixed cutoff and the bound C_*. On the overlap and $\widetilde{P}^{(1)} \times \Delta^n_{\widetilde{\rho}_0}$, write

$$\gamma_0(s,t,v) = f^\sharp(s,t,\cdot)^{-1}(F_B(s,t,v)) = v + c_0(s,t,v). \tag{A.5}$$

Write

$$e_0 = \|c_0\|_{\infty,(A\cap B)\times \widetilde{P}^{(1)} \times \Delta^n_{\rho_0}}.$$

Equation (A.3) and Cauchy estimates from the larger fiber polydisc make $c_0, D_v c_0$ in (A.5) satisfy (2.3), with $e = e_0$ and $C_0 = C_*$ by the preceding choice of η_j^{sp}, with $e_0 < \widetilde{\rho}_0 - \rho_0$ so that the transition on $\Delta^n_{\rho_0}$ takes values in the outer chart. The same choice ensures that the resulting corrections preserve the target-chart margins on the radial and sector interfaces. Applying Lemma 2.6, with working radius ρ_0 and the inverse chart of $f^\sharp$ on $\Delta^n_{\widetilde{\rho}_0}$, gives corrections a_0, b_0. For t near $P^{(1)}$, the resulting map and its distance to f satisfy

$$\begin{aligned} F_0(s,t) &= \begin{cases} f^\sharp(s,t,a_0(s,t)), & s \in A, \\ G_j(s,t,b_0(s,t)), & s \in B \cap U_j, \end{cases} \\ \sup_{K \times P^{(1)}} d_Y(F_0, f) &\le \omega_{f^\sharp, K \times P^{(1)}}(C_* e_0) < \epsilon_1. \end{aligned} \tag{A.6}$$

The two expressions in (A.6) agree on $A \cap B$ by Lemma 2.6. Choose an outer radius $R_* > R$ below the outer endpoint of every Γ_j; each segment therefore meets the circle $\{|s| = R_*\}$. Choose thin polar rectangular neighborhoods U_j^0 of these segments with

$$U_j^0 \Subset O \cap U_j, \qquad \Gamma_j \cap \{|s| \le R_*\} \subset U_j^0.$$

They are pairwise disjoint and thin enough that each lies in its own angular wedge. After a cyclic relabeling of the radial segments we may also assume that U_{j-1}^0 and U_j^0 bound the

sector S_j introduced below. For small corrections their images remain in $V_j \cap V_{j+1}$, and F_0 is holomorphic near the compact set

$$K_0 = K \cup \bigcup_{j=1}^{\ell} \overline{U_j^0}.$$

At this point F_0 is holomorphic near $K_0 \times P^{(1)}$, satisfies the estimate on $K \times P^{(1)}$ in (A.6), and maps each corridor U_j^0 into the corresponding intersection $V_j \cap V_{j+1}$. These properties permit the sector-filling step.

Filling the intervening sectors. Let S_j, $1 \le j \le \ell$, be the closed polar sector between Γ_{j-1} and Γ_j, of radial range $r \le |s| \le R_*$, with indices read cyclically. Write

$$K_j = K_0 \cup S_1 \cup \cdots \cup S_j;$$

then $S_i \cap S_j \subset K_0$ for $i \ne j$, $S_i \cap K_0 = J_i$, and $K_R \subset K_\ell$. Distinct sectors therefore meet only along radial edges contained in K_0, and the unfilled sectors have the fixed joining sets

$$J_j = K_0 \cap S_j = K_{j-1} \cap S_j.$$

The base arc of J_j lies in I_j and its radial portions lie in $U_{j-1}^0 \cup U_j^0$. By the preceding choice of the approximation errors,

$$F_0(J_j \times P^{(1)}) \Subset V_j \qquad (1 \le j \le \ell).$$

Claim. Let $1 \le j \le \ell$ and let $k = j+1$. Suppose that F_{j-1} is holomorphic near $K_{j-1} \times P^{(k-1)}$ and

$$F_{j-1}(J_i \times P^{(k-1)}) \Subset V_i \qquad (j \le i \le \ell).$$

Then there is a map F_j holomorphic near $K_j \times P^{(k)}$ such that

$$\sup_{K_{j-1} \times P^{(k)}} d_Y(F_j, F_{j-1}) < \epsilon_k \tag{A.7}$$

and

$$F_j(J_i \times P^{(k)}) \Subset V_i \qquad (j < i \le \ell).$$

Proof of the claim. Choose a thin U-shaped closed band D_j around J_j and a larger disc D_j' such that

$$S_j \cap K_{j-1} \subset \operatorname{int} D_j, \qquad F_{j-1} \text{ is holomorphic near } D_j \times P^{(k-1)},$$
$$S_j \subset \operatorname{int} D_j', \qquad D_j \subset D_j' \text{ is a special planar disc pair.}$$

Shrinking the band if necessary, we have

$$F_{j-1}(D_j \times P^{(k-1)}) \Subset V_j.$$

After reducing its fiber radius, a chart spray for F_{j-1} near K_{j-1} therefore satisfies

$$F_{j-1}^\sharp(D_j \times P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n) \Subset V_j.$$

Choose an open band $\mathcal{W}_j \Subset \operatorname{int} D_j$ containing $S_j \cap K_{j-1}$. The compact sets $K_{j-1} \setminus \mathcal{W}_j$ and $S_j \setminus \mathcal{W}_j$ are disjoint. A smooth cutoff separating them, followed by shrinking a bounded neighborhood O_j of K_j, gives bounded planar open sets $A_j, B_j \subset O_j$ such that

$$\begin{aligned} K_j \subset O_j = A_j \cup B_j, &\qquad \overline{A_j \setminus B_j} \cap \overline{B_j \setminus A_j} = \varnothing, \\ A_j \cap B_j \Subset \operatorname{int} D_j, &\qquad B_j \cap K_{j-1} \Subset \operatorname{int} D_j, \end{aligned} \tag{A.8}$$

with A_j compactly contained in the base domain of $F_{j-1}^\sharp$ and $B_j \Subset \operatorname{int} D_j'$. Fix the planar cutoff and the resulting splitting bound C_j at this point.

At this operation the n spray variables join the original q passive parameters. Prescribe the retained parameter domains

$$W_j = P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n, \qquad W_j' = P^{(k)} \times \overline{\Delta}_{\beta_j'}^n, \qquad W_j' \Subset \operatorname{int} W_j, \qquad 0 < \rho_j < \widetilde{\rho}_j < \beta_j' < \beta_j,$$

before applying LAP. If $j < \ell$, also choose $\mu_j > 0$ such that

$$\{y \in Y : d_Y(y, F_{j-1}(J_i \times P^{(k-1)})) < \mu_j\} \subset V_i \qquad (i > j)$$

for every $i > j$; such a common margin exists by the compact containments in the hypothesis of the claim. With C_j, the inverse-chart margins, and, when $j < \ell$, the margin μ_j fixed, choose $\eta_j^{\mathrm{sec}} > 0$ sufficiently small for the gluing estimates below. LAP gives a holomorphic map H_j near $D_j' \times W_j'$ with

$$\sup_{D_j \times W_j'} d_Y(H_j, F_{j-1}^\sharp) < \eta_j^{\mathrm{sec}}. \tag{A.9}$$

After choosing H_j, choose an open polydisc $\widetilde{P}^{(k)}$ with

$$P^{(k)} \Subset \widetilde{P}^{(k)} \Subset \operatorname{int} P^{(k-1)}.$$

By taking this polydisc sufficiently close to $P^{(k)}$, we ensure

$$\begin{aligned} A_j \times \widetilde{P}^{(k)} \times \Delta^n_{\widetilde{\rho}_j} &\subset \operatorname{dom} F_{j-1}^\sharp, \\ B_j \times \widetilde{P}^{(k)} \times \Delta^n_{\widetilde{\rho}_j} &\subset \operatorname{dom} H_j. \end{aligned} \tag{A.10}$$

After further shrinking $\widetilde{P}^{(k)}$ about $P^{(k)}$, we also have

$$\sup_{\overline{A_j \cap B_j} \times \widetilde{P}^{(k)} \times \overline{\Delta}^n_{\widetilde{\rho}_j}} d_Y(H_j, F_{j-1}^\sharp) < 2\eta_j^{\mathrm{sec}}.$$

Indeed, $\overline{A_j \cap B_j} \Subset \operatorname{int} D_j$ and $\widetilde{\rho}_j < \beta_j'$, so this follows from (A.9), continuity, and compactness. Apply Lemma 2.5 to (A_j, B_j) with passive parameter domain $\widetilde{P}^{(k)}$. Its splitting bound C_j is unchanged by shrinking this parameter domain.

On the overlap and $\widetilde{P}^{(k)} \times \Delta^n_{\widetilde{\rho}_j}$, the transition is

$$\gamma_j = (F_{j-1}^\sharp)^{-1} \circ H_j = \mathrm{id} + c_j,$$

where the inverse and composition act in the fiber variable. Uniform continuity of the fixed inverse charts makes c_j uniformly small on the larger fiber product as $\eta_j^{\mathrm{sec}} \to 0$. Cauchy estimates across the fixed gap $\widetilde{\rho}_j - \rho_j$ then make $D_v c_j$ uniformly small on the smaller product. Since C_j is fixed, the smallness conditions (2.3) hold on $\widetilde{P}^{(k)} \times \Delta^n_{\rho_j}$ for sufficiently small η_j^{sec}. Write

$$e_j = \|c_j\|_{\infty, (A_j \cap B_j) \times \widetilde{P}^{(k)} \times \Delta^n_{\rho_j}}.$$

Taking η_j^{sec} small enough also gives $e_j < \widetilde{\rho}_j - \rho_j$, so the transition on $\Delta^n_{\rho_j}$ takes values in the outer chart. Applying Lemma 2.6 with the splitting operators above, working radius ρ_j, and the inverse chart of $F_{j-1}^\sharp$ on $\Delta^n_{\widetilde{\rho}_j}$ gives corrections a_j, b_j such that

$$a_j = \gamma_j(s, t, b_j), \qquad \|a_j\|_{\infty, A_j \times P^{(k)}} + \|b_j\|_{\infty, B_j \times P^{(k)}} \le C_j e_j. \tag{A.11}$$

These corrections give the map near $K_j \times P^{(k)}$

$$F_j(s,t) = \begin{cases} F_{j-1}^\sharp(s, t, a_j(s,t)), & s \in A_j, \\ H_j(s, t, b_j(s,t)), & s \in B_j. \end{cases} \tag{A.12}$$

Equation (A.11) makes the two expressions agree on $A_j \cap B_j$. On $B_j \cap K_{j-1}$, (A.8) and (A.10) make (A.9) applicable. Hence the continuity modulus ω_j of the fixed old spray gives

$$\sup_{K_{j-1} \times P^{(k)}} d_Y(F_j, F_{j-1}) \le \eta_j^{\mathrm{sec}} + \omega_j(C_j e_j).$$

By the choice of η_j^{sec}, the preceding upper bound is less than ϵ_k and, when $j < \ell$, also less than $\mu_j/2$. This proves (A.7) and preserves all the remaining chart containments. □

Apply the claim successively for $j = 1, \dots, \ell$, starting with F_0. Its conclusion supplies the hypothesis for the next sector. Let $F = F_\ell$. Since $K_R \subset K_\ell$ and $P_0 \Subset \operatorname{int} P^{(N)}$, the map F is holomorphic near $K_R \times P_0$. Finally, (A.1), (A.6), and (A.7) give

$$\sup_{K\times P_0} d_Y(F, f) \le \epsilon_1 + \sum_{k=2}^{N} \epsilon_k < \epsilon.$$

□

Appendix B. The polydisc approximation argument

Proof of Proposition 2.9. We use the polynomial source coordinates and exhaustion of [56, Proposition 3.4]. The thin transverse fibers ensure that the boundary fibers lie in coordinate witness neighborhoods, as required by Lemma 2.7; for the final gluing we use the compositional splitting theorem [19, Theorem 4.1].

Polynomial source coordinates. An affine change of source variables reduces the problem to $Q = \overline{\Delta}_1^m$. Fix $R > 1$ and $\epsilon > 0$. For $m = 1$, compactness gives finitely many witness neighborhoods along the boundary circle, so Lemma 2.7 gives approximation on every prescribed larger disc. We may then pass directly to the exhaustion below. Hence assume $m \ge 2$.

Choose $b > 1$ such that f is holomorphic near $\overline{\Delta}_b^m$. Fix the data of Lemma 2.8 and $\ell > 0$ with

$$1 < r_m < \cdots < r_1 < r_1 + 3\ell < b.$$

Choose $r_1 + 2\ell < b' < b$. On the original polydisc source, Lemma 2.4 gives a chart spray $f^\sharp(x, v)$ near $\overline{\Delta}_{b'}^m \times \overline{\Delta}_\beta^n$, with $f^\sharp(x, 0) = f(x)$. This spray is fixed before the polynomial degree d is chosen. Fix also $R_+ > R$ and $J \ge 1$ with $a^J R_+ < 1$, and write

$$\Omega = \Delta_{R_+}^m, \qquad h_d = H_d^{-J} L^J.$$

Then $h_d(\Omega) \subset E_{d,J}$, and (2.13) gives $h_d \to \mathrm{id}$ near $\overline{\Delta}_1^m$.

Choose fixed numbers $r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell$ as in (2.10). On the annulus

$$\mathcal{A}_s = \{c_0 \le |s|^{d^{m-1}} \le c_3\},$$

the map $\Phi_d(s, \xi)$ lies in $\Delta_{b'}^m$ for large d, and its diameter as ξ ranges over Ξ_2 is $O(1/d)$ by (2.11). Cover the compact image $f(\overline{\Delta}_{b'}^m)$ by finitely many coordinate witness neighborhoods, using a smaller cover with a positive Lebesgue number. Uniform continuity of $f^\sharp$, the estimate (2.11), and a reduction of the spray radius ensure that, for each fixed s on $|s|^{d^{m-1}} = c_3$, all values

$$f^\sharp(\Phi_d(s, \xi), v), \qquad \xi \in \Xi_2, \quad v \in \overline{\Delta}_\beta^n,$$

lie compactly in one witness neighborhood. The neighborhood may depend on s. Increase d also so that

$$\sup_{\overline{\Delta}_1^m} d_Y(f \circ h_d, f) < \epsilon/2, \qquad \sup_{\overline{\Delta}_1^m} |h_d(z)_1| < c_1, \qquad r_j (C c_2)^{1/d} < b' \quad (j \ge 2),$$

where C is the constant in (2.7). Fix this degree and write $h = h_d$; all source geometry and the spray radius now remain fixed.

Apply Lemma 2.7 to $f^\sharp(\Phi_d(s, \xi), v)$ on $\mathcal{A}_s$, with (ξ, v) as passive parameters. Choose a retained transverse polydisc compactly inside Ξ_2 and still containing $\{|\xi_j| \le B\}$ in its interior, and retain a smaller fiber polydisc with positive margin. The lemma gives a holomorphic map $G(s, \xi, v)$ approximating the whole spray there, with arbitrarily small error, out to a prescribed finite outer radius in s. Choose that radius larger than $c_3^{1/d^{m-1}}$ and larger than

$$\sup\{|s(h(z))| : z \in \Omega,\ |h(z)_1| \ge c_1\}.$$

This supremum is finite by $|s(h(z))|^{d^{m-1}} = |h(z)_1|$ and boundedness of h on $\overline{\Omega}$; the requirement is vacuous if the set is empty. The inner radius is strictly below $c_1^{1/d^{m-1}}$.

On the open cover

$$A = \{z \in \Omega : |h(z)_1| < c_2\}, \qquad B' = \{z \in \Omega : |h(z)_1| > c_1\},$$

consider the sprays

$$F_A(z,v) = f^\sharp(h(z),v), \qquad F_{B'}(z,v) = G(s(h(z)),\xi(h(z)),v).$$

For $z \in A$, (2.7) gives $|h(z)_j| \le r_j(Cc_2)^{1/d} < b'$ for $j \ge 2$, while $|h(z)_1| < c_2 < b'$. Thus F_A is the pullback of the chart spray already fixed on the original polydisc. Lemma 2.8 and the retained margins give $F_{B'}$ on all of B'. On $A \cap B'$, $\Phi_d(s(h(z)),\xi(h(z))) = h(z)$, so $F_{B'}$ approximates F_A uniformly on a fixed fiber domain.

Fix compact source neighborhoods and concentric fiber balls $\overline{B}_\rho \Subset B_{\rho_+}$ in that domain, with a uniform margin for the inverse fiber charts of F_A. Improving the planar approximation gives the transition

$$\gamma(z,v) = F_A(z,\cdot)^{-1}(F_{B'}(z,v)) = v + c(z,v), \qquad \|c\|_\infty \longrightarrow 0,$$

on a fixed neighborhood of the compact overlap used below. The degree, h, Ω, the retained parameter domains, and the inverse-chart margins stay fixed while this error tends to zero.

Compositional gluing. Choose a compact convex set K and numbers d_1, d_2 such that

$$\overline{\Delta}_R^m \Subset \operatorname{int} K \Subset \Omega, \qquad c_1 < d_1 < d_2 < c_2,$$

and set

$$K_A = K \cap \{|h(z)_1| \le d_2\}, \qquad K_B = K \cap \{|h(z)_1| \ge d_1\}.$$

Then $K_A \cup K_B = K$,

$$\overline{K_A \setminus K_B} \cap \overline{K_B \setminus K_A} = \varnothing, \qquad K_A \cap K_B \Subset A \cap B'.$$

For the fixed radii $0 < \rho < \rho_+$ above, Cauchy estimates give

$$\sup_{(K_A \cap K_B)\times\overline{B}_\rho} \|D_v c\| \le C(\rho_+ - \rho)^{-1}\,\|c\|_\infty \longrightarrow 0.$$

Hence, for sufficiently accurate planar approximation,

$$\Gamma(z,v) = (z,\gamma(z,v)), \qquad |\gamma(z,v) - \gamma(z,w)| \ge \tfrac{1}{2}|v-w|$$

on a fixed smaller fiber ball. After reducing ρ if necessary, Γ is defined and biholomorphic onto its image on an open neighborhood of $(K_A \cap K_B) \times \overline{B}_\rho$. It preserves the vertical foliation $\{z = \text{constant}\}$.

The compact sets $K_A \times \overline{B}_\rho$ and $K_B \times \overline{B}_\rho$ have separated closed fringes, while their union $K \times \overline{B}_\rho$ is convex and hence has a basis of Stein neighborhoods. Thus the ordered pair $(K_B \times \overline{B}_\rho, K_A \times \overline{B}_\rho)$ satisfies the hypotheses of [19, Theorem 4.1]. Let $\omega_{A,\overline{\Delta}_1^m}$ be a continuity modulus of F_A over the unit polydisc. Choose $\eta_{\mathrm{gl}} > 0$ smaller than the reserved fiber margin and such that

$$\omega_{A,\overline{\Delta}_1^m}(\eta_{\mathrm{gl}}) < \epsilon/2.$$

At this point the degree, h, Ω, K_A, K_B, the source neighborhoods, and the fiber margins are fixed. The inverse-chart bounds, the Cauchy bounds, and the smallness threshold in the splitting theorem therefore remain fixed as the planar approximation error tends to zero. Making $\|c\|_\infty$ sufficiently small, the foliation-preserving form of the splitting theorem gives neighborhoods $U_A \supset K_A$, $U_B \supset K_B$ and fiber maps α_A, α_B satisfying

$$\gamma(z,\alpha_B(z,v)) = \alpha_A(z,v), \qquad \|\alpha_A - \mathrm{id}\|_\infty + \|\alpha_B - \mathrm{id}\|_\infty < \eta_{\mathrm{gl}}.$$

After restricting $U_A \Subset A$ and $U_B \Subset B'$, the map

$$\widetilde{f}(z) = \begin{cases} F_A(z,\alpha_A(z,0)), & z \in U_A, \\ F_{B'}(z,\alpha_B(z,0)), & z \in U_B \end{cases}$$

is holomorphic near $\overline{\Delta}_R^m$. Since $\overline{\Delta}_1^m \subset K_A$, we have

$$\sup_{\overline{\Delta}_1^m} d_Y(\widetilde{f}, f) \le \omega_{A,\overline{\Delta}_1^m}(\eta_{\mathrm{gl}}) + \sup_{\overline{\Delta}_1^m} d_Y(f \circ h, f) < \epsilon.$$

Exhaustion. This proves extension to every prescribed finite radius for all $m \ge 1$. Set $f_0 = f$. As in the final exhaustion in the proof of [56, Proposition 3.4], choose radii $1 = R_0 < R_1 < \cdots \to \infty$ and, after an affine rescaling at each stage, maps f_j holomorphic near $\overline{\Delta}_{R_j}^m$ such that, for $j \ge 1$,

$$\sup_{\overline{\Delta}_{R_{j-1}}^m} d_Y(f_j, f_{j-1}) < \epsilon 2^{-j-1}.$$

Completeness of d_Y gives a limit F, uniformly on compact subsets of $\mathbb{C}^m$. Moreover, $\sup_{\overline{\Delta}_1^m} d_Y(F, f) < \epsilon$. In a target chart, local uniform convergence and the scalar uniform limit theorem give $F \in \mathscr{O}(\mathbb{C}^m, Y)$. □

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Statement on the use of artificial intelligence

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References

[1] A. Alarcón and F. Forstnerič. Oka-1 manifolds. *Mathematische Zeitschrift* 311 (2025), no. 2, Paper No. 33, 34 pp. doi:10.1007/s00209-025-03833-4.

[2] A. Alzati, E. Ballico, and G. Ottaviani. The theorem of Mather on generic projections for singular varieties. *Geometriae Dedicata* 85 (2001), 113–117. doi:10.1023/A:1010383721612. https://people.dimai.unifi.it/ottaviani/pubblic/singmather.pdf.

[3] S. An and S.-Y. Xie. *Every smooth complex Fano threefold is Oka.* Private note dated October 6, 2026, 62 pp. https://xiesongyan.github.io/papers/an_xie_every_smooth_complex_fano_threefold_oka_2026.pdf.

[4] F. Campana. Connexité rationnelle des variétés de Fano. *Annales scientifiques de l'École Normale Supérieure* (4) 25 (1992), 539–545. doi:10.24033/asens.1658.

[5] F. Campana and J. Winkelmann. Dense entire curves in rationally connected manifolds. With an appendix by J. Kollár. *Algebraic Geometry* 10 (2023), 521–553. doi:10.14231/AG-2023-018.

[6] J. Cao and A. Höring. Manifolds with nef anticanonical bundle. *Journal für die reine und angewandte Mathematik* 724 (2017), 203–244. Author version: arXiv:1305.1018v2; author's erratum (13 September 2015).

[7] J. Carlson and P. Griffiths. A defect relation for equidimensional holomorphic mappings between algebraic varieties. *Annals of Mathematics* (2) 95 (1972), 557–584. doi:10.2307/1970871. https://publications.ias.edu/sites/default/files/defectrelation.pdf.

[8] Q. Chen and Y. Zhu. $\mathbb{A}^1$-curves on log smooth varieties. *Journal für die reine und angewandte Mathematik* 756 (2019), 1–35. doi:10.1515/crelle-2017-0028. https://math-zhu.github.io/papers/CZ_A1_pair.pdf.

[9] O. Debarre. *Rational curves on algebraic varieties.* Lecture notes for GAeL XVIII and the Strasbourg Semaine spéciale, version November 14, 2020. https://perso.imj-prg.fr/wp-content/uploads/debarre-pub/NotesGAEL.pdf.

[10] O. Debarre. Variétés rationnellement connexes (d'après T. Graber, J. Harris, J. Starr et A. J. de Jong). *Astérisque* 290 (2003), Séminaire Bourbaki, Exp. No. 905, 243–266. https://www.numdam.org/item/SB_2001-2002__44__243_0/.

[11] J.-P. Demailly, T. Peternell, and M. Schneider. Compact complex manifolds with numerically effective tangent bundles. *Journal of Algebraic Geometry* 3 (1994), no. 2, 295–345.

[12] Y.-H. Du. A Levi-index criterion for Oka complements of compact domains. Preprint, 2026. https://duyunheng.github.io/papers/A_Levi-index_criterion_for_Oka_complements_of_compact_domains.pdf. Accessed October 11, 2026.

[13] Y.-H. Du, B. Guo, P.-C. Wang, and S.-Y. Xie. *Periodic transport and Oka complements in* $\mathbb{C}^n$. Preprint, 2026. arXiv:2609.28089v2.

[14] Y.-H. Du, B. Guo, and S.-Y. Xie. *Analytic and algebraic Oka-1 approximation for smooth projective morphisms with rationally connected fibers*. Preprint, 2026. arXiv:2609.11883v1.

[15] Y.-H. Du, B. Guo, and S.-Y. Xie. *Analytic construction of rational curves on Fano manifolds*. Preprint, 2026. arXiv:2609.11612.

[16] B. Farb and J. Wolfson. Topology and arithmetic of resultants, II: the resultant = 1 hypersurface. With an appendix by C. Cazanave. *Algebraic Geometry* 4 (2017), no. 3, 337–352. doi:10.14231/AG-2017-019.

[17] J. E. Fornæss, F. Forstnerič, and E. F. Wold. Holomorphic approximation: the legacy of Weierstrass, Runge, Oka–Weil, and Mergelyan. In D. Breaz and M. Th. Rassias (eds.), *Advancements in Complex Analysis: From Theory to Practice*, 133–192. Springer, Cham, 2020. doi:10.1007/978-3-030-40120-7_5.

[18] F. Forstnerič. The Oka principle for sections of subelliptic submersions. *Mathematische Zeitschrift* 241 (2002), no. 3, 527–551. doi:10.1007/s00209-002-0429-3.

[19] F. Forstnerič. Noncritical holomorphic functions on Stein manifolds. *Acta Mathematica* 191 (2003), 143–189. doi:10.1007/BF02392963.

[20] F. Forstnerič. Extending holomorphic mappings from subvarieties in Stein manifolds. *Annales de l'Institut Fourier* 55 (2005), no. 3, 733–751. doi:10.5802/aif.2112.

[21] F. Forstnerič. Runge approximation on convex sets implies the Oka property. *Annals of Mathematics* 163 (2006), no. 2, 689–707. doi:10.4007/annals.2006.163.689.

[22] F. Forstnerič. Oka manifolds. *Comptes Rendus Mathématique* 347 (2009), nos. 17–18, 1017–1020. doi:10.1016/j.crma.2009.07.005.

[23] F. Forstnerič. *Stein Manifolds and Holomorphic Mappings: The Homotopy Principle in Complex Analysis*. First edition. Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, vol. 56. Springer, 2011. doi:10.1007/978-3-642-22250-4.

[24] F. Forstnerič. Oka manifolds: From Oka to Stein and back. With an appendix by F. Lárusson. *Annales de la Faculté des Sciences de Toulouse. Mathématiques (6)* 22 (2013), no. 4, 747–809. doi:10.5802/afst.1388.

[25] F. Forstnerič. *Stein Manifolds and Holomorphic Mappings: The Homotopy Principle in Complex Analysis*. Second edition. Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, vol. 56. Springer, 2017. doi:10.1007/978-3-319-61058-0. See also the author's errata, January 23, 2023.

[26] F. Forstnerič. Surjective holomorphic maps onto Oka manifolds. In D. Angella, C. Medori, and A. Tomassini (eds.), *Complex and Symplectic Geometry*, 73–84. Springer INdAM Series, vol. 21. Springer, Cham, 2017. doi:10.1007/978-3-319-62914-8_6. https://users.fmf.uni-lj.si/forstneric/papers/2017INdAM.pdf.

[27] F. Forstnerič. Recent developments on Oka manifolds. *Indagationes Mathematicae (N.S.)* 34 (2023), no. 2, 367–417. doi:10.1016/j.indag.2023.01.005. https://users.fmf.uni-lj.si/forstneric/papers/2023Indag.pdf.

[28] F. Forstnerič. From Stein manifolds to Oka manifolds: the h-principle in complex analysis. In *Proceedings of the International Congress of Mathematicians 2026*, vol. 4, 657–677. SIAM, 2026. doi:10.1137/25M1802808.

[29] F. Forstnerič and Y. Kusakabe. Oka tubes in holomorphic line bundles. *Mathematische Annalen* 391 (2025), no. 4, 5265–5292. doi:10.1007/s00208-024-03051-z. https://arxiv.org/abs/2310.14871v4.

[30] F. Forstnerič and F. Lárusson. Survey of Oka theory. *New York Journal of Mathematics* 17A (2011), 11–38. https://nyjm.albany.edu/j/2011/17a-2p.pdf.

[31] F. Forstnerič and F. Lárusson. Holomorphic flexibility properties of compact complex surfaces. *International Mathematics Research Notices* 2014 (2014), no. 13, 3714–3734. doi:10.1093/imrn/rnt044.

[32] F. Forstnerič and F. Lárusson. Oka-1 manifolds: new examples and properties. *Mathematische Zeitschrift* 309 (2025), Paper No. 26, 16 pp. doi:10.1007/s00209-024-03649-8.

[33] F. Forstnerič and F. Lárusson. Every projective Oka manifold is elliptic. *Mathematical Research Letters* 33 (2026), no. 1, 297–312. doi:10.4310/MRL.260503003231. https://arxiv.org/abs/2502.20028v6.

[34] F. Forstnerič and T. Ritter. Oka properties of ball complements. *Mathematische Zeitschrift* 277 (2014), nos. 1–2, 325–338. doi:10.1007/s00209-013-1258-2. https://users.fmf.uni-lj.si/forstneric/papers/2014MZ.pdf.

[35] F. Forstnerič and E. F. Wold. Fibrations and Stein neighborhoods. *Proceedings of the American Mathematical Society* 138 (2010), 2037–2042. doi:10.1090/S0002-9939-09-10223-X.

[36] F. Forstnerič and E. F. Wold. Oka domains in Euclidean spaces. *International Mathematics Research Notices* 2024 (2024), no. 3, 1801–1824. doi:10.1093/imrn/rnac347.
[37] F. Forstnerič and E. F. Wold. Holomorphic families of Fatou–Bieberbach domains and applications to Oka manifolds. *Mathematical Research Letters* 27 (2020), no. 6, 1697–1706. https://users.fmf.uni-lj.si/forstneric/papers/2020MRL.pdf.
[38] H. Grauert. Analytische Faserungen über holomorph-vollständigen Räumen. *Mathematische Annalen* 135 (1958), 263–273. EuDML.
[39] M. Gromov. Oka's principle for holomorphic sections of elliptic bundles. *Journal of the American Mathematical Society* 2 (1989), no. 4, 851–897. doi:10.1090/S0894-0347-1989-1001851-9.
[40] B. Guo and S.-Y. Xie. *Smooth quartic hypersurfaces of dimension at least three in projective space are Oka*. Private note dated September 30, 2026, 24 pp. https://xiesongyan.github.io/papers/guo_xie_smooth_quartic_hypersurfaces_oka_2026.pdf.
[41] A. Hanysz. Holomorphic flexibility properties of the space of cubic rational maps. *The Journal of Geometric Analysis* 25 (2015), 1620–1649. doi:10.1007/s12220-014-9487-0.
[42] J. Kollár. Rationally connected varieties over local fields. *Annals of Mathematics* 150 (1999), 357–367. doi:10.2307/121107.
[43] J. Kollár, Y. Miyaoka, and S. Mori. Rational connectedness and boundedness of Fano manifolds. *Journal of Differential Geometry* 36 (1992), 765–779. doi:10.4310/jdg/1214453188.
[44] Y. Kusakabe. Dense holomorphic curves in spaces of holomorphic maps and applications to universal maps. *International Journal of Mathematics* 28 (2017), no. 4, 1750028, 15 pp. doi:10.1142/S0129167X17500288. https://arxiv.org/abs/1702.08022v1.
[45] Y. Kusakabe. An implicit function theorem for sprays and applications to Oka theory. *International Journal of Mathematics* 31 (2020), no. 9, 2050071, 9 pp. doi:10.1142/S0129167X20500718. https://arxiv.org/abs/2004.12397v1.
[46] Y. Kusakabe. Oka complements of countable sets and nonelliptic Oka manifolds. *Proceedings of the American Mathematical Society* 148 (2020), no. 3, 1233–1238. doi:10.1090/proc/14832. https://arxiv.org/abs/1907.12024v1.
[47] Y. Kusakabe. Elliptic characterization and localization of Oka manifolds. *Indiana University Mathematics Journal* 70 (2021), no. 3, 1039–1054. doi:10.1512/iumj.2021.70.8454. https://arxiv.org/abs/1808.06290v1.
[48] Y. Kusakabe. Elliptic characterization and unification of Oka maps. *Mathematische Zeitschrift* 298 (2021), nos. 3–4, 1735–1750. doi:10.1007/s00209-020-02661-y. https://arxiv.org/abs/2003.03757v1.
[49] Y. Kusakabe. Thom's jet transversality theorem for regular maps. *The Journal of Geometric Analysis* 31 (2021), no. 6, 6031–6041. doi:10.1007/s12220-020-00515-x. https://arxiv.org/abs/2004.13539v1.
[50] Y. Kusakabe. *Surjective morphisms onto subelliptic varieties*. Preprint, 2022. arXiv:2212.06412v1.
[51] Y. Kusakabe. Oka properties of complements of holomorphically convex sets. *Annals of Mathematics* (2) 199 (2024), no. 2, 899–917. doi:10.4007/annals.2024.199.2.7. https://arxiv.org/abs/2005.08247v2.
[52] Y. Kusakabe and S. Matsumura. *Atiyah classes and ellipticity of Oka manifolds*. Preprint, 2026. arXiv:2610.12175v1.
[53] S. Matsumura. *Fundamental groups of compact Kähler manifolds with semi-positive holomorphic sectional curvature*. *Forum of Mathematics, Sigma*, to appear. arXiv:2502.00367v1.
[54] K. Nomizu and H. Ozeki. The existence of complete Riemannian metrics. *Proceedings of the American Mathematical Society* 12 (1961), no. 6, 889–891. doi:10.1090/S0002-9939-1961-0133785-8.
[55] K. Oka. Sur les fonctions analytiques de plusieurs variables. III. Deuxième problème de Cousin. *Journal of Science of the Hiroshima University, Series A* 9 (1939), 7–19. doi:10.32917/HMJ/1558490525.
[56] OpenAI. *Every complex K3 surface is Oka*. Preprint dated September 23, 2026, 56 pp.; repository version of October 6, 2026, commit `adc7f1241b42`. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Every-complex-K3-surface-is-Oka-September-23-2026/paper.pdf.
[57] M.-H. Schwartz. *Lectures on Stratification of Complex Analytic Sets*. Tata Institute of Fundamental Research, Bombay, 1966. https://mathweb.tifr.res.in/Documents/Publications/Lectures/tifr38.pdf.
[58] The Stacks Project Authors. *The Stacks Project*, Tag 062Y. https://stacks.math.columbia.edu/tag/062Y. Accessed October 10, 2026.
[59] P.-C. Wang, S.-Y. Xie, and S. Zhao. *Oka properties of genus-one K3 surfaces and generalized Kummer surfaces*. Private note dated October 6, 2026, 43 pp. https://xiesongyan.github.io/papers/wang_xie_zhao_oka_genus_one_k3_kummer_2026.pdf.
[60] S.-Y. Xie. *Elliptic complements of cubic hypersurfaces*. Preprint, 2026. arXiv:2608.02413v3.
[61] S.-Y. Xie and S. Zhao. *Periods of Oka K3 surfaces are dense*. Preprint, 2026. arXiv:2608.24392v2.

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